

**SIKSHA 'O' ANUSANDHAN
DEEMED TO BE UNIVERSITY**



DevOps Summer Training Program 2026

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**Department of Computer Science & Engineering
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Assignment 1

TOPIC : Basic Linux Commands

Date: 08/07/2026

Question 01: Go through Chapter 2 of B.A. Forouzan and clearly understand TCP and IP protocols.

Answer: TCP (Transmission Control Protocol) is a reliable, connection-oriented protocol that ensures data is delivered correctly and in order. IP (Internet Protocol) is responsible for addressing and routing data packets from the source to the destination over a network. Together TCP/IP enables reliable communication between devices over the Internet.

Question 02: Use the manual page and go through all the options for commands such as: ls, mv, cp, rm, sudo, chmod, vi, nano, touch, ps, kill, curl and wget.

Answer :

man ls :

```
LS(1)                                                    General Commands Manual                               LS(1)

NAME
  ls - list directory contents

SYNOPSIS
  ls [-@BCFGHILOPRSTUWabcdeghiklmnopqrstuvwxy%N,] [--color=when] [-D format] [file ...]

DESCRIPTION
  For each operand that names a file of a type other than directory, ls displays its name as well as any requested, associated information. For each operand that names a file of type directory, ls displays the names of files contained within that directory, as well as any requested, associated information.

  If no operands are given, the contents of the current directory are displayed. If more than one operand is given, non-directory operands are displayed first; directory and non-directory operands are sorted separately and in lexicographical order.

  The following options are available:

  -@      Display extended attribute keys and sizes in long (-l) output.

  -A      Include directory entries whose names begin with a dot ('.') except for . and .. . Automatically set for the super-user unless -I is specified.

  -B      Force printing of non-printable characters (as defined by ctype(3) and current locale settings) in file names as \xxx, where xxx is the numeric value of the character in octal. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

  -C      Force multi-column output; this is the default when output is to a terminal.

  -D format
      When printing in the long (-l) format, use format to format the date and time output. The argument format is a string used by strftime(3). Depending on the choice of format string, this may result in a different number of columns in the output. This option overrides the -T option. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

  -F      Display a slash ('/') immediately after each pathname that is a directory, an asterisk ('*') after each that is executable, an at sign ('@') after each symbolic link, an equals sign ('=') after each socket, a percent sign ('%') after each whiteout, and a vertical bar ('|') after each that is a FIFO.

  -G      Enable colorized output. This option is equivalent to defining CLICOLOR or COLORTERM in the environment and setting --color=auto. (See below.) This functionality can be compiled out by removing the definition of COLORLS. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

  -H      Symbolic links on the command line are followed. This option is assumed if none of the -F, -d, or -l options are specified.

  -I      Prevent -A from being automatically set for the super-user. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

  -L      Follow all symbolic links to final target and list the file or directory the link references rather than the link itself. This option cancels the -P option.

  -O      Include the file flags in a long (-l) output. This option is incompatible with IEEE Std 1003.1-2008 ("POSIX.1"). See chflags(1) for a list of file flags and their meanings.

  -P      If argument is a symbolic link, list the link itself rather than the object the link references. This option cancels the -H and -L options.

  -R      Recursively list subdirectories encountered.

  -S      Sort by size (largest file first) before sorting the operands in lexicographical order.

  -T      When printing in the long (-l) format, display complete time information for the file, including month, day, hour, minute, second, and year. The -D option gives even more control over the output format. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

  -U      Use time when file was created for sorting or printing. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

  -W      Display whiteouts when scanning directories. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

  -a      Include directory entries whose names begin with a dot ('.').

  -b      As -B, but use C escape codes whenever possible. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

  -c      Use time when file status was last changed for sorting or printing.

  --color=when
```

man mv :

MV(1)	General Commands Manual	MV(1)
NAME		
mv -- move files		
SYNOPSIS		
mv [-f -i -n] [-hv] <u>source</u> <u>target</u>		
mv [-f -i -n] [-v] <u>source</u> ... <u>directory</u>		
DESCRIPTION		
In its first form, the mv utility renames the file named by the <u>source</u> operand to the destination path named by the <u>target</u> operand. This form is assumed when the last operand does not name an already existing directory.		
In its second form, mv moves each file named by a <u>source</u> operand to a destination file in the existing directory named by the <u>directory</u> operand. The destination path for each operand is the pathname produced by the concatenation of the last operand, a slash, and the final pathname component of the named file.		
The following options are available:		
-f Do not prompt for confirmation before overwriting the destination path. (The -f option overrides any previous -i or -n options.)		
-h If the <u>target</u> operand is a symbolic link to a directory, do not follow it. This causes the mv utility to rename the file <u>source</u> to the destination path <u>target</u> rather than moving <u>source</u> into the directory referenced by <u>target</u> .		
-i Cause mv to write a prompt to standard error before moving a file that would overwrite an existing file. If the response from the standard input begins with the character 'y' or 'Y', the move is attempted. (The -i option overrides any previous -f or -n options.)		
-n Do not overwrite an existing file. (The -n option overrides any previous -f or -i options.)		
-v Cause mv to be verbose, showing files after they are moved.		
It is an error for the <u>source</u> operand to specify a directory if the target exists and is not a directory.		
If the destination path does not have a mode which permits writing, mv prompts the user for confirmation as specified for the -i option.		
As the rename(2) call does not work across file systems, mv uses cp(1) and rm(1) to accomplish the move. The effect is equivalent to:		
rm -f destination_path && \ cp -prp source_file destination && \ rm -rf source_file		
EXIT STATUS		
The mv utility exits 0 on success, and >0 if an error occurs.		
The command "mv dir/afile dir" will abort with an error message.		
LEGACY DIAGNOSTICS		
In legacy mode, the command "mv dir/afile dir" will fail silently, returning an exit code of 0.		
For more information about legacy mode, see compat(5).		
EXAMPLES		
Rename file <u>foo</u> to <u>bar</u> , overwriting <u>bar</u> if it already exists:		
\$ mv -f foo bar		
COMPATIBILITY		
The -h, -n, and -v options are non-standard and their use in scripts is not recommended.		
The mv utility now supports HFS+ Finder and Extended Attributes and resource forks. The mv utility will no longer strip resource forks off of HFS files. For an alternative method, refer to cp(1).		
SEE ALSO		
cp(1), rm(1), symlink(7)		
:		

Command	Important options	Purpose
ls	-l, -a, -h	List files, show hidden files, human-readable sizes
mv	-i, -v	Ask before overwrite, show operation
cp	-r, -i, -v	Copy directories, ask before overwrite, show operation
rm	-r, -i, -f	Remove directories, confirm deletion, force removal
sudo	-u, -i	Run as another user, start login shell
chmod	-R, -v	Change permissions recursively, show changes
vi	-R, -r	Read-only mode, recover file
nano	-l, -m	Show line numbers, enable mouse
touch	-a, -m	Change access/modification time
ps	-f, -e	Full-format listing, show all processes
kill	-9, -15	Force terminate, terminate gracefully
curl	-I, -O	Show headers, save file with remote name
wget	-O, -q	Specify output file, quiet mode

Question 03: Add your username to the sudoers list.

Command / Answer:

```
skpandey@Ss-MacBook-Air ~ % whoami
skpandey
skpandey@Ss-MacBook-Air ~ % sudo visudo
Password:
/etc/sudoers:60:34: syntax error
includedir /private/etc/sudoers.d
^
What now? ^Cvisudo exiting due to signal: Interrupt: 2
skpandey@Ss-MacBook-Air ~ % sudo visudo
/etc/sudoers:61:34: syntax error
includedir /private/etc/sudoers.d
^
What now? ^Cvisudo exiting due to signal: Interrupt: 2
skpandey@Ss-MacBook-Air ~ % sudo sh -c 'echo "skpandey ALL=(ALL) ALL" >> /etc/sudoers'
skpandey@Ss-MacBook-Air ~ % sudo whoami
root
skpandey@Ss-MacBook-Air ~ % █
```

Question 04: Create a file named notes.txt using touch. Rename notes.txt to final_report.txt and move it into a new directory named archive using mv. Modify the permissions of archive/final_report.txt so the owner has read/write access, the group has read-only access, and others have no access (-rw-r-----). Add some unsorted text to final_report.txt, display its contents using cat, and copy it to another file using output redirection.

Command / Answer:

```
skpandey@Ss-MacBook-Air ~ % touch notes.txt
skpandey@Ss-MacBook-Air ~ % mv notes.txt final_report.txt
skpandey@Ss-MacBook-Air ~ % mkdir archive
mkdir: archive: File exists
skpandey@Ss-MacBook-Air ~ % mv final_report.txt archive/
skpandey@Ss-MacBook-Air ~ % echo -e "Orange\nApple\nMango\nBanana" > archive/final_report.txt
skpandey@Ss-MacBook-Air ~ % cat archive/final_report.txt
Orange
Apple
Mango
Banana
skpandey@Ss-MacBook-Air ~ % chmod 640 archive/final_report.txt
skpandey@Ss-MacBook-Air ~ % ls -l archive/final_report.txt
-rw-r----- 1 skpandey staff 26 Aug 21 12:31 archive/final_report.txt
skpandey@Ss-MacBook-Air ~ % cat archive/final_report.txt > archive/copy.txt
skpandey@Ss-MacBook-Air ~ % cat archive/final_report.txt > archive/copy.txt
skpandey@Ss-MacBook-Air ~ % cat archive/copy.txt
Orange
Apple
Mango
Banana
skpandey@Ss-MacBook-Air ~ % █
```

Question 05: Using output redirection (>) create server.conf with SERVER_NAME=localhost, PORT=8080, DEBUG_MODE=true and MAX_CONN=100. Create /etc/test_config, move the file there using sudo if required, change DEBUG_MODE=true to DEBUG_MODE=false, save and exit, then secure the file so only root can read and write. Verify and report the permissions.

Command / Answer:

```
skpandey@Ss-MacBook-Air ~ % touch notes.txt
skpandey@Ss-MacBook-Air ~ % mv notes.txt final_report.txt
skpandey@Ss-MacBook-Air ~ % mkdir archive
mkdir: archive: File exists
skpandey@Ss-MacBook-Air ~ % mv final_report.txt archive/
skpandey@Ss-MacBook-Air ~ % echo -e "Orange\nApple\nMango\nBanana" > archive/final_report.txt
skpandey@Ss-MacBook-Air ~ % cat archive/final_report.txt
Orange
Apple
Mango
Banana
skpandey@Ss-MacBook-Air ~ % chmod 640 archive/final_report.txt
skpandey@Ss-MacBook-Air ~ % ls -l archive/final_report.txt
-rw-r----- 1 skpandey staff 26 Aug 21 12:31 archive/final_report.txt
skpandey@Ss-MacBook-Air ~ % cat archive/final_report.txt > archive/copy.txt
skpandey@Ss-MacBook-Air ~ % cat archive/final_report.txt > archive/copy.txt
skpandey@Ss-MacBook-Air ~ % cat archive/copy.txt
Orange
Apple
Mango
Banana
skpandey@Ss-MacBook-Air ~ % █
```

Question 06: Run: ping 127.0.0.1 > /dev/null &. Use ps to find the Process ID (PID) of the background task and use kill to stop it.

```

skpandey@MacBook-Air ~ % ping 127.0.0.1 > /dev/null &
[1] 14983
skpandey@MacBook-Air ~ % ps
  PID TTY          TIME CMD
14625 ttys000    0:00.11 -zsh
14983 ttys000    0:00.01 ping 127.0.0.1
skpandey@MacBook-Air ~ % kill 14983
[1] + terminated ping 127.0.0.1 > /dev/null
skpandey@MacBook-Air ~ % ps
  PID TTY          TIME CMD
14625 ttys000    0:00.12 -zsh
skpandey@MacBook-Air ~ % █

```

Command / Answer:

[illegible]

Assignment 2

TOPIC : Bash Scripting

Date: 10/07/2026

Question 01: Write a bash script to find the latest file in any given folder. The folder should be presented as a command line argument. Make a copy of the file and find the most occurring word in the file.

Answer:

```
skpandey@Ss-MacBook-Air ~ % nano q1.sh
skpandey@Ss-MacBook-Air ~ % chmod +x q1.sh
skpandey@Ss-MacBook-Air ~ % ./q1.sh .
Latest file: q1.sh
6 |
skpandey@Ss-MacBook-Air ~ % echo "apple mango apple banana apple mango" > test.txt
skpandey@Ss-MacBook-Air ~ % ./q1.sh .
Latest file: test.txt
3 apple
skpandey@Ss-MacBook-Air ~ % █
```

Question 02: Write a bash script to print the name of the person who last logged into the system. From the home folder find out (if any) files for which this person is the owner. Can you find the total login time of this user.

Answer:

```
skpandey@Ss-MacBook-Air ~ % nano q1.sh
skpandey@Ss-MacBook-Air ~ % chmod +x q1.sh
skpandey@Ss-MacBook-Air ~ % ./q1.sh .
Latest file: q1.sh
6 |
skpandey@Ss-MacBook-Air ~ % echo "apple mango apple banana apple mango" > test.txt
skpandey@Ss-MacBook-Air ~ % ./q1.sh .
Latest file: test.txt
3 apple
skpandey@Ss-MacBook-Air ~ % clear
|
skpandey@Ss-MacBook-Air ~ % nano q2.sh
skpandey@Ss-MacBook-Air ~ % chmod +x q2.sh
skpandey@Ss-MacBook-Air ~ % ./q2.sh
Last logged in user: skpandey
Files owned by skpandey:
/Users/skpandey
/Users/skpandey/ec2_nginx.sh
/Users/skpandey/s3tf
/Users/skpandey/s3tf/terraform.tfstate.backup
/Users/skpandey/s3tf/terraform.tfstate
/Users/skpandey/s3tf/main.tf
/Users/skpandey/s3tf/index.html
/Users/skpandey/s3tf/.terraform
/Users/skpandey/s3tf/.terraform/providers
/Users/skpandey/s3tf/.terraform/providers/registry.terraform.io
/Users/skpandey/s3tf/.terraform/providers/registry.terraform.io/hashicorp
/Users/skpandey/s3tf/.terraform/providers/registry.terraform.io/hashicorp/aws
/Users/skpandey/s3tf/.terraform/providers/registry.terraform.io/hashicorp/aws/6.57.1
/Users/skpandey/s3tf/.terraform/providers/registry.terraform.io/hashicorp/aws/6.57.1/darwin_arm64
/Users/skpandey/s3tf/.terraform/providers/registry.terraform.io/hashicorp/aws/6.57.1/darwin_arm64/terraform-provider-aws_v6.57.1_x5
/Users/skpandey/s3tf/.terraform/providers/registry.terraform.io/hashicorp/aws/6.57.1/darwin_arm64/LICENSE.txt
/Users/skpandey/s3tf/error.html
/Users/skpandey/s3tf/.terraform.lock.hcl
/Users/skpandey/file.sh
/Users/skpandey/.eclipse
/Users/skpandey/.eclipse/org.eclipse.oomph.jreinfo
/Users/skpandey/.eclipse/org.eclipse.oomph.jreinfo/defaults.properties
/Users/skpandey/.eclipse/org.eclipse.oomph.jreinfo/infos.txt
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.update
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.update/platform.xml
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.mainData.1
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.contributors.1
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.namespaces.1
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.contributions.1
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.table.1
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.manager
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.manager/.fileTableLock
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.manager/.fileTable.8
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.manager/.fileTable.7
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.extraData.1
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.core.runtime/.orphans.1
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.osgi
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.osgi/157
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.osgi/157/0
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.osgi/157/0/.cp
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.osgi/157/0/.cp/icons
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.osgi/157/0/.cp/icons/install_wiz.png
/Users/skpandey/.eclipse/1738435388_macosx_cocoa_x86_64/configuration/org.eclipse.osgi/166
```

```

/Users/skpandey/Downloads/WhatsApp Image 2023-08-24 at 9.07.27 AM.jpeg
/Users/skpandey/Downloads/CN_Project6_report.docx
/Users/skpandey/Downloads/Ch-5_lecture_2.pdf
/Users/skpandey/Downloads/Untitled document (6).pdf
/Users/skpandey/Downloads/XIIMarksheet_2.pdf
/Users/skpandey/Downloads/9788770046985_previewpdf.pdf
/Users/skpandey/Downloads/Shrutii Pandey iClass Student Enrollment form.pdf
/Users/skpandey/Downloads/Creativity and Design Modelling Abstract Shapes Affirmations Wraparound Mug.png
/Users/skpandey/Downloads/CDA(EET2211)Assignment1 (1).pdf
/Users/skpandey/Downloads/AnyScanner_03_23_2026(1).pdf
/Users/skpandey/Downloads/_ (2).jpeg
/Users/skpandey/Downloads/WhatsApp Image 2023-08-24 at 9.47.29 AM.jpeg
/Users/skpandey/Downloads/IM_Chapter_25_Production and Growth_01842025.pdf
/Users/skpandey/Downloads/Assignment_2.pdf
/Users/skpandey/Downloads/CSE 3652 Mid-Sem Question April 2025.doc
/Users/skpandey/Downloads/2024_Dec_PPWC_End_Sem_Question_set_1.pdf
/Users/skpandey/Downloads/in.gov.cbse-HSCER-126463592023 (1).pdf
/Users/skpandey/Downloads/CSE 3652 End-Sem Question_2025.pdf
/Users/skpandey/Downloads/Worksheet.pdf
/Users/skpandey/Downloads/Brown Wood Minimalist Profile LinkedIn Banner (1).png
/Users/skpandey/Downloads/Bakhuda Tumhi ho lyrics.mp3
/Users/skpandey/Downloads/6a238bf5-252c-4577-9405-c929fb2208af.jpeg
/Users/skpandey/.cache
/Users/skpandey/.cache/tooling
/Users/skpandey/.cache/tooling/gradle
/Users/skpandey/.cache/tooling/gradle/versions.json
/Users/skpandey/.aws
/Users/skpandey/.aws/config
/Users/skpandey/.aws/cli
/Users/skpandey/.aws/cli/cache
/Users/skpandey/.aws/cli/cache/session.db
/Users/skpandey/.aws/credentials
/Users/skpandey/Shruu
/Users/skpandey/Shruu/twoSum.java
/Users/skpandey/cve_extracted.csv
/Users/skpandey/nginx_script.sh
/Users/skpandey/output.txt
/Users/skpandey/.viminfo
/Users/skpandey/.zsh_sessions
/Users/skpandey/.zsh_sessions/05E1405A-822D-40DC-A559-A20AED6501D8.history
/Users/skpandey/.zsh_sessions/BC36B5D0-2544-480D-B5D3-563A3D02957C.session
/Users/skpandey/.zsh_sessions/CD2B4F8B-1ABF-40BC-A47E-E2908021D7F1.session
/Users/skpandey/.zsh_sessions/078FFC58-E516-409B-B5EA-831A5078A300.historynew
/Users/skpandey/.zsh_sessions/05E1405A-822D-40DC-A559-A20AED6501D8.session
/Users/skpandey/.zsh_sessions/_expiration_check.timestamp
/Users/skpandey/.zsh_sessions/CD2B4F8B-1ABF-40BC-A47E-E2908021D7F1.history
/Users/skpandey/.zsh_sessions/BC36B5D0-2544-480D-B5D3-563A3D02957C.history
/Users/skpandey/.zsh_sessions/138EF0EC-B676-4F2F-807B-9E6C1136816A.history
/Users/skpandey/.zsh_sessions/326D8421-A172-43BF-A0E8-0F93BE3A1690.session
/Users/skpandey/.zsh_sessions/68BBAB15-4A43-49BC-B2C0-B8BE24EC4A2A.session
/Users/skpandey/.zsh_sessions/3AB0D67F-20D0-43B7-8E70-F76102F51391.history
/Users/skpandey/.zsh_sessions/12788B3E-20DA-40A7-8B70-C5F5F294D384.session
/Users/skpandey/.zsh_sessions/F0B2F1D0-817A-48DC-83C9-065815F670EC.history
/Users/skpandey/.zsh_sessions/326D8421-A172-43BF-A0E8-0F93BE3A1690.history
/Users/skpandey/.zsh_sessions/68BBAB15-4A43-49BC-B2C0-B8BE24EC4A2A.history
/Users/skpandey/.zsh_sessions/138EF0EC-B676-4F2F-807B-9E6C1136816A.session
/Users/skpandey/.zsh_sessions/F0B2F1D0-817A-48DC-83C9-065815F670EC.session
/Users/skpandey/.zsh_sessions/3AB0D67F-20D0-43B7-8E70-F76102F51391.session
/Users/skpandey/.zsh_sessions/12788B3E-20DA-40A7-8B70-C5F5F294D384.history
/Users/skpandey/policy.sh
/Users/skpandey/awk.txt
/Users/skpandey/blackjack.js
Login time:
skpandey ttys000 Fri Aug 21 12:22 - 12:22 (00:00)
skpandey@Sk-MacBook-Air ~ %

```

Question 03: Write a bash script to use find command to find the path of conf.d file. From the path replace all “/” symbols and replace them with “-” and print the path.

Answer:

```

skpandey@Sk-MacBook-Air ~ % nano q3.sh
skpandey@Sk-MacBook-Air ~ % chmod +x q3.sh
skpandey@Sk-MacBook-Air ~ % ./q3.sh
find: /Users/skpandey/Library/Application Support/CallHistoryTransactions: Operation not permitted
find: /Users/skpandey/Library/Application Support/CloudDocs/session/db: Operation not permitted
find: /Users/skpandey/Library/Application Support/com.apple.sharedfilelist: Operation not permitted
find: /Users/skpandey/Library/Application Support/AddressBook: Operation not permitted
find: /Users/skpandey/Library/Application Support/com.apple.TCC: Operation not permitted
find: /Users/skpandey/Library/Application Support/FileProvider: Operation not permitted
find: /Users/skpandey/Library/Application Support/AddressBook: Operation not permitted
find: /Users/skpandey/Library/Application Support/com.apple.avfoundation/Frecents: Operation not permitted
find: /Users/skpandey/Library/Application Support/CallHistoryDB: Operation not permitted
find: /Users/skpandey/Library/Assistant/SiriVocabulary: Operation not permitted
find: /Users/skpandey/Library/Autosave Information: Operation not permitted
find: /Users/skpandey/Library/IdentityServices: Operation not permitted
find: /Users/skpandey/Library/Calendars: Operation not permitted
find: /Users/skpandey/Library/Messages: Operation not permitted
find: /Users/skpandey/Library/HomeKit: Operation not permitted
find: /Users/skpandey/Library/Sharing: Operation not permitted
find: /Users/skpandey/Library/Mail: Operation not permitted
find: /Users/skpandey/Library/DuetExpertCenter: Operation not permitted
find: /Users/skpandey/Library/Accounts: Operation not permitted
find: /Users/skpandey/Library/Safari: Operation not permitted
find: /Users/skpandey/Library/Bioime: Operation not permitted
find: /Users/skpandey/Library/Shortcuts: Operation not permitted
find: /Users/skpandey/Library/Suggestions: Operation not permitted
find: /Users/skpandey/Library/Group Containers/group.com.apple.secure-control-center-preferences: Operation not permitted
find: /Users/skpandey/Library/Group Containers/group.com.apple.notes: Operation not permitted
find: /Users/skpandey/Library/Containers/com.apple.archiveutility: Operation not permitted
find: /Users/skpandey/Library/Containers/com.apple.Safari: Operation not permitted
find: /Users/skpandey/Library/Containers/com.apple.CloudDocs.MobileDocumentsFileProvider: Operation not permitted
find: /Users/skpandey/Library/Containers/com.apple.mail: Operation not permitted
find: /Users/skpandey/Library/Containers/com.apple.Notes: Operation not permitted
find: /Users/skpandey/Library/Containers/com.apple.corerecents.recents/Data/Library/Recents: Operation not permitted
find: /Users/skpandey/Library/Containers/com.apple.stocks: Operation not permitted
find: /Users/skpandey/Library/PersonalizationPortrait: Operation not permitted
find: /Users/skpandey/Library/Metadata/CoreSpotlight: Operation not permitted
find: /Users/skpandey/Library/Reminders: Operation not permitted
find: /Users/skpandey/Library/Cookies: Operation not permitted
find: /Users/skpandey/Library/CoreFollowUp: Operation not permitted
find: /Users/skpandey/Library/StatusKit: Operation not permitted
find: /Users/skpandey/Library/Caches/com.apple.HomeKit: Operation not permitted
find: /Users/skpandey/Library/Caches/CloudKit: Operation not permitted
find: /Users/skpandey/Library/Caches/com.apple.Safari: Operation not permitted
find: /Users/skpandey/Library/Caches/FamilyCircle: Operation not permitted
find: /Users/skpandey/Library/Caches/com.apple.homed: Operation not permitted
find: /Users/skpandey/Library/Caches/com.apple.adprivacyd: Operation not permitted
find: /Users/skpandey/.Trash: Operation not permitted
find: /Users/skpandey/Documents: Operation not permitted
Path: /Users/skpandey/conf.d
~Users/skpandey-conf.d
skpandey@Sk-MacBook-Air ~ %

```

Assignment 3

TOPIC : String Manipulation Commands

Date: 17/07/2026

Question 01: Create a directory logs_practice and populate it with these sample files,

```
mkdir logs_practice && cd logs_practice
```

```
cat > sales_jan.csv << 'EOF'
```

```
101,Rahul Verma,North,45000,Active
```

```
102,Sneha Reddy,South,52000,Active
```

```
103,Amit Joshi,East,38000,Inactive
```

```
104,Divya Nair,West,61000,Active
```

```
105,Karan Mehta,North,29000,Inactive
```

```
EOF
```

```
cat > sales_feb.csv << 'EOF'
```

```
106,Pooja Iyer,South,55000,Active
```

```
107,Vikram Rao,East,42000,Active
```

```
108,Neha Gupta,West,33000,Inactive
```

```
EOF
```

```
touch summary.txt notes.md archive.tar.gz
```

Answer the following using the specified tools:

1. (ls) List all files in logs_practice and count how many of them are .csv files, without opening any file.
2. (wc) For each .csv file, print the number of lines (records) it contains. Then find the total number of records across both CSV files in a single command.
3. (cut) From sales_jan.csv, extract only the name and region (2nd and 3rd fields) for every record.
4. (awk) Print full records from both CSV files combined where the status is Active and salary is greater than 40000.
5. (awk + regex) Print only the names of employees whose name starts with a vowel (A, E, I, O, U — case-insensitive).
6. (sed) In a single command, replace all occurrences of Inactive with Terminated across both CSV files, printing the result to the terminal (do not modify the original files).
7. (cut + sort + wc) Extract just the region field from both files combined, and count how many unique regions appear in total.
8. (Pipeline: cat + awk + sed + wc) Combine both CSV files, use awk to filter only Active employees, use sed to strip out the employee ID (1st field), and finally use wc -l to count how many such trimmed records remain.

Answer:


```

skpandey@Ss-MacBook-Air ~ % mkdir logs_practice
skpandey@Ss-MacBook-Air ~ % cd logs_practice
skpandey@Ss-MacBook-Air logs_practice % cat > sales_jan.csv
101,Rahul Verma,North,45000,Active
102,Sneha Reddy,South,52000,Active
103,Amit Joshi,East,38000,Inactive
104,Divya Nair,West,61000,Active
105,Karan Mehta,North,29000,Inactive
skpandey@Ss-MacBook-Air logs_practice % cat > sales_jan.csv
101,Rahul Verma,North,45000,Active
102,Sneha Reddy,South,52000,Active
103,Amit Joshi,East,38000,Inactive
104,Divya Nair,West,61000,Active
105,Karan Mehta,North,29000,Inactive
skpandey@Ss-MacBook-Air logs_practice % cat > sales_feb.csv
106,Pooja Iyer,South,55000,Active
107,Vikram Rao,East,42000,Active
108,Neha Gupta,West,33000,Inactive
skpandey@Ss-MacBook-Air logs_practice % touch summary.txt notes.md archive.tar.gz
skpandey@Ss-MacBook-Air logs_practice % ls
archive.tar.gz  notes.md          sales_feb.csv  sales_jan.csv  summary.txt
skpandey@Ss-MacBook-Air logs_practice % ls
archive.tar.gz  notes.md          sales_feb.csv  sales_jan.csv  summary.txt
skpandey@Ss-MacBook-Air logs_practice % ls *.csv | wc -l
2
skpandey@Ss-MacBook-Air logs_practice % wc -l sales_jan.csv
5 sales_jan.csv
skpandey@Ss-MacBook-Air logs_practice % wc -l sales_feb.csv
3 sales_feb.csv
skpandey@Ss-MacBook-Air logs_practice % wc -l sales_jan.csv sales_feb.csv
5 sales_jan.csv
3 sales_feb.csv
8 total
skpandey@Ss-MacBook-Air logs_practice % cut -d ',' -f 2,3 sales_jan.csv
Rahul Verma,North
Sneha Reddy,South
Amit Joshi,East
Divya Nair,West
Karan Mehta,North
skpandey@Ss-MacBook-Air logs_practice % awk -F ',' '$5=="Active" && $4>40000' sales_jan.csv sales_feb.csv
101,Rahul Verma,North,45000,Active
102,Sneha Reddy,South,52000,Active
104,Divya Nair,West,61000,Active
106,Pooja Iyer,South,55000,Active
107,Vikram Rao,East,42000,Active
skpandey@Ss-MacBook-Air logs_practice % awk -F ',' '$2 ~ /^[AEIOUaeiou]/ {print $2}' sales_jan.csv sales_feb.csv
Amit Joshi
skpandey@Ss-MacBook-Air logs_practice % sed 's/Inactive/Terminated/g' sales_jan.csv sales_feb.csv
101,Rahul Verma,North,45000,Active
102,Sneha Reddy,South,52000,Active
103,Amit Joshi,East,38000,Terminated
104,Divya Nair,West,61000,Active
105,Karan Mehta,North,29000,Terminated
106,Pooja Iyer,South,55000,Active
107,Vikram Rao,East,42000,Active
108,Neha Gupta,West,33000,Terminated
skpandey@Ss-MacBook-Air logs_practice % cut -d ',' -f 3 sales_jan.csv sales_feb.csv | sort | uniq | wc -l
4
skpandey@Ss-MacBook-Air logs_practice % cat sales_jan.csv sales_feb.csv | awk -F ',' '$5=="Active" | sed 's/^[^,]*,/' | wc -l
5

```

Question 02: Print all lines in /etc/passwd containing the word root.

1. Use -i to case-insensitively search for the string ROOT, and compare the output with question 1 are they the same? Why or why not?
2. Use -c to count how many lines contain /bin/bash as the login shell (without printing the actual lines).
3. Use -n to print matching lines along with their line numbers for any user whose shell is /usr/sbin/nologin. Use -v to print all lines that do NOT contain nologin — i.e., users with an actual usable shell.
4. Use -E (extended regex) to find all lines where the shell is either /bin/bash or /bin/sh (use the | alternation operator).
5. Use -w to search for the exact word bin (matching only the whole word, not substrings like/sbin or bin inside another word).
6. Use grep -E with a regex to find lines where the UID (3rd field) is a 3-digit number (e.g., 100–999), by matching the pattern around the colons.
7. Use grep with ^ and \$ anchors to find lines where the username starts with s and print only those lines.
8. Use ls -l /etc/ piped into grep to list only files/directories whose names contain the word conf.
9. Use cat /etc/passwd piped into grep, then piped further into wc -l, to count how many total user accounts exist on the system. Combine ps aux | grep to find all running processes

related to ssh (hint: this will also match the grep process itself — how would you exclude it from the output?).

10. Use `history | grep` (or `cat /etc/passwd | grep`) combined with `cut` to extract just the usernames of accounts matching a grep filter for shell `/bin/bash`.

11. Explain the difference between `grep`, `egrep`, and `grep -E`. Are they functionally equivalent today?

Answer:

```
5
skpandey@Ss-MacBook-Air logs_practice % grep root /etc/passwd
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
_cvmsroot:*:212:212:CVMS Root:/var/empty:/usr/bin/false
skpandey@Ss-MacBook-Air logs_practice % grep -i ROOT /etc/passwd
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
_cvmsroot:*:212:212:CVMS Root:/var/empty:/usr/bin/false
skpandey@Ss-MacBook-Air logs_practice % grep -c "/bin/bash" /etc/passwd
1
skpandey@Ss-MacBook-Air logs_practice % grep -n "/usr/sbin/nologin" /etc/passwd
skpandey@Ss-MacBook-Air logs_practice % grep -v nologin /etc/passwd
##
# User Database
#
# Note that this file is consulted directly only when the system is running
# in single-user mode. At other times this information is provided by
# Open Directory.
#
# See the opendirectoryd(8) man page for additional information about
# Open Directory.
##
nobody:*:2:2:Unprivileged User:/var/empty:/usr/bin/false
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
_uucp:*:4:4:Unix to Unix Copy Protocol:/var/spool/uucp:/usr/sbin/uucico
_taskgated:*:13:13:Task Gate Daemon:/var/empty:/usr/bin/false
_networkd:*:24:24:Network Services:/var/networkd:/usr/bin/false
_installassistant:*:25:25:Install Assistant:/var/empty:/usr/bin/false
_lp:*:26:26:Printing Services:/var/spool/cups:/usr/bin/false
_postfix:*:27:27:Postfix Mail Server:/var/spool/postfix:/usr/bin/false
_scsd:*:31:31:Service Configuration Service:/var/empty:/usr/bin/false
_ces:*:32:32:Certificate Enrollment Service:/var/empty:/usr/bin/false
_appstore:*:33:33:Mac App Store Service:/var/db/appstore:/usr/bin/false
_mcxair:*:54:54:MCX AppLaunch:/var/empty:/usr/bin/false
_appleevents:*:55:55:AppleEvents Daemon:/var/empty:/usr/bin/false
_geod:*:56:56:Geo Services Daemon:/var/db/geod:/usr/bin/false
_devdocs:*:59:59:Developer Documentation:/var/empty:/usr/bin/false
_sandbox:*:60:60:Seatbelt:/var/empty:/usr/bin/false
_mdnsresponder:*:65:65:MDNSResponder:/var/empty:/usr/bin/false
_ard:*:67:67:Apple Remote Desktop:/var/empty:/usr/bin/false
_www:*:70:70:World Wide Web Server:/Library/WebServer:/usr/bin/false
_eppc:*:71:71:Apple Events User:/var/empty:/usr/bin/false
_cvs:*:72:72:CVS Server:/var/empty:/usr/bin/false
_svn:*:73:73:SVN Server:/var/empty:/usr/bin/false
_mysql:*:74:74:MySQL Server:/var/empty:/usr/bin/false
_sshd:*:75:75:sshd Privilege separation:/var/empty:/usr/bin/false
_qtss:*:76:76:QuickTime Streaming Server:/var/empty:/usr/bin/false
_cyrus:*:77:77:Cyrus Administrator:/var/imap:/usr/bin/false
_mailman:*:78:78:Mailman List Server:/var/empty:/usr/bin/false
_appserver:*:79:79:Application Server:/var/empty:/usr/bin/false
_clamav:*:82:82:ClamAV Daemon:/var/virusmails:/usr/bin/false
_amavisd:*:83:83:AMaViS Daemon:/var/virusmails:/usr/bin/false
_jabber:*:84:84:Jabber XMPP Server:/var/empty:/usr/bin/false
_appowner:*:87:87:Application Owner:/var/empty:/usr/bin/false
_windowserver:*:88:88:WindowServer:/var/empty:/usr/bin/false
_spotlight:*:89:89:Spotlight:/var/empty:/usr/bin/false
_tokend:*:91:91:Token Daemon:/var/empty:/usr/bin/false
_securityagent:*:92:92:SecurityAgent:/var/db/securityagent:/usr/bin/false
_calendar:*:93:93:Calendar:/var/empty:/usr/bin/false
_teamsserver:*:94:94:TeamsServer:/var/teamsserver:/usr/bin/false
_update_sharing:*:95:95:Update Sharing:/var/empty:/usr/bin/false
_installer:*:96:96:Installer:/var/empty:/usr/bin/false
_atsserver:*:97:97:ATS Server:/var/empty:/usr/bin/false
```

`grep root /etc/passwd` and `grep -i ROOT /etc/passwd` are usually same for root because the matching text is already lowercase. `-i` simply makes the search case-insensitive.

`grep`, `egrep`, and `grep -E`:

`grep` is used to search for patterns in text. `egrep` was used for extended regular expressions. `grep -E` also supports extended regular expressions. Today `grep -E` is preferred because `egrep` is considered an older form of the same functionality.

```

_coremediaiod:*236:236:Core Media IO Daemon:/var/empty:/usr/bin/false
_launchservicesd:*239:239:_launchservicesd:/var/empty:/usr/bin/false
_iconservices:*240:240:IconServices:/var/empty:/usr/bin/false
_distnote:*241:241:DistNote:/var/empty:/usr/bin/false
_nsurlsessiond:*242:242:NSURLSession Daemon:/var/db/nsurlsessiond:/usr/bin/false
_displaypolicyd:*244:244:Display Policy Daemon:/var/empty:/usr/bin/false
_astris:*245:245:Astris Services:/var/db/astris:/usr/bin/false
_krbfast:*246:246:Kerberos FAST Account:/var/empty:/usr/bin/false
_gamecontrollerd:*247:247:Game Controller Daemon:/var/empty:/usr/bin/false
_mbssetupuser:*248:248:Setup User:/var/setup/bin/bash
_ondemand:*249:249:On Demand Resource Daemon:/var/db/ondemand:/usr/bin/false
_xserverdocs:*251:251:macOS Server Documents Service:/var/empty:/usr/bin/false
_wwwproxy:*252:252:WWW Proxy:/var/empty:/usr/bin/false
_mobileasset:*253:253:MobileAsset User:/var/ma:/usr/bin/false
_findmydevice:*254:254:Find My Device Daemon:/var/db/findmydevice:/usr/bin/false
_datadetectord:*257:257:DataDetectors:/var/db/datadetectors:/usr/bin/false
_captiveagent:*258:258:captiveagent:/var/empty:/usr/bin/false
_ctkd:*259:259:ctkd Account:/var/empty:/usr/bin/false
_applepay:*260:260:applepay Account:/var/db/applepay:/usr/bin/false
_hid:*261:261:HID Service User:/var/db/hidd:/usr/bin/false
_cmiodalassistants:*262:262:CoreMedia IO Assistants User:/var/db/cmiodalassistants:/usr/bin/false
_analyticisd:*263:263:Analytics Daemon:/var/db/analyticisd:/usr/bin/false
_fpsd:*265:265:FPS Daemon:/var/db/fpsd:/usr/bin/false
_timed:*266:266:Time Sync Daemon:/var/db/timed:/usr/bin/false
_nearbyd:*268:268:Proximity and Ranging Daemon:/var/db/nearbyd:/usr/bin/false
_reportmemoryexception:*269:269:ReportMemoryException:/var/db/reportmemoryexception:/usr/bin/false
_driverkit:*270:270:DriverKit:/var/empty:/usr/bin/false
_diskimagesiod:*271:271:DiskImages IO Daemon:/var/db/diskimagesiod:/usr/bin/false
_logd:*272:272:Log Daemon:/var/db/diagnostics:/usr/bin/false
_appinstalld:*273:273:App Install Daemon:/var/db/appinstalld:/usr/bin/false
_installcoordinationd:*274:274:Install Coordination Daemon:/var/db/installcoordinationd:/usr/bin/false
_demod:*275:275:Demo Daemon:/var/empty:/usr/bin/false
_rmd:*277:277:Remote Management Daemon:/var/db/rmd:/usr/bin/false
_accessoryupdater:*278:278:Accessory Update Daemon:/var/db/accessoryupdater:/usr/bin/false
_knowledgegraphd:*279:279:Knowledge Graph Daemon:/var/db/knowledgegraphd:/usr/bin/false
_coreml:*280:280:CoreML Services:/var/empty:/usr/bin/false
_sntp:*281:281:SNTP Server Daemon:/var/empty:/usr/bin/false
_trustd:*282:282:trustd:/var/empty:/usr/bin/false
_darwindaemon:*284:284:Darwin Daemon:/var/db/darwindaemon:/usr/bin/false
_notification_proxy:*285:285:Notification Proxy:/var/empty:/usr/bin/false
_oahd:*441:441:OAH Daemon:/var/empty:/usr/bin/false
skpandey@Ss-MacBook-Air logs_practice % ls -l /etc/ | grep conf
skpandey@Ss-MacBook-Air logs_practice % ls -l /etc/ | grep conf
-rw-r--r-- 1 root wheel 1051 Oct 7 2022 asl.conf
-rw-r--r-- 1 root wheel 1935 Oct 7 2022 autofs.conf
-rw-r--r-- 1 root wheel 8 Jul 14 2022 kern.loader.conf
-rw-r--r-- 1 root wheel 4627 Oct 7 2022 man.conf
-rw-r--r-- 1 root wheel 1318 Oct 7 2022 newsyslog.conf
-rw-r--r-- 1 root wheel 43 Oct 7 2022 nfs.conf
-rw-r--r-- 1 root wheel 557 Oct 7 2022 notify.conf
-rw-r--r-- 1 root wheel 22 Nov 28 2022 ntp.conf
-rw-r--r-- 1 root wheel 23 Oct 7 2022 ntp_opendirectory.conf
-rw-r--r-- 1 root wheel 1027 Oct 7 2022 pf.conf
lrwxr-xr-x 1 root wheel 22 Jul 14 2022 resolv.conf -> ../var/run/resolv.conf
-rw-r--r-- 1 root wheel 891 Oct 7 2022 rtadvd.conf
-rw-r--r-- 1 root wheel 96 Oct 7 2022 syslog.conf
drwxr-xr-x 3 root wheel 96 Aug 21 12:40 test.config
skpandey@Ss-MacBook-Air logs_practice % cat /etc/passwd | grep . | wc -l
123
skpandey@Ss-MacBook-Air logs_practice % ps aux | grep ssh
skpandey 3448 0.0 0.0 407986728 968 ?? S 5Aug26 0:00.02 /usr/bin/ssh-agent -l
skpandey 16331 0.0 0.0 408626368 1648 s000 S+ 8:58PM 0:00.00 grep ssh
skpandey@Ss-MacBook-Air logs_practice % grep "/bin/bash" /etc/passwd | cut -d ':' -f 1
_mbssetupuser
skpandey@Ss-MacBook-Air logs_practice % █

```

Question 03: Create a sample log file with 20 lines:

Answer the following:

1. (head) Print the first 5 lines of events.log.
2. (tail) Print the last 5 lines of events.log.
3. (head with negative-style usage) Print all lines except the last 3 using head.
4. (tail with +N) Print all lines starting from line 15 to the end, using tail.
5. (tail -f, conceptual + practical) What does tail -f events.log do differently from a plain tail events.log? When would you use it in real-world sysadmin work?
6. (Combining head and tail) Using a pipeline of head and tail together, print only line 10 of the file. (Combining with grep) Print the last 3 FAIL events in the file (hint: filter first, then take the tail).
7. (wc + tail) Without hardcoding the number 20, write a command that prints the last 25% of lines in the file dynamically (hint: use wc -l to compute the count first).

Answer:

```
skpandey@Ss-MacBook-Air logs_practice % cat > events.log
1 INFO Server started
2 INFO User login
3 FAIL Login failed
4 INFO File uploaded
5 INFO User logout
6 FAIL Connection failed
7 INFO Server running
8 INFO User login
9 FAIL Database error
10 INFO File downloaded
11 INFO User login
12 FAIL Timeout error
13 INFO Server running
14 INFO File uploaded
15 FAIL Login failed
16 INFO User logout
17 INFO Server running
18 FAIL Connection failed
19 INFO User login
20 INFO Server stopped
skpandey@Ss-MacBook-Air logs_practice % head -5 events.log
1 INFO Server started
2 INFO User login
3 FAIL Login failed
4 INFO File uploaded
5 INFO User logout
skpandey@Ss-MacBook-Air logs_practice % tail -5 events.log
16 INFO User logout
17 INFO Server running
18 FAIL Connection failed
19 INFO User login
20 INFO Server stopped
skpandey@Ss-MacBook-Air logs_practice % head -n -3 events.log
head: illegal line count --- -3
skpandey@Ss-MacBook-Air logs_practice % tail -n +15 events.log
15 FAIL Login failed
16 INFO User logout
17 INFO Server running
18 FAIL Connection failed
19 INFO User login
20 INFO Server stopped
skpandey@Ss-MacBook-Air logs_practice % tail -f events.log
11 INFO User login
12 FAIL Timeout error
13 INFO Server running
14 INFO File uploaded
15 FAIL Login failed
16 INFO User logout
17 INFO Server running
18 FAIL Connection failed
19 INFO User login
20 INFO Server stopped
^C
skpandey@Ss-MacBook-Air logs_practice % head -10 events.log | tail -1
10 INFO File downloaded
skpandey@Ss-MacBook-Air logs_practice % grep FAIL events.log | tail -3
12 FAIL Timeout error
15 FAIL Login failed
18 FAIL Connection failed
skpandey@Ss-MacBook-Air logs_practice % tail -n $((wc -l < events.log) / 4) events.log
16 INFO User logout
17 INFO Server running
18 FAIL Connection failed
19 INFO User login
```

`tail -f` continuously monitors a file and displays new lines as they are added. It is useful for monitoring log files in real time.

Assignment 4

TOPIC : Aws Assignment IAM

Date: 21/07/2026

Question 01: Given a CSV file (users.csv) listing new hires who need AWS access. Write a single bash script (provision_iam.sh) that reads this file, validates it, creates IAM users, creates a custom policy, and attaches it with no manual AWS Console steps.

Answer:

```
skpandey@Ss-MacBook-Air ~ % nano iamq1.sh
skpandey@Ss-MacBook-Air ~ % chmod +x iamq1.sh
skpandey@Ss-MacBook-Air ~ % ./q1.sh
Latest file: iamq1.sh
cp: /iamq1.sh: No such file or directory
cat: /iamq1.sh: No such file or directory
skpandey@Ss-MacBook-Air ~ % ./iamq1.sh
{
  "User": {
    "Path": "/",
    "UserName": "Test",
    "UserId": "AIDAXJE44KPP044NVV4CT",
    "Arn": "arn:aws:iam::500692702174:user/Test",
    "CreateDate": "2026-08-21T16:22:52+00:00"
  }
}
Test user created and Administrator access added
skpandey@Ss-MacBook-Air ~ % aws iam list-attached-user-policies --user-name Test
[
  {
    "AttachedPolicies": [
      {
        "PolicyName": "AdministratorAccess",
        "PolicyArn": "arn:aws:iam::aws:policy/AdministratorAccess"
      }
    ]
  }
]
skpandey@Ss-MacBook-Air ~ % nano iamq1.sh
skpandey@Ss-MacBook-Air ~ % █
```

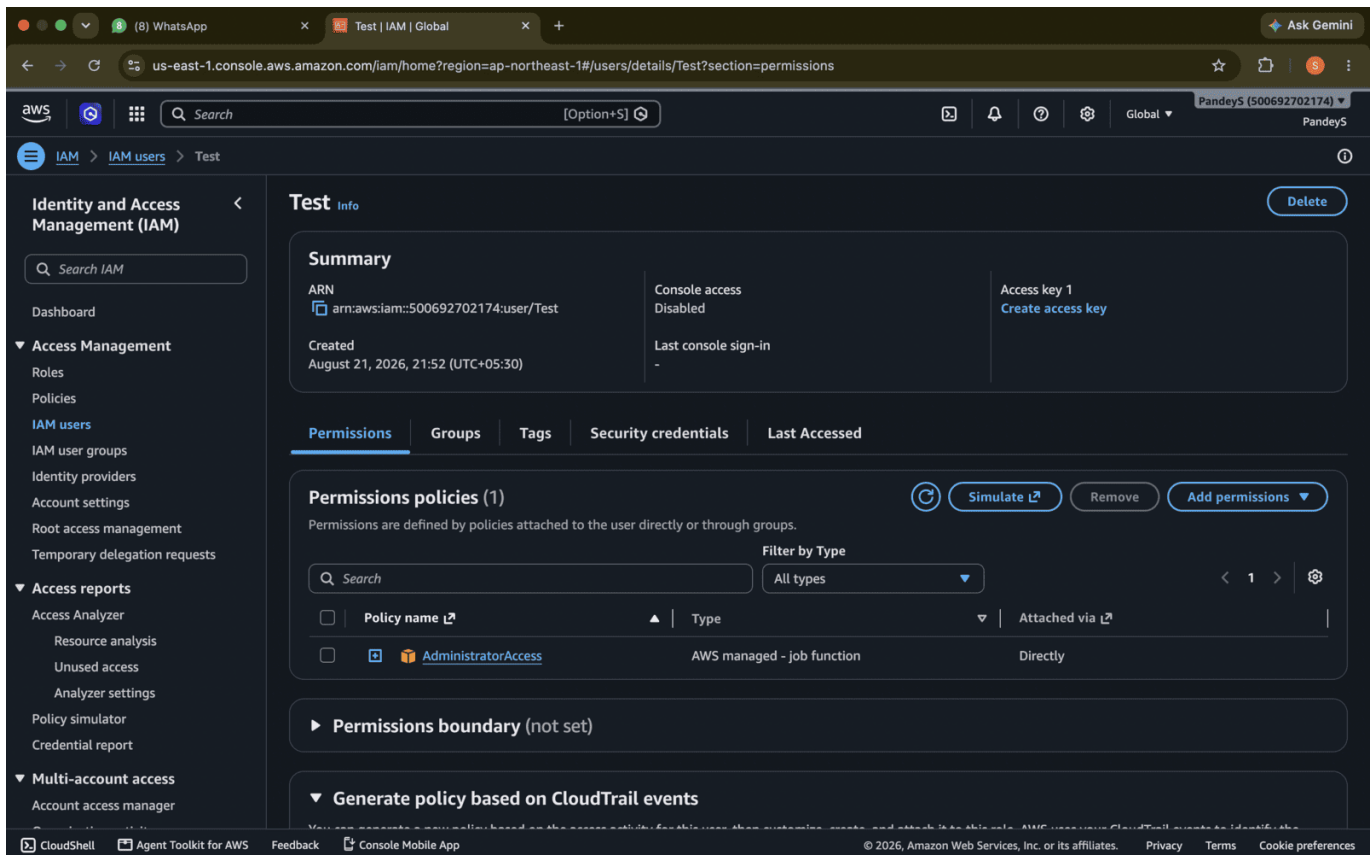
UW PICO 5.09 File: iamq1.sh

#!/bin/bash

aws iam create-user --user-name Test

aws iam attach-user-policy --user-name Test --policy-arn arn:aws:iam::aws:policy/AdministratorAccess

echo "Test user created and Administrator access added"



Question 02: Write a Bash script to create an IAM role named Tester-role. In the same script create a Access policy for Tester-Role named Tester-Policy. The policy should provide readonly access to s3. Attach the role to the policy.

Answer:

```
skpandey@Ss-MacBook-Air ~ % nano iamq2.sh
skpandey@Ss-MacBook-Air ~ % chmod +x iamq2.sh
skpandey@Ss-MacBook-Air ~ % ./iamq2.sh
{
  "Role": {
    "Path": "/",
    "RoleName": "Tester-role",
    "RoleId": "AROAXJE44KPPL4UX4EXI4",
    "Arn": "arn:aws:iam::500692702174:role/Tester-role",
    "CreateDate": "2026-08-21T16:42:02+00:00",
    "AssumeRolePolicyDocument": {
      "Version": "2012-10-17",
      "Statement": [
        {
          "Effect": "Allow",
          "Principal": {
            "Service": "ec2.amazonaws.com"
          },
          "Action": "sts:AssumeRole"
        }
      ]
    }
  }
}

{
  "Policy": {
    "PolicyName": "Tester-Policy",
    "PolicyId": "ANPAXJE44KPFPQMOPNRVD",
    "Arn": "arn:aws:iam::500692702174:policy/Tester-Policy",
    "Path": "/",
    "DefaultVersionId": "v1",
    "AttachmentCount": 0,
    "PermissionsBoundaryUsageCount": 0,
    "IsAttachable": true,
    "CreateDate": "2026-08-21T16:42:06+00:00",
    "UpdateDate": "2026-08-21T16:42:06+00:00"
  }
}
Role and policy created successfully
skpandey@Ss-MacBook-Air ~ % aws iam list-attached-role-policies --role-name Tester-role
{
  "AttachedPolicies": [
    {
      "PolicyName": "Tester-Policy",
      "PolicyArn": "arn:aws:iam::500692702174:policy/Tester-Policy"
    }
  ]
}
```

UW PICO 5.09

File: iamq2.sh

```

#!/bin/bash

aws iam create-role --role-name Tester-role \
--assume-role-policy-document '{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": {"Service": "ec2.amazonaws.com"},
    "Action": "sts:AssumeRole"
  }]
}'

aws iam create-policy --policy-name Tester-Policy \
--policy-document '{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": ["s3:GetObject", "s3:ListBucket"],
    "Resource": "*"
  }]
}'

aws iam attach-role-policy \
--role-name Tester-role \
--policy-arn arn:aws:iam::500692702174:policy/Tester-Policy

echo "Role and policy created successfully"

```

us-east-1.console.aws.amazon.com/iam/home?region=ap-northeast-1#/roles/details/Tester-role

aws

Search

[Option+S]

Global

PandeyS (500692702174)

IAM > Roles > Tester-role

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

Access Analyzer

Resource analysis

Unused access

Analyzer settings

Policy simulator

Credential report

Multi-account access

Account access manager

Tester-role Info

Summary

Creation date

August 21, 2026, 22:12 (UTC+05:30)

ARN

arn:aws:iam::500692702174:role/Tester-role

Last activity

-

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (1) Info

Simulate

Remove

Add permissions

You can attach up to 20 managed policies.

Search

Filter by Type

All types

☐	Policy name	Type	Attached entities
☐	Tester-Policy	Customer managed	1

Permissions boundary (not set)

Generate policy based on CloudTrail events

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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Question 03: Extend the above bash script to create another user named Test2. Add Test1 and Test2 to a user group named testing(which should also be created through the script). Attach Tester-Role to all users in the user group testing.

Answer :

```

skpandey@Ss-MacBook-Air ~ % nano iamq3.sh
skpandey@Ss-MacBook-Air ~ % chmod +x iamq3.sh
skpandey@Ss-MacBook-Air ~ % ./iamq3.sh
{
  "User": {
    "Path": "/",
    "UserName": "Test2",
    "UserId": "AIDAXJE44KPPIMTAVZYRN",
    "Arn": "arn:aws:iam::500692702174:user/Test2",
    "CreateDate": "2026-08-21T16:53:29+00:00"
  }
}
{
  "Group": {
    "Path": "/",
    "GroupName": "testing",
    "GroupId": "AGPAXJE44KPPGNXLG74CU",
    "Arn": "arn:aws:iam::500692702174:group/testing",
    "CreateDate": "2026-08-21T16:53:31+00:00"
  }
}
}

```

UW PICO 5.09 File: iamq3.sh

```
#!/bin/bash
```

```
aws iam create-user --user-name Test2

aws iam create-group --group-name testing

aws iam add-user-to-group --user-name Test --group-name testing

aws iam add-user-to-group --user-name Test2 --group-name testing

aws iam attach-group-policy --group-name testing \
--policy-arn arn:aws:iam::508692726214:policy/Tester-Policy

echo "Users added to testing group"
```

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Assignment 5

TOPIC : Aws Assignment EC2

Date: 21/07/2026

Question 01: Write a bash script to create an instance and display its public IP and DNS.

Answer:

```
skpandey@Ss-MacBook-Air ~ % nano ec2q1.sh
skpandey@Ss-MacBook-Air ~ % chmod +x ec2q1.sh
skpandey@Ss-MacBook-Air ~ % ./ec2q1.sh
Instance ID: i-0c67eb074386f7ca7
```

```
-----
| DescribeInstances |
|-----|
| 35.79.22.141 |
| ec2-35-79-22-141.ap-northeast-1.compute.amazonaws.com |
|-----|
skpandey@Ss-MacBook-Air ~ %
```

```
UW PICO 5.09 File: ec2q1.sh
#!/bin/bash
id=$(aws ec2 run-instances --image-id ami-0f7e90d3283d2e250 --instance-type t3.micro --count 1 --query 'Instances[0].InstanceId' --output text)
echo "Instance ID: $id"
sleep 10
aws ec2 describe-instances --instance-ids $id --query 'Reservations[0].Instances[0].[PublicIpAddress,PublicDnsName]' --output table
```

The screenshot displays the AWS IAM console interface. The left sidebar shows the navigation menu with 'IAM user groups' selected. The main content area shows the details for the 'testing' user group. The 'Summary' section includes the user group name 'testing', creation time 'August 21, 2026, 22:23 (UTC+05:30)', and ARN 'arn:aws:iam::500692702174:group/testing'. The 'Users (2)' tab is active, showing a table with two users: 'Test' and 'Test2'. The table columns are 'User name', 'Groups', 'Last activity', and 'Creation time'. The footer of the console shows the date '© 2026, Amazon Web Services, Inc. or its affiliates.' and links for 'Privacy', 'Terms', and 'Cookie preferences'.

User name	Groups	Last activity	Creation time
Test	1	None	41 minutes ago
Test2	1	None	10 minutes ago

Question 02: Write a bash script to create an instance and check whether it has a elastic IP assigned to it or not.

Answer:

```
skpandey@Ss-MacBook-Air ~ % nano ec2q2.sh
skpandey@Ss-MacBook-Air ~ % chmod +x ec2q2.sh
skpandey@Ss-MacBook-Air ~ % ./ec2q2.sh
Instance ID: i-00b7ce09f512d595e
No Elastic IP is assigned
skpandey@Ss-MacBook-Air ~ %
```

The image displays two screenshots. The top screenshot is a terminal window titled 'UW PICO 5.09' with a file named 'ec2q2.sh'. It shows the execution of a bash script that creates an EC2 instance and checks for an Elastic IP. The script output shows the instance ID 'i-00b7ce09f512d595e' and confirms that 'No Elastic IP is assigned'.

The bottom screenshot is a screenshot of the AWS Management Console, specifically the 'Instance details' page for the instance 'i-00b7ce09f512d595e' (ec2q2). The console shows the instance is in a 'Running' state. Key details include:

- Instance ID:** i-00b7ce09f512d595e
- Public IPv4 address:** 13.196.163.243
- Private IPv4 addresses:** 172.31.23.98
- Instance state:** Running
- Private IP DNS name (IPv4 only):** ip-172-31-23-98.ap-northeast-1.compute.internal
- Instance type:** t3.micro
- VPC ID:** vpc-08c782a59c8ecfa38
- Subnet ID:** subnet-004a8538d44da7f9d
- Instance ARN:** arn:aws:ec2:ap-northeast-1:500692702174:instance/i-00b7ce09f512d595e
- Auto-assigned IP address:** 13.196.163.243 [Public IP]
- Hostname type:** IP name: ip-172-31-23-98.ap-northeast-1.compute.internal
- Answer private resource DNS name:** -
- IAM role:** -
- IMDSv2:** Required
- Operator:** -
- Public DNS:** ec2-13-196-163-243.ap-northeast-1.compute.amazonaws.com
- Elastic IP addresses:** -
- AWS Compute Optimizer finding:** Opt-in to AWS Compute Optimizer for recommendations.
- Auto Scaling Group name:** -
- Managed:** false

Assignment 6

TOPIC : Aws Assignment VPC

Date: 24/07/2026

Question 01: Create a VPC with CIDR 10.0.0.0/16 using the AWS CLI. Tag it appropriately and capture the returned VpcId into vars.env. Verify with if the VPC was created and report

Answer:

```
skpandey@Ss-MacBook-Air ~ % aws ec2 create-vpc --cidr-block 10.0.0.0/16 --tag-specifications 'ResourceType=vpc,Tags=[{Key=Name,Value=Assignment6-VPC}]' --query 'Vpc.VpcId' --output text
vpc-026c7aa8ab36d3940
skpandey@Ss-MacBook-Air ~ % echo "VPC_ID=vpc-026c7aa8ab36d3940" > vars.env
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-vpcs --vpc-ids vpc-0a1b2c3d4e5f67890 --query 'Vpcs[0].[VpcId,CidrBlock,State]' --output table
aws: [ERROR]: An error occurred (InvalidVpcID.NotFound) when calling the DescribeVpcs operation: The vpc ID 'vpc-0a1b2c3d4e5f67890' does not exist
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-vpcs --vpc-ids vpc-026c7aa8ab36d3940 --query 'Vpcs[0].[VpcId,CidrBlock,State]' --output table
+-----+-----+
| DescribeVpcs |
+-----+-----+
| vpc-026c7aa8ab36d3940 |
| 10.0.0.0/16 |
| available |
+-----+-----+
skpandey@Ss-MacBook-Air ~ % cat vars.env
VPC_ID=vpc-026c7aa8ab36d3940
skpandey@Ss-MacBook-Air ~ %
```

The screenshot displays the AWS Management Console interface for a VPC named 'Assignment6-VPC' (VPC ID: vpc-026c7aa8ab36d3940) in the Asia Pacific (Tokyo) region. The console shows the VPC's state as 'Available' and provides details on its configuration, including DNS resolution, DHCP options, and CIDR blocks. The 'Resource map' section at the bottom highlights the VPC, its subnets (0), and its route tables (1).

VPC dashboard < vpc-026c7aa8ab36d3940 / Assignment6-VPC

Filter by VPC: [v]

Details Info

Property	Value
VPC ID	vpc-026c7aa8ab36d3940
State	Available
DNS resolution	Enabled
Main network ACL	acl-09fc2fbb97f07e8dd
IPv4 CIDR (Network border group)	-
Encryption control ID	-
Tenancy	default
Default VPC	No
Network Address Usage metrics	Disabled
Encryption control mode	-
Block Public Access	Off
DHCP option set	dopt-01b4955bc82d09f96
IPv4 CIDR	10.0.0.0/16
Route 53 Resolver DNS Firewall rule groups	-
DNS hostnames	Disabled
Main route table	rtb-0b70da1d022972915
IPv6 pool	-
Owner ID	500692702174

Resource map Info

VPC
Your AWS virtual network
Assignment6-VPC

Subnets (0)
Subnets within this VPC

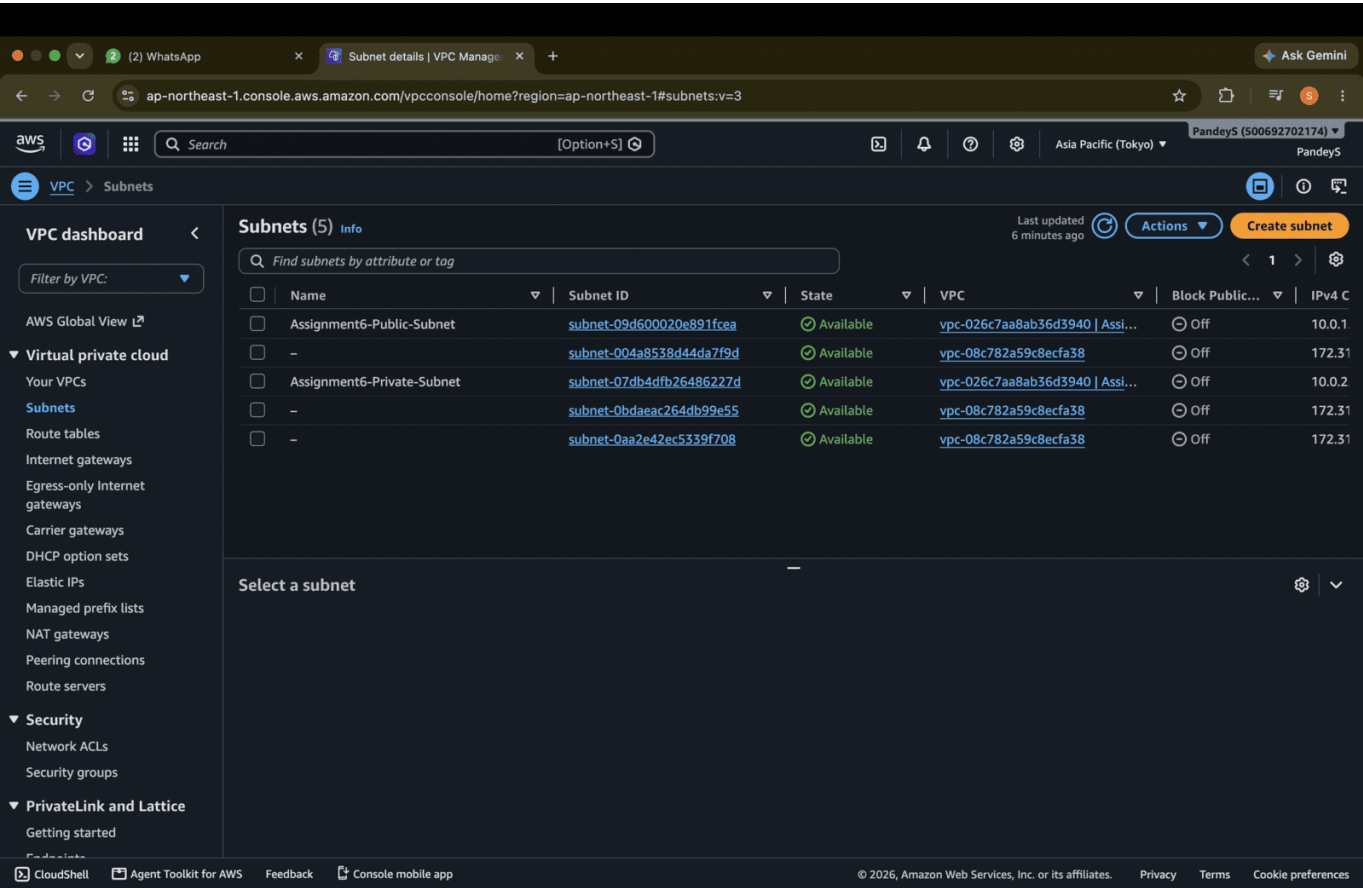
Route tables (1)
Route network traffic to resources
rtb-0b70da1d022972915

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Question 02: Create Public and Private Subnets inside the VPC from 1, create a public subnet (10.0.1.0/24) and a private subnet (10.0.2.0/24) in the same Availability Zone. Capture both SubnetIds into vars.env file

Answer:

```
skpandey@MacBook-Air ~ % aws ec2 create-subnet --vpc-id vpc-026c7aa8ab36d3940 --cidr-block 10.0.1.0/24 --availability-zone ap-northeast-1a --tag-specifications 'ResourceType=subnet,Tags=[{Key=Name,Value=Assignment6-Public-Subnet}]' --query 'Subnet.SubnetId' --output text
subnet-09d600020e891fcea
skpandey@MacBook-Air ~ % echo "PUBLIC_SUBNET_ID=subnet-09d600020e891fcea" >> vars.env
skpandey@MacBook-Air ~ % aws ec2 create-subnet --vpc-id vpc-026c7aa8ab36d3940 --cidr-block 10.0.2.0/24 --availability-zone ap-northeast-1a --tag-specifications 'ResourceType=subnet,Tags=[{Key=Name,Value=Assignment6-Private-Subnet}]' --query 'Subnet.SubnetId' --output text
subnet-07db4dfb26486227d
skpandey@MacBook-Air ~ % echo "PRIVATE_SUBNET_ID=subnet-07db4dfb26486227d" >> vars.env
skpandey@MacBook-Air ~ % aws ec2 describe-subnets --filters "Name=vpc-id,Values=vpc-026c7aa8ab36d3940" --query 'Subnets[*].{SubnetId,CidrBlock,AvailabilityZone,Tags[?Key==`Name`].Value[0]}' --output table
+-----+-----+-----+-----+
|                                     DescribeSubnets                                     |
+-----+-----+-----+-----+
| subnet-09d600020e891fcea | 10.0.1.0/24 | ap-northeast-1a | Assignment6-Public-Subnet |
| subnet-07db4dfb26486227d | 10.0.2.0/24 | ap-northeast-1a | Assignment6-Private-Subnet |
+-----+-----+-----+-----+
skpandey@MacBook-Air ~ %
```



Question 03: Create an Internet Gateway and attach it to the VPC. This is the first half of giving the public subnet internet access explain in one line why an IGW alone isn't enough yet.

Answer:

```
skpandey@Ss-MacBook-Air ~ % aws ec2 create-internet-gateway --tag-specifications 'ResourceType=internet-gateway,Tags=[{Key=Name,Value=Assignment6-IGW}]' --query 'InternetGateway.InternetGatewayId' --output text
igw-0cc30cd25daf36ce8
skpandey@Ss-MacBook-Air ~ % aws ec2 attach-internet-gateway --internet-gateway-id 0cc30cd25daf36ce8 --vpc-id vpc-026c7aa8ab36d3940

aws: [ERROR]: An error occurred (InvalidInternetGatewayId.Malformed) when calling the AttachInternetGateway operation: Invalid id: "0cc30cd25daf36ce8" (expecting "igw-...")
skpandey@Ss-MacBook-Air ~ % aws ec2 attach-internet-gateway --internet-gateway-id igw-0cc30cd25daf36ce8 --vpc-id vpc-026c7aa8ab36d3940
skpandey@Ss-MacBook-Air ~ % echo "IGW_ID=0cc30cd25daf36ce8" >> vars.env
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-internet-gateways --internet-gateway-ids 0cc30cd25daf36ce8 --query 'InternetGateways[0].[InternetGatewayId,Attachments[0].VpcId,Attachments[0].State]' --output table
le
aws: [ERROR]: An error occurred (InvalidInternetGatewayID.NotFound) when calling the DescribeInternetGateways operation: The internetGateway ID '0cc30cd25daf36ce8' does not exist
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-internet-gateways --internet-gateway-ids igw-0cc30cd25daf36ce8 --query 'InternetGateways[0].[InternetGatewayId,Attachments[0].VpcId,Attachments[0].State]' --output table
-----
|DescribeInternetGateways|
+-----+
| igw-0cc30cd25daf36ce8 |
| vpc-026c7aa8ab36d3940 |
| available             |
+-----+
skpandey@Ss-MacBook-Air ~ % cat vars.env
VPC_ID=vpc-026c7aa8ab36d3940
PUBLIC_SUBNET_ID=subnet-09d60020e891fcea
PRIVATE_SUBNET_ID=subnet-07db4dfb26486227d
IGW_ID=igw-0cc30cd25daf36ce8
skpandey@Ss-MacBook-Air ~ % █
```

An Internet Gateway alone is not enough because the subnet also needs a route table with a default route (0.0.0.0/0) pointing to the IGW.

Question 04: Create a route table, add a route 0.0.0.0/0 → IGW, and associate it with the public subnet. Verify the association using CLI commands (and queries) to print only the route table ID and destination CIDR.

Answer:

```
skpandey@Ss-MacBook-Air ~ % aws ec2 create-route-table --vpc-id vpc-026c7aa8ab36d3940 --tag-specifications 'ResourceType=route-table,Tags=[{Key=Name,Value=Assignment6-Public-RT}]' --query 'RouteTable.RouteTableId' --output text
rtb-0979a4feaf93907b6
skpandey@Ss-MacBook-Air ~ % echo "PUBLIC_RT_ID=0979a4feaf93907b6" >> vars.env
skpandey@Ss-MacBook-Air ~ % aws ec2 create-route --route-table-id rtb-0979a4feaf93907b6 --destination-cidr-block 0.0.0.0/0 --gateway-id igw-0cc30cd25daf36ce8
{
  "Return": true
}
skpandey@Ss-MacBook-Air ~ % aws ec2 associate-route-table --route-table-id rtb-0979a4feaf93907b6 --subnet-id subnet-09d60020e891fcea
{
  "AssociationId": "rtbassoc-0d5e21aa009c76802",
  "AssociationState": {
    "State": "associated"
  }
}
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-route-tables --route-table-ids rtb-0979a4feaf93907b6 --query 'RouteTables[0].[RouteTableId,Routes[?DestinationCidrBlock=`0.0.0.0/0`].DestinationCidrBlock|[0]]' --output table
-----
| DescribeRouteTables |
+-----+
| rtb-0979a4feaf93907b6 |
| 0.0.0.0/0             |
+-----+
skpandey@Ss-MacBook-Air ~ % █
```

ap-northeast-1.console.aws.amazon.com/vpconsole/home?region=ap-northeast-1#RouteTables:

Search [Option+S]

Asia Pacific (Tokyo) PandeyS (500692702174)

VPC > Route tables

VPC dashboard

Filter by VPC: AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Route tables (1/4) Info

Last updated 1 minute ago

Actions

Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	Assignment6-Public-RT	rtb-0979a4feaf93907b6	subnet-09d600020e891f...	-	No	vpc-026c7aa8ab36d3...
☐	-	rtb-0cf923e30b2601df4	-	-	Yes	vpc-08c782a59c8ecfa
☐	-	rtb-0b70da1d022972915	-	-	Yes	vpc-026c7aa8ab36d3...
☐	-	rtb-0e15dcdb87b430538	-	-	Yes	vpc-05e234be7f1beb

rtb-0979a4feaf93907b6 / Assignment6-Public-RT

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Details

Route table ID

rtb-0979a4feaf93907b6

VPC

vpc-026c7aa8ab36d3940 | Assignment6-VPC

Main

No

Owner ID

500692702174

Explicit subnet associations

subnet-09d600020e891fcea / Assignment6-Public-Subnet

Edge associations

-

CloudShell

Agent Toolkit for AWS

Feedback

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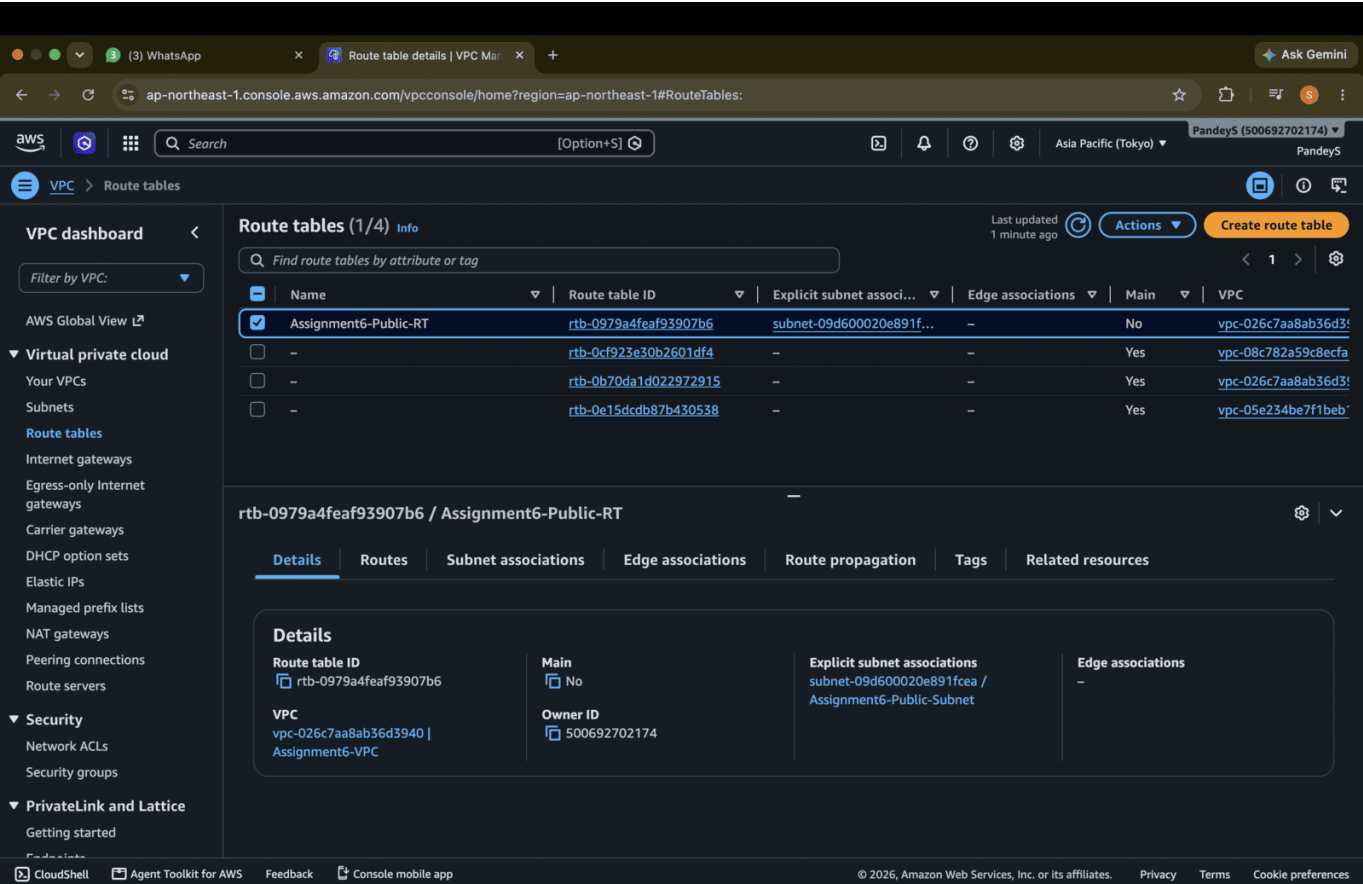
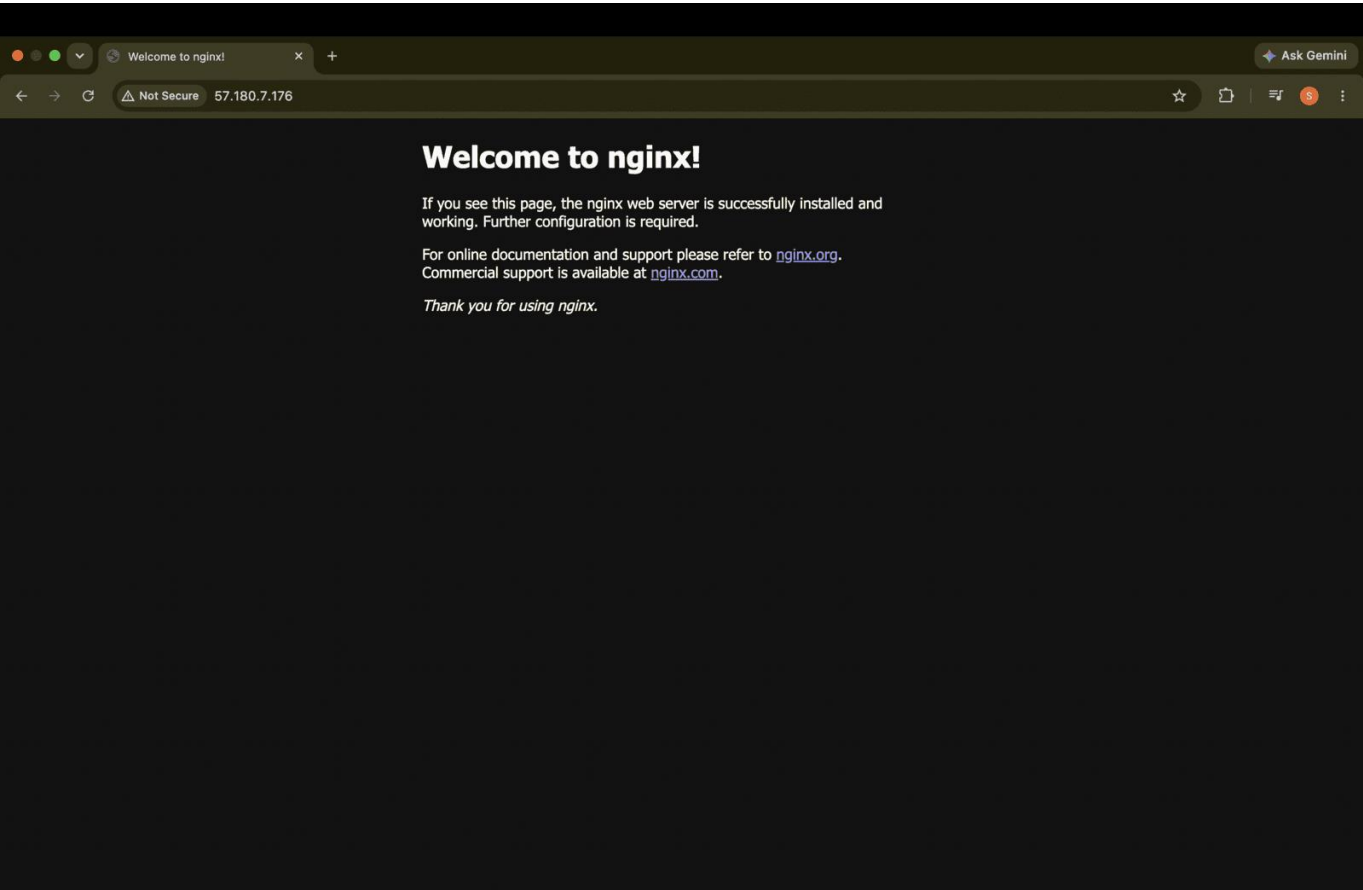
Terms

Cookie preferences

Question 05: Launch a t3.micro in the public subnet with auto-assign public IP enabled, using user-data to install and start nginx. Confirm it's reachable over HTTP from your machine

Answer:

```
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-images --owners 099720109477 --filters "Name=name,Values=ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-*" "Name=state,Values=available" "Name=architecture,Values=x86_64" --query 'Images | sort_by(@,&CreationDate)[-1].ImageId' --output text
ami-0e239f7f4b3226722
skpandey@Ss-MacBook-Air ~ % aws ec2 create-security-group --group-name Assignment6-Public-SG --description "Assignment 6 public instance security group" --vpc-id vpc-026c7aa8ab36d3940 --query 'GroupId' --output text
sg-0b69d519855fcbe8d
skpandey@Ss-MacBook-Air ~ % aws ec2 authorize-security-group-ingress --group-id sg-0b69d519855fcbe8d --protocol tcp --port 22 --cidr 0.0.0.0/0
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-0b11860e94ead6eab",
      "GroupId": "sg-0b69d519855fcbe8d",
      "GroupOwnerId": "500692702174",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 22,
      "ToPort": 22,
      "CidrIpv4": "0.0.0.0/0",
      "SecurityGroupRuleArn": "arn:aws:ec2:ap-northeast-1:500692702174:security-group-rule/sgr-0b11860e94ead6eab"
    }
  ]
}
skpandey@Ss-MacBook-Air ~ % aws ec2 authorize-security-group-ingress --group-id sg-0b69d519855fcbe8d --protocol tcp --port 80 --cidr 0.0.0.0/0
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-04529586389ef0fbb",
      "GroupId": "sg-0b69d519855fcbe8d",
      "GroupOwnerId": "500692702174",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 80,
      "ToPort": 80,
      "CidrIpv4": "0.0.0.0/0",
      "SecurityGroupRuleArn": "arn:aws:ec2:ap-northeast-1:500692702174:security-group-rule/sgr-04529586389ef0fbb"
    }
  ]
}
skpandey@Ss-MacBook-Air ~ % aws ec2 run-instances --image-id ami-0e239f7f4b3226722 --instance-type t3.micro --subnet-id subnet-09d600020e891fcea --security-group-ids sg-0b69d519855fcbe8d --associate-public-ip
[address --user-data '#!/bin/bash
apt-get update -y
apt-get install -y nginx
systemctl enable nginx
systemctl start nginx' --tag-specifications 'ResourceType=instance,Tags=[{Key=Name,Value=Assignment6-Public-Nginx}]' --query 'Instances[0].InstanceId' --output text
i-08f18b64e7f06c1b3
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-instances --instance-ids i-08f18b64e7f06c1b3 --query 'Reservations[0].Instances[0].PublicIpAddress' --output text
57.180.7.176
skpandey@Ss-MacBook-Air ~ % curl http://57.180.7.176
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>
```

Question 06: Allocate an Elastic IP, create a NAT Gateway in the public subnet, then create a second route table with 0.0.0.0/0 → NAT Gateway, associated with the private subnet. This mirrors Bit 3–4 but for outbound-only private traffic.

Answer:

```
skpandey@Ss-MacBook-Air ~ % aws ec2 allocate-address --domain vpc --query 'AllocationId' --output text
eipalloc-0e6d15bbe001abf3a
skpandey@Ss-MacBook-Air ~ % aws ec2 create-nat-gateway --subnet-id subnet-09d60020e091fcea --allocation-id eipalloc-0e6d15bbe001abf3a --tag-specifications 'ResourceType=natgateway,Tags=[{Key=Name,Value=Assignment6-NAT}]' --query 'NatGateway.NatGatewayId' --output text
nat-06d9c60b3cc0b5107
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-nat-gateways --nat-gateway-ids nat-06d9c60b3cc0b5107 --query 'NatGateways[0].{NatGatewayId,State}' --output table
+-----+
| DescribeNatGateways |
+-----+
| nat-06d9c60b3cc0b5107 |
| pending               |
+-----+
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-nat-gateways --nat-gateway-ids nat-06d9c60b3cc0b5107 --query 'NatGateways[0].{NatGatewayId,State}' --output table
+-----+
| DescribeNatGateways |
+-----+
| nat-06d9c60b3cc0b5107 |
| pending               |
+-----+
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-nat-gateways --nat-gateway-ids nat-06d9c60b3cc0b5107 --query 'NatGateways[0].{NatGatewayId,State}' --output table
+-----+
| DescribeNatGateways |
+-----+
| nat-06d9c60b3cc0b5107 |
| available            |
+-----+
skpandey@Ss-MacBook-Air ~ % aws ec2 create-route-table --vpc-id vpc-026c7aa8ab36d3940 --tag-specifications 'ResourceType=route-table,Tags=[{Key=Name,Value=Assignment6-Private-RT}]' --query 'RouteTable.RouteTableId' --output text
rtb-09f5443f2fc4b333d
skpandey@Ss-MacBook-Air ~ % aws ec2 create-route --route-table-id rtb-09f5443f2fc4b333d --destination-cidr-block 0.0.0.0/0 --nat-gateway-id nat-06d9c60b3cc0b5107
{
  "Return": true
}
skpandey@Ss-MacBook-Air ~ % aws ec2 associate-route-table --route-table-id rtb-09f5443f2fc4b333d --subnet-id subnet-07db4dfb26486227d
{
  "AssociationId": "rtbassoc-071cbef0854e623cf",
  "AssociationState": {
    "State": "associated"
  }
}
skpandey@Ss-MacBook-Air ~ % aws ec2 describe-route-tables --route-table-ids rtb-09f5443f2fc4b333d --query 'RouteTables[0].{RouteTableId,Routes[?DestinationCidrBlock=='0.0.0.0/0'].NatGatewayId|[0]}' --output table
+-----+
| DescribeRouteTables |
+-----+
| rtb-09f5443f2fc4b333d |
| nat-06d9c60b3cc0b5107 |
+-----+
skpandey@Ss-MacBook-Air ~ % █
```

The screenshot shows the AWS Management Console interface for NAT gateways. The top navigation bar includes the AWS logo, search bar, and region (Asia Pacific (Tokyo)). The left sidebar shows the VPC dashboard and various VPC resources. The main content area displays the 'NAT gateways (1/1)' page. A table lists the NAT gateway 'Assignment6-NAT' with its ID 'nat-06d9c60b3cc0b5107', connectivity type 'Public', and state 'Available'. Below the table, the details for the selected NAT gateway are shown, including its ARN, primary public IPv4 address, and state message.

Name	NAT gateway ID	Connectivity...	State	State message	Availability ...	Route table ID
Assignment6-NAT	nat-06d9c60b3cc0b5107	Public	Available	-	Zonal	-

Details

NAT gateway ID
nat-06d9c60b3cc0b5107

NAT gateway ARN
arn:aws:ec2:ap-northeast-1:500692702174:natgateway/nat-06d9c60b3cc0b5107

VPC
vpc-026c7aa8ab36d3940 / Assignment6-VPC

Connectivity type
Public

Primary public IPv4 address
35.75.227.214

Subnet
subnet-09d600020e891fcea / Assignment6-Public-Subnet

State
Available

Primary private IPv4 address
10.0.1.168

Created
Sunday, 23 August 2026 at 12:21:26 GMT+5:30

State message
-

Primary network interface ID
eni-090febb837b79f10c

Deleted
-

The screenshot shows the AWS Management Console interface for a VPC. The left sidebar contains navigation links for VPC dashboard, Virtual private cloud, and Security. The main content area shows the details for VPC 'rtb-09f5443f2fc4b333d'. The 'Routes' tab is active, displaying a table of routes. The table has columns for Destination, Target, Status, Propagated, and Route Origin. Two routes are listed: one for destination 0.0.0.0/0 with target nat-06d9c60b3cc0b5107, and another for destination 10.0.0.0/16 with target local. Both routes are marked as 'Active'.

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-06d9c60b3cc0b5107	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Question 07: Launch a t2.micro in the private subnet with **no** public IP. From the Bit 5 public instance, SSH-hop into it (bastion pattern) and confirm it can reach the internet (e.g., curl to a public endpoint) via the NAT Gateway but has no direct inbound path.

Answer:

```
ubuntu@ip-10-0-2-140:~$ chmod 400 ~/terraform-key.pem && ssh -i ~/terraform-key.pem ubuntu@10.0.2.140
chmod: cannot access '/home/ubuntu/terraform-key.pem': No such file or directory
ubuntu@ip-10-0-2-140:~$ curl -I https://example.com
HTTP/2 200
date: Sun, 23 Aug 2026 07:27:14 GMT
content-type: text/html
server: cloudflare
last-modified: Tue, 18 Aug 2026 20:06:42 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 10560
cf-cache-status: HIT
cf-ray: a2f864c0fab3ec10-NRT
ubuntu@ip-10-0-2-140:~$ █
```

Question 08: Automate with a Skeleton Script Consolidate Bits 1–8 into provision_vpc.sh using the function-stub pattern below, making it idempotent (each function checks via describe-* before creating):

```
create_vpc() { ;; }
create_subnets() { ;; }
setup_igw_and_route() { ;; }
setup_nat_and_route() { ;; }
launch_instances() { ;; }
skpandey@Ss-MacBook-Air ~ % nano provision_vpc.sh
skpandey@Ss-MacBook-Air ~ % chmod +x provision_vpc.sh
skpandey@Ss-MacBook-Air ~ % ./provision_vpc.sh
VPC already exists: vpc-026c7aa8ab36d3940
Subnets already exist
{
  "InternetGateways": [
    {
      "Attachments": [
        {
          "State": "available",
          "VpcId": "vpc-026c7aa8ab36d3940"
        }
      ],
      "InternetGatewayId": "igw-0cc30cd25daf36ce8",
      "OwnerId": "500692702174",
      "Tags": [
        {
          "Key": "Name",
          "Value": "Assignment6-IGW"
        }
      ]
    }
  ]
}
Internet Gateway already exists
NAT Gateway already exists
EC2 instances already exist
skpandey@Ss-MacBook-Air ~ % █
```

```
UW PICO 5.09 File: provision_vpc.sh

#!/bin/bash

source vars.env

create_vpc() {
    if aws ec2 describe-vpcs --vpc-ids "$VPC_ID" >/dev/null 2>&1; then
        echo "VPC already exists: $VPC_ID"
    else
        VPC_ID=$(aws ec2 create-vpc --cidr-block 10.0.0.0/16 --query "Vpc.VpcId" --output
text)
        echo "VPC_ID=$VPC_ID" >> vars.env
        echo "VPC created: $VPC_ID"
    fi
}

create_subnets() {
    if aws ec2 describe-subnets --filters "Name=vpc-id,Values=$VPC_ID" >/dev/null 2>&1; then
        echo "Subnets already exist"
    else
        echo "Subnets need to be created"
    fi
}

setup_igw_and_route() {
    if aws ec2 describe-internet-gateways --filters "Name=attachment.vpc-id,Values=$VPC_ID"
>/dev/null 2>&1; then
        echo "Internet Gateway already exists"
    else
        echo "Internet Gateway needs to be created"
    fi
}

setup_nat_and_route() {
    if aws ec2 describe-nat-gateways --filter "Name=vpc-id,Values=$VPC_ID" >/dev/null 2>&1;
then
        echo "NAT Gateway already exists"
    else
        echo "NAT Gateway needs to be created"
    fi
}

launch_instances() {
    if aws ec2 describe-instances --filters "Name=vpc-id,Values=$VPC_ID" >/dev/null 2>&1;
then
        echo "EC2 instances already exist"
    else
        echo "EC2 instances need to be created"
    fi
}

create_vpc
create_subnets
setup_igw_and_route
setup_nat_and_route
launch_instances
```

Get Help

Exit

WriteOut

Justify

Read File

Where is

Prev Pg

Next Pg

Cut Text

UnCut Text

Cur Pos

To Spell

Question 09: Write `audit_vpc.sh`, scheduled via cron every 15 minutes, that logs NAT Gateway/instance state to `/var/log/vpc_audit.log`. Then, given a sample `access.log` from the Bit 5 nginx server, answer using `wc`, `awk`, `sort`, `uniq`, `sed`, `grep`:

- a) Total requests and unique IPs
- b) Top 5 requesting IPs
- c) All HTTP 500 responses
- d) IP-redacted version of the log (via `sed`) safe to share externally

Answer:

```
skpandey@Ss-MacBook-Air ~ % cat > audit_vpc.sh <<'EOF'
#!/bin/bash
source vars.env
echo "==== VPC AUDIT $(date) =====> vpc_audit.log
echo "NAT Gateway:"> vpc_audit.log
aws ec2 describe-nat-gateways --nat-gateway-ids nat-06d9c60b3cc0b5107 --output text >> vpc_audit.log
echo "Instances:"> vpc_audit.log
aws ec2 describe-instances --filters "Name=vpc-id,Values=$VPC_ID" --output text >> vpc_audit.log
echo "===== " >> vpc_audit.log
EOF
skpandey@Ss-MacBook-Air ~ % chmod +x audit_vpc.sh
skpandey@Ss-MacBook-Air ~ % ./audit_vpc.sh
skpandey@Ss-MacBook-Air ~ % cat vpc_audit.log
===== VPC AUDIT Sun Aug 23 13:10:09 IST 2026 =====
NAT Gateway:
Instances:
=====
===== VPC AUDIT Sun Aug 23 13:10:51 IST 2026 =====
NAT Gateway:
===== VPC AUDIT Sun Aug 23 13:11:27 IST 2026 =====
NAT Gateway:
===== VPC AUDIT Sun Aug 23 13:12:14 IST 2026 =====
NAT Gateway:
NATGATEWAYS      zonal    public  2026-08-23T06:51:26+00:00    nat-06d9c60b3cc0b5107    available subnet-09d600020e891fcea vpc-026c7aa8ab36d3940
NATGATEWAYADDRESSES  eipalloc-0e6d15bbe01abf3a    eipassoc-0ea8c36c20852fa9e    True    eni-070febb837b79f10c    10.0.1.168    35.75.227.214    succeeded
TAGS      Name      Assignment6-NAT
Instances:
RESERVATIONS  500692702174    r-00f4a04d5d434189b
INSTANCES      0    x86_64    uefi-preferred    076e9306-cbc2-4051-8588-f814e8576e23    uefi    False    True    xen    ami-0e239f7f4b3226722    i-00a8a8b14bc7e953e    t3.micro    terraform-key    20
26-08-23T07:22:01+00:00    Linux/UNIX    ip-10-0-1-196.ap-northeast-1.compute.internal    10.0.1.196    52.194.214.94    /dev/sda1    ebs    True    subnet-09d600020e891fcea    RunI
nstances      2026-08-23T07:22:01+00:00    hvm    vpc-026c7aa8ab36d3940
BLOCKDEVICESMAPPINGS    /dev/sda1
EBS      2026-08-23T07:22:02+00:00    True    0    attached    vol-05c19b4b5b161da23
CAPACITYRESERVATIONSPECIFICATION    open
CPUOPTIONS      1    2
ENCLAVEOPTIONS    False
HIBERNATIONOPTIONS    False
MAINTENANCEOPTIONS    default default
METADATATOPTIONS    enabled disabled    2    required    disabled    applied
MONITORING    disabled
NETWORKINTERFACES    interface    06:f3:f1:3e:d2:0d    eni-01d66e3d3880509c5    500692702174    10.0.1.196    True    in-use    subnet-09d600020e891fcea    vpc-026c7aa8ab36d3940
ASSOCIATION    amazon    52.194.214.94
ATTACHMENT    2026-08-23T07:22:01+00:00    eni-attach-03300762161cadd31    True    0    0 attached
GROUPS    sg-0b69d519855fcb8d    Assignment6-Public-SG
OPERATOR    False
PRIVATEIPADDRESSES    True    10.0.1.196
ASSOCIATION    amazon    52.194.214.94
NETWORKPERFORMANCEOPTIONS    default
OPERATOR    False    False
PLACEMENT    ap-northeast-1a apn1-az4    default
PRIVATEDNSNAMEOPTIONS    False    False    ip-name
SECURITYGROUPS    sg-0b69d519855fcb8d    Assignment6-Public-SG
STATE    16    running
TAGS      Name      Assignment6-Bastion
RESERVATIONS  500692702174    r-0557460c6920f66a9
INSTANCES      0    x86_64    uefi-preferred    221da060-7ad9-46d0-8bbb-2284a816dad2    uefi    False    True    xen    ami-0e239f7f4b3226722    i-0bc8c8c1441953cad    t3.micro    terraform-key    20
26-08-23T07:25:50+00:00    Linux/UNIX    ip-10-0-2-140.ap-northeast-1.compute.internal    10.0.2.140    /dev/sda1    ebs    True    subnet-07db4dfb26486227d    RunInstances    2026
-08-23T07:25:50+00:00    hvm    vpc-026c7aa8ab36d3940
BLOCKDEVICESMAPPINGS    /dev/sda1
EBS      2026-08-23T07:25:51+00:00    True    0    attached    vol-083739448fccc1e8e
CAPACITYRESERVATIONSPECIFICATION    open
CPUOPTIONS      1    2
ENCLAVEOPTIONS    False
HIBERNATIONOPTIONS    False
MAINTENANCEOPTIONS    default default

skpandey@Ss-MacBook-Air ~ % cat > access.log <<'EOF'
192.168.1.10 -- [23/Aug/2026:10:00:01 +0530] "GET / HTTP/1.1" 200 612
192.168.1.11 -- [23/Aug/2026:10:01:12 +0530] "GET /about HTTP/1.1" 200 512
192.168.1.10 -- [23/Aug/2026:10:02:20 +0530] "GET /login HTTP/1.1" 500 421
192.168.1.12 -- [23/Aug/2026:10:03:05 +0530] "GET /products HTTP/1.1" 200 823
192.168.1.10 -- [23/Aug/2026:10:04:33 +0530] "POST /login HTTP/1.1" 500 311
192.168.1.13 -- [23/Aug/2026:10:05:41 +0530] "GET /contact HTTP/1.1" 200 601
192.168.1.11 -- [23/Aug/2026:10:06:10 +0530] "GET / HTTP/1.1" 200 612
192.168.1.14 -- [23/Aug/2026:10:07:25 +0530] "GET /admin HTTP/1.1" 500 250
192.168.1.12 -- [23/Aug/2026:10:08:37 +0530] "GET /cart HTTP/1.1" 200 734
192.168.1.10 -- [23/Aug/2026:10:09:50 +0530] "GET /logout HTTP/1.1" 200 421
EOF
skpandey@Ss-MacBook-Air ~ % cat access.log
192.168.1.10 -- [23/Aug/2026:10:00:01 +0530] "GET / HTTP/1.1" 200 612
192.168.1.11 -- [23/Aug/2026:10:01:12 +0530] "GET /about HTTP/1.1" 200 512
192.168.1.10 -- [23/Aug/2026:10:02:20 +0530] "GET /login HTTP/1.1" 500 421
192.168.1.12 -- [23/Aug/2026:10:03:05 +0530] "GET /products HTTP/1.1" 200 823
192.168.1.10 -- [23/Aug/2026:10:04:33 +0530] "POST /login HTTP/1.1" 500 311
192.168.1.13 -- [23/Aug/2026:10:05:41 +0530] "GET /contact HTTP/1.1" 200 601
192.168.1.11 -- [23/Aug/2026:10:06:10 +0530] "GET / HTTP/1.1" 200 612
192.168.1.14 -- [23/Aug/2026:10:07:25 +0530] "GET /admin HTTP/1.1" 500 250
192.168.1.12 -- [23/Aug/2026:10:08:37 +0530] "GET /cart HTTP/1.1" 200 734
192.168.1.10 -- [23/Aug/2026:10:09:50 +0530] "GET /logout HTTP/1.1" 200 421
skpandey@Ss-MacBook-Air ~ % wc -l access.log
10 access.log
skpandey@Ss-MacBook-Air ~ % awk '{print $1}' access.log | sort | uniq | wc -l
5
skpandey@Ss-MacBook-Air ~ % awk '{print $1}' access.log | sort | uniq -c | sort -nr | head -5
[  4 192.168.1.10
  2 192.168.1.12
  2 192.168.1.11
  1 192.168.1.14
  1 192.168.1.13
skpandey@Ss-MacBook-Air ~ % awk '$9 == 500' access.log
192.168.1.10 -- [23/Aug/2026:10:02:20 +0530] "GET /login HTTP/1.1" 500 421
192.168.1.10 -- [23/Aug/2026:10:04:33 +0530] "POST /login HTTP/1.1" 500 311
192.168.1.14 -- [23/Aug/2026:10:07:25 +0530] "GET /admin HTTP/1.1" 500 250
skpandey@Ss-MacBook-Air ~ % sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/XXX.XXX.XXX.XXX/' access.log
XXX.XXX.XXX.XXX -- [23/Aug/2026:10:00:01 +0530] "GET / HTTP/1.1" 200 612
XXX.XXX.XXX.XXX -- [23/Aug/2026:10:01:12 +0530] "GET /about HTTP/1.1" 200 512
XXX.XXX.XXX.XXX -- [23/Aug/2026:10:02:20 +0530] "GET /login HTTP/1.1" 500 421
XXX.XXX.XXX.XXX -- [23/Aug/2026:10:03:05 +0530] "GET /products HTTP/1.1" 200 823
XXX.XXX.XXX.XXX -- [23/Aug/2026:10:04:33 +0530] "POST /login HTTP/1.1" 500 311
XXX.XXX.XXX.XXX -- [23/Aug/2026:10:05:41 +0530] "GET /contact HTTP/1.1" 200 601
XXX.XXX.XXX.XXX -- [23/Aug/2026:10:06:10 +0530] "GET / HTTP/1.1" 200 612
XXX.XXX.XXX.XXX -- [23/Aug/2026:10:07:25 +0530] "GET /admin HTTP/1.1" 500 250
XXX.XXX.XXX.XXX -- [23/Aug/2026:10:08:37 +0530] "GET /cart HTTP/1.1" 200 734
XXX.XXX.XXX.XXX -- [23/Aug/2026:10:09:50 +0530] "GET /logout HTTP/1.1" 200 421
skpandey@Ss-MacBook-Air ~ %
```


Report Questions :

1. Why does the private route table point to the NAT Gateway, not the IGW? (Bits 3 vs 6)

Answer: The private subnet uses the NAT Gateway for outbound internet access while preventing direct inbound internet connections; the IGW is used by the public subnet.

2. Why does the S3 Gateway Endpoint reduce cost/security exposure vs. routing through NAT? (Bit 8)

Answer: An S3 Gateway Endpoint keeps S3 traffic inside the AWS network, avoiding NAT Gateway data-processing charges and reducing exposure to the public internet.

3. If your `awk` HTTP-500 filter returns nothing despite errors existing, what's the likely field/delimiter issue? (Bit 10)

Answer: The HTTP status code may not be in field \$9 because the log format or delimiter is different, so the `awk` field number needs to be adjusted.

4. What would you see in `describe-route-tables` output if someone mistakenly added 0.0.0.0/0 → IGW to the *private* route table? How would your Bit 10 audit script catch it?

Answer: `describe-route-tables` would show 0.0.0.0/0 with the Internet Gateway ID in the private route table. The audit script can detect this by checking the private route table's default route and reporting if its target is an IGW instead of the NAT Gateway.

Assignment 7

TOPIC : AWS Assignment Reverse Proxy

Date: 27/07/2026

Question 01: Launch and configure multiple EC2 instances on AWS. Deploy a simple web application on backend instances. Configure NGINX as a reverse proxy to route and load-balance traffic. Verify and troubleshoot a reverse proxy setup end to end.

Answer:

```
-
skpandey@Ss-MacBook-Air ~ % aws ec2 create-security-group --group-name Assignment7-Web1-SG --description "Security group for Web Server 1" --vpc-id vpc-026c7aa8ab36d3940 --query "GroupId" --output text
sg-0a90fffd84d0162b70
skpandey@Ss-MacBook-Air ~ % aws ec2 create-security-group --group-name Assignment7-Web2-SG --description "Security group for Web Server 2" --vpc-id vpc-026c7aa8ab36d3940 --query "GroupId" --output text
sg-0464596ce89f96a4b
skpandey@Ss-MacBook-Air ~ % aws ec2 authorize-security-group-ingress --group-id sg-0a90fffd84d0162b70 --protocol tcp --port 80 --source-group sg-0dc19ef8fdb2614b7
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-0ffb8c41a0cddc15b",
      "GroupId": "sg-0a90fffd84d0162b70",
      "GroupOwnerId": "500692702174",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 80,
      "ToPort": 80,
      "ReferencedGroupInfo": {
        "GroupId": "sg-0dc19ef8fdb2614b7",
        "UserId": "500692702174"
      },
      "SecurityGroupRuleArn": "arn:aws:ec2:ap-northeast-1:500692702174:security-group-rule/sgr-0ffb8c41a0cddc15b"
    }
  ]
}
skpandey@Ss-MacBook-Air ~ % aws ec2 authorize-security-group-ingress --group-id sg-0464596ce89f96a4b --protocol tcp --port 80 --source-group sg-0dc19ef8fdb2614b7
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-07f1795ea71a20b74",
      "GroupId": "sg-0464596ce89f96a4b",
      "GroupOwnerId": "500692702174",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 80,
      "ToPort": 80,
      "ReferencedGroupInfo": {
        "GroupId": "sg-0dc19ef8fdb2614b7",
        "UserId": "500692702174"
      },
      "SecurityGroupRuleArn": "arn:aws:ec2:ap-northeast-1:500692702174:security-group-rule/sgr-07f1795ea71a20b74"
    }
  ]
}
skpandey@Ss-MacBook-Air ~ % aws ec2 authorize-security-group-ingress --group-id sg-0a90fffd84d0162b70 --protocol tcp --port 22 --cidr 0.0.0.0/0
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-033747e0c78659675",
      "GroupId": "sg-0a90fffd84d0162b70",
      "GroupOwnerId": "500692702174",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 22,
      "ToPort": 22,
      "CidrIpv4": "0.0.0.0/0",
      "SecurityGroupRuleArn": "arn:aws:ec2:ap-northeast-1:500692702174:security-group-rule/sgr-033747e0c78659675"
    }
  ]
}
skpandey@Ss-MacBook-Air ~ % aws ec2 authorize-security-group-ingress --group-id sg-0464596ce89f96a4b --protocol tcp --port 22 --cidr 0.0.0.0/0
{
  "Return": true,
```

```

Selecting previously unselected package ssl-cert.
Preparing to unpack .../9-ssl-cert_1.1.2ubuntu1_all.deb ...
Unpacking ssl-cert (1.1.2ubuntu1) ...
Setting up ssl-cert (1.1.2ubuntu1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ssl-cert.service → /usr/lib/systemd/system/ssl-cert.service.
Setting up libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Setting up liblua5.4-0:amd64 (5.4.6-3build2) ...
Setting up apache2-data (2.4.58-1ubuntu8.15) ...
Setting up libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...
Setting up libaprutil1-ldap:amd64 (1.6.3-1.1ubuntu7) ...
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7) ...
Setting up apache2-utils (2.4.58-1ubuntu8.15) ...
Setting up apache2-bin (2.4.58-1ubuntu8.15) ...
Setting up apache2 (2.4.58-1ubuntu8.15) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-1-211:~$ sudo systemctl start apache2
ubuntu@ip-10-0-1-211:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-10-0-1-211:~$ echo "<h1>Response from Web Server 1</h1>" | sudo tee /var/www/html/index.html
<h1>Response from Web Server 1</h1>
ubuntu@ip-10-0-1-211:~$ curl localhost
<h1>Response from Web Server 1</h1>
ubuntu@ip-10-0-1-211:~$ █

```

```

Selecting previously unselected package ssl-cert.
Preparing to unpack .../9-ssl-cert_1.1.2ubuntu1_all.deb ...
Unpacking ssl-cert (1.1.2ubuntu1) ...
Setting up ssl-cert (1.1.2ubuntu1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ssl-cert.service → /usr/lib/systemd/system/ssl-cert.service.
Setting up libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Setting up liblua5.4-0:amd64 (5.4.6-3build2) ...
Setting up apache2-data (2.4.58-1ubuntu8.15) ...
Setting up libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...
Setting up libaprutil1-ldap:amd64 (1.6.3-1.1ubuntu7) ...
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7) ...
Setting up apache2-utils (2.4.58-1ubuntu8.15) ...
Setting up apache2-bin (2.4.58-1ubuntu8.15) ...
Setting up apache2 (2.4.58-1ubuntu8.15) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-1-215:~$ sudo systemctl start apache2
ubuntu@ip-10-0-1-215:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-10-0-1-215:~$ echo "<h1>Response from Web Server 2</h1>" | sudo tee /var/www/html/index.html
<h1>Response from Web Server 2</h1>
ubuntu@ip-10-0-1-215:~$ curl localhost
<h1>Response from Web Server 2</h1>
ubuntu@ip-10-0-1-215:~$ █

```

```
63 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-10-0-1-231:~$ sudo apt install -y nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
Need to get 568 kB of archives.
After this operation, 1602 kB of additional disk space will be used.
Get:1 http://ap-northeast-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.17 [45.1 kB]
Get:2 http://ap-northeast-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.17 [523 kB]
Fetched 568 kB in 0s (25.3 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 72595 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.17_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.17) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.17_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.17) ...
Setting up nginx-common (1.24.0-2ubuntu7.17) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.17) ...
  * Upgrading binary nginx [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-1-231:~$ sudo systemctl start nginx
ubuntu@ip-10-0-1-231:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-10-0-1-231:~$ sudo systemctl status nginx --no-pager
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-08-23 11:12:38 UTC; 20s ago
     Docs: man:nginx(8)
    Main PID: 1948 (nginx)
      Tasks: 3 (limit: 1013)
    Memory: 2.4M (peak: 5.0M)
       CPU: 29ms
    CGroup: /system.slice/nginx.service
            └─1948 *nginx: master process /usr/sbin/nginx -g daemon on; master_process on;*
              └─1951 *nginx: worker process*
                └─1952 *nginx: worker process*
```

Aug 23 11:12:38 ip-10-0-1-231 systemd[1]: Starting nginx.service - A high performance web se.ver...

Aug 23 11:12:38 ip-10-0-1-231 systemd[1]: Started nginx.service - A high performance web ser.erver.

Hint: Some lines were ellipsized, use -l to show in full.

ubuntu@ip-10-0-1-231:~\$ █


```
After this operation, 1602 kB of additional disk space will be used.
Get:1 http://ap-northeast-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.17 [45.1 kB]
Get:2 http://ap-northeast-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.17 [523 kB]
Fetched 568 kB in 0s (25.3 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 72595 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.17_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.17) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.17_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.17) ...
Setting up nginx-common (1.24.0-2ubuntu7.17) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.17) ...
  * Upgrading binary nginx [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-1-231:~$ sudo systemctl start nginx
ubuntu@ip-10-0-1-231:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-10-0-1-231:~$ sudo systemctl status nginx --no-pager
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-08-23 11:12:38 UTC; 20s ago
     Docs: man:nginx(8)
    Main PID: 1948 (nginx)
      Tasks: 3 (limit: 1013)
    Memory: 2.4M (peak: 5.0M)
       CPU: 29ms
    CGroup: /system.slice/nginx.service
            └─1948 *nginx: master process /usr/sbin/nginx -g daemon on; master_process on;*
              └─1951 *nginx: worker process*
                └─1952 *nginx: worker process*
```

Aug 23 11:12:38 ip-10-0-1-231 systemd[1]: Starting nginx.service - A high performance web se.ver...

Aug 23 11:12:38 ip-10-0-1-231 systemd[1]: Started nginx.service - A high performance web ser.erver.

Hint: Some lines were ellipsized, use -l to show in full.

ubuntu@ip-10-0-1-231:~\$ sudo nano /etc/nginx/conf.d/reverse-proxy.conf

ubuntu@ip-10-0-1-231:~\$ sudo nginx -t

2026/08/23 11:13:56 [warn] 2171#2171: conflicting server name "-" on 0.0.0.0:80, ignored

nginx: the configuration file /etc/nginx/nginx.conf syntax is ok

nginx: configuration file /etc/nginx/nginx.conf test is successful

ubuntu@ip-10-0-1-231:~\$ sudo systemctl reload nginx

ubuntu@ip-10-0-1-231:~\$ sudo rm /etc/nginx/sites-enabled/default

ubuntu@ip-10-0-1-231:~\$ sudo nginx -t

nginx: the configuration file /etc/nginx/nginx.conf syntax is ok

nginx: configuration file /etc/nginx/nginx.conf test is successful

ubuntu@ip-10-0-1-231:~\$ sudo systemctl reload nginx

ubuntu@ip-10-0-1-231:~\$ curl http://localhost

<h1>Response from Web Server 1</h1>

ubuntu@ip-10-0-1-231:~\$ █

```

Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-1-231:~$ sudo systemctl start nginx
ubuntu@ip-10-0-1-231:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-10-0-1-231:~$ sudo systemctl status nginx --no-pager
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-08-23 11:12:38 UTC; 20s ago
     Docs: man:nginx(8)
    Main PID: 1948 (nginx)
      Tasks: 5 (limit: 1013)
    Memory: 2.4M (peak: 5.0M)
       CPU: 29ms
    CGroup: /system.slice/nginx.service
            └─1948 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─1951 "nginx: worker process"
                  └─1952 "nginx: worker process"

Aug 23 11:12:38 ip-10-0-1-231 systemd[1]: Starting nginx.service - A high performance web se...
Aug 23 11:12:38 ip-10-0-1-231 systemd[1]: Started nginx.service - A high performance web ser...
Hint: Some lines were ellipsized, use -l to show in full.
ubuntu@ip-10-0-1-231:~$ sudo nano /etc/nginx/conf.d/reverse-proxy.conf
ubuntu@ip-10-0-1-231:~$ sudo nginx -t
2026/08/23 11:13:56 [warn] 2171#2171: conflicting server name "_" on 0.0.0.0:80, ignored
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-10-0-1-231:~$ sudo systemctl reload nginx
ubuntu@ip-10-0-1-231:~$ sudo rm /etc/nginx/sites-enabled/default
ubuntu@ip-10-0-1-231:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-10-0-1-231:~$ sudo systemctl reload nginx
ubuntu@ip-10-0-1-231:~$ curl http://localhost
<h1>Response from Web Server 1</h1>
ubuntu@ip-10-0-1-231:~$ exit
logout
Connection to 18.181.161.173 closed.
skpandey@Ss-MacBook-Air ~ % curl http://18.181.161.173
<h1>Response from Web Server 1</h1>
skpandey@Ss-MacBook-Air ~ % for i in {1..6}; do curl -s http://18.181.161.173; echo; done
<h1>Response from Web Server 2</h1>

<h1>Response from Web Server 1</h1>

<h1>Response from Web Server 2</h1>

<h1>Response from Web Server 1</h1>

<h1>Response from Web Server 2</h1>

<h1>Response from Web Server 1</h1>
skpandey@Ss-MacBook-Air ~ % █

```

Questions to Answer in Your Report

- What was the exact NGINX directive that enabled load balancing across the two backend servers?
Answer: The upstream backend_servers block with the two server directives enabled NGINX to load-balance requests between the backend servers using round-robin by default.
- What is the purpose of the `X-Forwarded-For` header, and what would break without it?
Answer: X-Forwarded-For passes the original client's IP address to the backend server. Without it, the backend would see the proxy's IP instead of the actual client's IP.
- What did you observe in the security group setup that specifically prevents clients from bypassing the proxy?
Answer: The backend security groups allow HTTP (port 80) only from the proxy security group's traffic, so clients cannot directly access the backend web servers.
- If `web-server-1` were stopped, what would happen to requests going through the proxy? Test it and describe the actual behavior you observed.
Answer: NGINX continues sending requests to the available Web Server 2, so the website remains accessible. Requests are served by Web Server 2 after Web Server 1 becomes unavailable.
- Name one difference between what NGINX is doing here and what Route 53 would do if it were placed in front of this same setup.
Answer: NGINX performs application-level HTTP reverse proxying and load balancing between backend servers, while Route 53 is a DNS service that routes users to IP addresses or AWS resources before the HTTP connection is made.

Assignment 8

TOPIC : AWS Assignment Docker

Date: 10/08/2026

Question 01: Pull and run the hello-world image

Answer:

```
ubuntu@ip-10-0-1-215:~$ sudo systemctl start docker
ubuntu@ip-10-0-1-215:~$ sudo systemctl enable docker
ubuntu@ip-10-0-1-215:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3-24.04.2
ubuntu@ip-10-0-1-215:~$ sudo docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
4f55086f7d00: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbbd48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
ubuntu@ip-10-0-1-215:~$ sudo docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

ubuntu@ip-10-0-1-215:~$
```

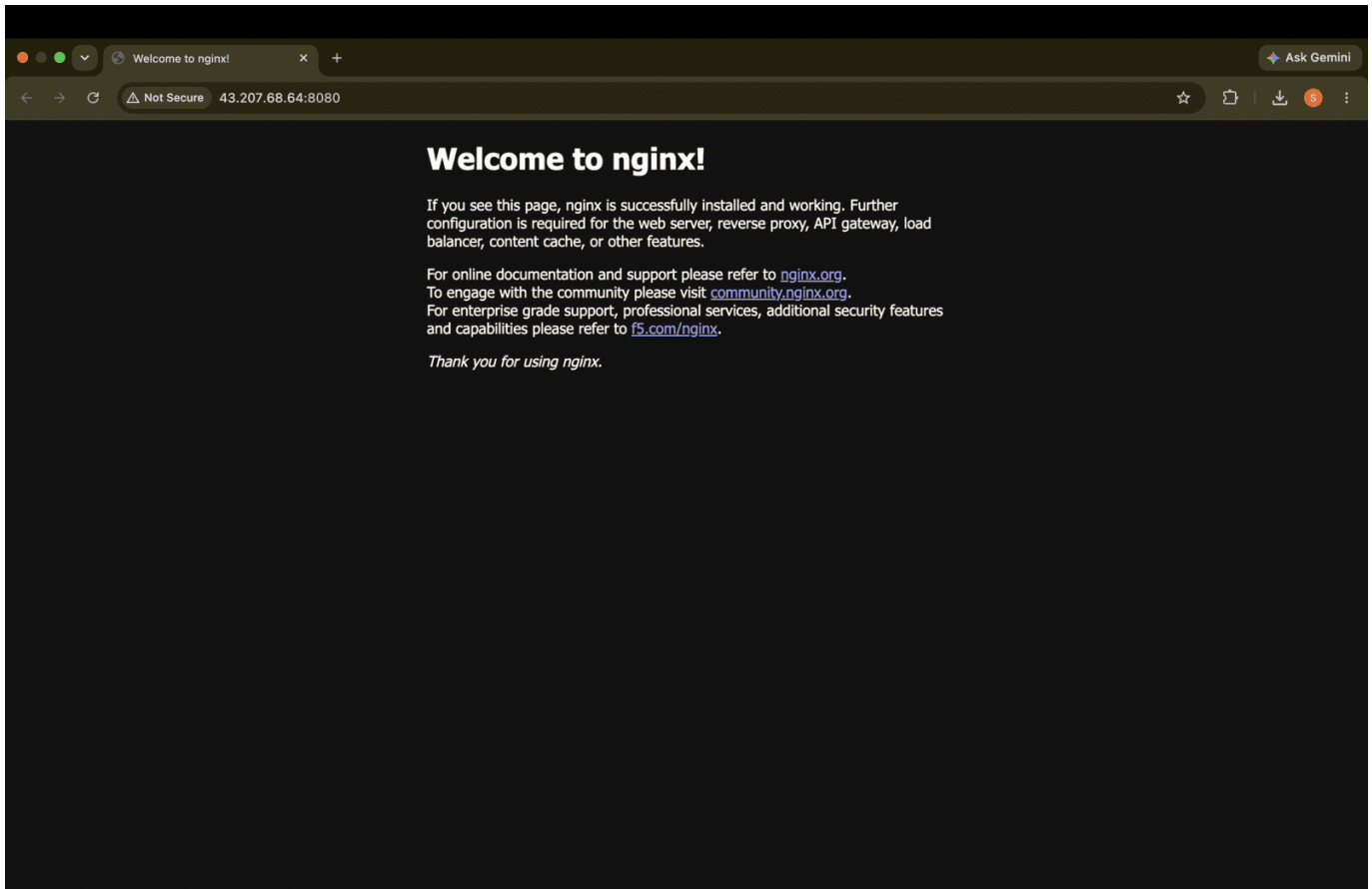
Question 02: Pull and run the nginx image, check if the webpage is actively hosted.

Answer:

```
ubuntu@ip-10-0-1-215:~$ sudo docker run -d --name mynginx -p 8080:80 nginx
275079f48be043884f093061273fd6d7d678eaff79afe58d04290fd5ebacb871
ubuntu@ip-10-0-1-215:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                    NAMES
275079f48be0   nginx    "/docker-entrypoint. ..."  5 seconds ago Up 4 seconds    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   mynginx
ubuntu@ip-10-0-1-215:~$ curl localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
ubuntu@ip-10-0-1-215:~$ exit
^
```

Question 03: Pull and run mysql container from hub. Can you log into the tty provided using -it option.

Answer:

```
ubuntu@ip-10-0-1-215:~$ sudo docker pull mysql
Using default tag: latest
latest: Pulling from library/mysql
71fa527c6c68: Pull complete
c4e766e27938: Pull complete
0bb65eb170f9: Pull complete
30627cea5424: Pull complete
7e887550dd4: Pull complete
35475b275575: Pull complete
27683f99b921: Pull complete
1791ad7fecf: Pull complete
e489d0cc782ea: Pull complete
808810e42188: Download complete
81ee3e1129f4: Download complete
Digest: sha256:66aec17cd21a956029b83f083b813073859e8355dc1a0e55df6ba02f0e32345
Status: Downloaded newer image for mysql:latest
docker.io/library/mysql:latest
ubuntu@ip-10-0-1-215:~$ sudo docker run -it --name mysql-container mysql
2026-08-24 15:59:29+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 26.7.0-1.el9 started.
2026-08-24 15:59:30+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-08-24 15:59:30+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 26.7.0-1.el9 started.
2026-08-24 15:59:30+00:00 [ERROR] [Entrypoint]: Database is uninitialized and password option is not specified
You need to specify one of the following as an environment variable:
- MYSQL_ROOT_PASSWORD
- MYSQL_ALLOW_EMPTY_PASSWORD
- MYSQL_RANDOM_ROOT_PASSWORD
ubuntu@ip-10-0-1-215:~$ sudo docker rm mysql-container
mysql-container
ubuntu@ip-10-0-1-215:~$ sudo docker run -it --name mysql-container -e MYSQL_ROOT_PASSWORD=admin123 mysql
2026-08-24 16:00:10+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 26.7.0-1.el9 started.
2026-08-24 16:00:11+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-08-24 16:00:11+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 26.7.0-1.el9 started.
2026-08-24 16:00:11+00:00 [Note] [Entrypoint]: Initializing database files
2026-08-24 16:00:11.657172Z 0 [System] [MY-015017] [Server] MySQL Server Initialization - start.
2026-08-24 16:00:11.671071Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 26.7.0) initializing of server in progress as process 79
2026-08-24 16:00:11.683941Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-08-24 16:00:12.309410Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-08-24 16:00:14.441832Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching off the --initialize-insecure option.
2026-08-24 16:00:17.315550Z 0 [System] [MY-015018] [Server] MySQL Server Initialization - end.
2026-08-24 16:00:17+00:00 [Note] [Entrypoint]: Database files initialized
2026-08-24 16:00:17+00:00 [Note] [Entrypoint]: Starting temporary server
mysqld will log errors to /var/lib/mysql/a18904dc4530.err
mysqld is running as pid 118
2026-08-24 16:00:19+00:00 [Note] [Entrypoint]: Temporary server started.
'/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
2026-08-24 16:00:21+00:00 [Note] [Entrypoint]: Stopping temporary server
2026-08-24 16:00:24+00:00 [Note] [Entrypoint]: Temporary server stopped
2026-08-24 16:00:24+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.
2026-08-24 16:00:24.219176Z 0 [System] [MY-015015] [Server] MySQL Server - start.
2026-08-24 16:00:24.528898Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 26.7.0) starting as process 1
2026-08-24 16:00:24.528920Z 0 [System] [MY-015603] [Server] MySQL Server has access to 2 logical CPUs.
2026-08-24 16:00:24.528936Z 0 [System] [MY-015603] [Server] MySQL Server has access to 955523072 bytes of physical memory.
2026-08-24 16:00:24.554064Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-08-24 16:00:25.054143Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-08-24 16:00:25.463001Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-08-24 16:00:25.463083Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-08-24 16:00:25.463102Z 0 [Warning] [MY-015696] [Server] TLS channel 'mysql_main' is not configured to require post-quantum key exchange algorithms; future TLS sessions may use a non-post-quantum key exchange algorithm because force_pqc is disabled
2026-08-24 16:00:25.471671Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different director y.
2026-08-24 16:00:25.520248Z 0 [Warning] [MY-015696] [Server] Plugin mysqlx reported: 'TLS channel 'mysqlx' is not configured to require post-quantum key exchange algorithms; future TLS sessions may use a non-
```

```

71fa527c6c68: Pull complete
c4e76ae27938: Pull complete
9bb65eb170f9: Pull complete
30627cea5424: Pull complete
7e887550bdc4: Pull complete
35475b276575: Pull complete
27683f999921: Pull complete
1791a4d7fecf: Pull complete
e480dcc782ea: Pull complete
808810e42158: Download complete
81ee3e1129f4: Download complete
Digest: sha256:60aec17cd21a956029b83f083b813073859e8355dca0e0e55df6ba02f0e32345
Status: Downloaded newer image for mysql:latest
docker.io/library/mysql:latest
ubuntu@ip-10-0-1-215:~$ sudo docker run -it --name mysql-container mysql
2026-08-24 15:59:29+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 26.7.0-1.el9 started.
2026-08-24 15:59:30+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-08-24 15:59:30+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 26.7.0-1.el9 started.
2026-08-24 15:59:30+00:00 [ERROR] [Entrypoint]: Database is uninitialized and password option is not specified
You need to specify one of the following as an environment variable:
- MYSQL_ROOT_PASSWORD
- MYSQL_ALLOW_EMPTY_PASSWORD
- MYSQL_RANDOM_ROOT_PASSWORD
ubuntu@ip-10-0-1-215:~$ sudo docker rm mysql-container
mysql-container
ubuntu@ip-10-0-1-215:~$ sudo docker run -it --name mysql-container -e MYSQL_ROOT_PASSWORD=admin123 mysql
2026-08-24 16:00:10+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 26.7.0-1.el9 started.
2026-08-24 16:00:11+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-08-24 16:00:11+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 26.7.0-1.el9 started.
2026-08-24 16:00:11+00:00 [Note] [Entrypoint]: Initializing database files
2026-08-24T16:00:11.667177Z 0 [System] [MY-015017] [Server] MySQL Server Initialization - start.
2026-08-24T16:00:11.671071Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 26.7.0) initializing of server in progress as process 79
2026-08-24T16:00:11.683941Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-08-24T16:00:12.309410Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-08-24T16:00:14.441832Z 6 [Warning] [MY-010463] [Server] root@localhost is created with an empty password ! Please consider switching off the --initialize-insecure option.
2026-08-24T16:00:17.315556Z 0 [System] [MY-015018] [Server] MySQL Server Initialization - end.
2026-08-24 16:00:17+00:00 [Note] [Entrypoint]: Database files initialized
2026-08-24 16:00:17+00:00 [Note] [Entrypoint]: Starting temporary server
mysqld will log errors to /var/lib/mysql/a18904dc46530.err
mysqld is running as pid 118
2026-08-24 16:00:19+00:00 [Note] [Entrypoint]: Temporary server started.
'/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'

2026-08-24 16:00:21+00:00 [Note] [Entrypoint]: Stopping temporary server
2026-08-24 16:00:24+00:00 [Note] [Entrypoint]: Temporary server stopped

2026-08-24 16:00:24+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.

2026-08-24T16:00:24.219176Z 0 [System] [MY-015015] [Server] MySQL Server - start.
2026-08-24T16:00:24.528982Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 26.7.0) starting as process 1
2026-08-24T16:00:24.528920Z 0 [System] [MY-015603] [Server] MySQL Server has access to 2 logical CPUs.
2026-08-24T16:00:24.528936Z 0 [System] [MY-015603] [Server] MySQL Server has access to 955523072 bytes of physical memory.
2026-08-24T16:00:24.554064Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-08-24T16:00:25.054143Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-08-24T16:00:25.463001Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-08-24T16:00:25.463003Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-08-24T16:00:25.463102Z 0 [Warning] [MY-015696] [Server] TLS channel 'mysql_main' is not configured to require post-quantum key exchange algorithms; future TLS sessions may use a non-post-quantum key exch
ange algorithm because force_pqc is disabled
2026-08-24T16:00:25.471671Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different director
y.
2026-08-24T16:00:25.520248Z 0 [Warning] [MY-015696] [Server] Plugin mysqlx reported: 'TLS channel 'mysqlx' is not configured to require post-quantum key exchange algorithms; future TLS sessions may use a non-
post-quantum key exchange algorithm because force_pqc is disabled'
2026-08-24T16:00:25.520828Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '26.7.0' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
2026-08-24T16:00:25.520601Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock

```

Question 04: Write a Bash script to create an EC2 instance using terraform. Use file provisioner to send another bash script to the newly created instance. The script contains instructions to install nginx and mysql containers. Use remote-exec feature of terraform to run this script when the resource is being created.

Answer:

```

ubuntu@ip-172-31-28-42:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                    NAMES
363d550c4cba   mysql    "docker-entrypoint.s..." 3 minutes ago  Up 3 minutes  3306/tcp, 33060/tcp      mysql
cb3921c41a81   nginx    "/docker-entrypoint. ..." 3 minutes ago  Up 3 minutes  80/tcp                  nginx
ubuntu@ip-172-31-28-42:~$

```

```
#!/bin/bash
```

```
sudo apt update -y  
sudo apt install docker.io -y
```

```
sudo systemctl start docker
```

```
sudo docker pull nginx  
sudo docker pull mysql
```

```
sudo docker run -d --name nginx nginx  
sudo docker run -d --name mysql -e MYSQL_ROOT_PASSWORD=root mysql
```

```
⌘ Get Help  
⌘ Exit
```

```
⌘ WriteOut  
⌘ Justify
```

```
⌘ Read File  
⌘ Where is
```

```
⌘ Prev Pg  
⌘ Next Pg
```

```
⌘ Cut Text  
⌘ UnCut Text
```

```
⌘ Cur Pos  
⌘ To Spell
```

```

  UW PICO 5.09                                     File: main.tf
}
provider "aws" {
  region = "ap-northeast-1"
}

resource "aws_instance" "web" {
  ami           = "ami-0e239f7f4b3226722"
  instance_type = "t3.micro"
  key_name      = "terraform-key"

  provisioner "file" {
    source      = "install.sh"
    destination = "/home/ubuntu/install.sh"

    connection {
      type      = "ssh"
      user      = "ubuntu"
      private_key = file("/Users/skpandey/Downloads/terraform-key.pem")
      host      = self.public_ip
    }
  }

  provisioner "remote-exec" {
    inline = [
      "chmod +x /home/ubuntu/install.sh",
      "sudo /home/ubuntu/install.sh"
    ]
  }

  connection {
    type      = "ssh"
    user      = "ubuntu"
    private_key = file("/Users/skpandey/Downloads/terraform-key.pem")
    host      = self.public_ip
  }
}
}

}

^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Pg      ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify       ^W Where is      ^V Next Pg      ^U UnCut Text    ^T To Spell
```

Question 05: Pull the nginx image from docker hub, rename the image to mynginx and upload it to docker hub.

Answer:

skpandey@Ss-MacBook-Air assignment8 % ssh -i /Users/skpandey/Downloads/terraform-key.pem ubuntu@57.183.19.79
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1019-aws x86_64)

- * Documentation: <https://help.ubuntu.com>
- * Management: <https://landscape.canonical.com>
- * Support: <https://ubuntu.com/pro>

System information as of Mon Aug 24 16:45:59 UTC 2026

System load:	0.0	Temperature:	-273.1 C
Usage of /:	57.8% of 6.71GB	Processes:	123
Memory usage:	77%	Users logged in:	0
Swap usage:	0%	IPv4 address for ens5:	172.31.28.42

Expanded Security Maintenance for Applications is not enabled.

72 updates can be applied immediately.
64 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

Last login: Mon Aug 24 16:41:55 2026 from 117.195.124.73
ubuntu@ip-172-31-28-42:~\$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:0d4374c710a9649200e84f8ef8dbdd4fa76c0c107839cd50f1e42a63916b0f2e
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-28-42:~\$ sudo docker tag nginx mynginx
ubuntu@ip-172-31-28-42:~\$ sudo docker images

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA	In Use
mynginx:latest	0d4374c710a9	241MB	66.2MB	U	
mysql:latest	66aec17cd21a	1.3GB	290MB	U	
nginx:latest	0d4374c710a9	241MB	66.2MB	U	

USING WEB-BASED LOGIN

 [Info](#) → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: HHKQ-VNFB
Press ENTER to open your browser or submit your device code here: <https://login.docker.com/activate>
Waiting for authentication in the browser..

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded
ubuntu@ip-172-31-28-42:~\$ docker info --format '{{.Name}}'

Expanded Security Maintenance for Applications is not enabled.

72 updates can be applied immediately.
64 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

Last login: Mon Aug 24 16:41:55 2026 from 117.195.124.73
ubuntu@ip-172-31-28-42:~\$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:0d4374c710a9649200e84f8ef8dbdd4fa76c0c107839cd50f1e42a63916b0f2e
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-28-42:~\$ sudo docker tag nginx mynginx
ubuntu@ip-172-31-28-42:~\$ sudo docker images

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA	In Use
mynginx:latest	0d4374c710a9	241MB	66.2MB	U	
mysql:latest	66aec17cd21a	1.3GB	290MB	U	
nginx:latest	0d4374c710a9	241MB	66.2MB	U	

USING WEB-BASED LOGIN

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Waiting for authentication in the browser..

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded
ubuntu@ip-172-31-28-42:~\$ docker info --format '{{.Name}}'

permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
ubuntu@ip-172-31-28-42:~\$ sudo docker tag mynginx pandeysdoc/mynginx:latest
ubuntu@ip-172-31-28-42:~\$ sudo docker push pandeysdoc/mynginx:latest
The push refers to repository [docker.io/pandeysdoc/mynginx]
f530c3e421fc: Mounted from library/nginx
70b55399d6de: Mounted from library/nginx
5d480233f531: Mounted from library/nginx
26c307b5e35a: Mounted from library/nginx
746b934a8960: Mounted from library/nginx
5508f6432d3e: Mounted from library/nginx
128fcc7b2300: Mounted from library/nginx
latest: digest: sha256:c2e385ef468149bdc3297621cea453b47b095816fec4fc7be6ff837bce8deb7d size: 2290

 [Info](#) → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:0d4374c710a9649200e84f8ef8dbdd4fa76c0c107839cd50f1e42a63916b0f2e -> sha256:c2e385ef468149bdc3297621cea453b47b095816fec4fc7be6ff837bce8deb7d
ubuntu@ip-172-31-28-42:~\$

Assignment 9

TOPIC : AWS Assignment TerraForm

Date: 10/08/2026

1. create a vpc using terraform, attach a subnet to that vpc and attach a routing table to the subnet with 0.0.0.0 path/route. create an ec2 instance inside the subnet.

Answer:

```
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
skpandey@Ss-MacBook-Air terraform-q1 % terraform show
# aws_instance.main:
resource "aws_instance" "main" {
  ami              = "ami-0e239f7f4b3226722"
  arn              = "arn:aws:ec2:ap-northeast-1:508692702174:instance/i-04fd6f3ba24991111"
  associate_public_ip_address = true
  availability_zone = "ap-northeast-1a"
  disable_api_stop  = false
  disable_api_termination = false
  ebs_optimized      = false
  force_destroy      = false
  get_password_data  = false
  hibernation        = false
  host_id            = null
  iam_instance_profile = null
  id                 = "i-04fd6f3ba24991111"
  instance_initiated_shutdown_behavior = "stop"
  instance_lifecycle = null
  instance_state     = "running"
  instance_type      = "t3.micro"
  ipv6_address_count = 0
  ipv6_addresses     = []
  key_name            = null
  monitoring          = false
  outpost_arn         = null
  password_data       = null
  placement_group     = null
  placement_group_id  = null
  placement_partition_number = 0
  primary_network_interface_id = "eni-06cc67f959faafe00"
  private_dns         = "ip-10-0-1-218.ap-northeast-1.compute.internal"
  private_ip          = "10.0.1.218"
  public_dns          = null
  public_ip           = "54.65.250.183"
  region              = "ap-northeast-1"
  secondary_private_ips = []
  security_groups     = []
  source_dest_check    = true
  spot_instance_request_id = null
  subnet_id           = "subnet-0932d341a834ceea9"
  tags                = {
    "Name" = "terraform-ec2"
  }
  tags_all           = {
    "Name" = "terraform-ec2"
  }
  tenancy             = "default"
  user_data_replace_on_change = false
  vpc_security_group_ids = [
    "sg-077c0ad33d177d7df",
  ]
  capacity_reservation_specification {
    capacity_reservation_preference = "open"
  }
  cpu_options {
    amd_sev_snp = null
    core_count  = 1
    nested_virtualization = null
    threads_per_core = 2
  }
  credit_specification {
```

```

    metadata_options {
      http_endpoint      = "enabled"
      http_protocol_ipv6 = "disabled"
      http_put_response_hop_limit = 2
      http_tokens        = "required"
      instance_metadata_tags = "disabled"
    }

    primary_network_interface {
      delete_on_termination = true
      network_interface_id = "eni-86cc67f959faafe00"
    }

    private_dns_name_options {
      enable_resource_name_dns_a_record = false
      enable_resource_name_dns_aaaa_record = false
      hostname_type                     = "ip-name"
    }

    root_block_device {
      delete_on_termination = true
      device_name           = "/dev/sda1"
      encrypted             = false
      iops                  = 3000
      kms_key_id            = null
      tags                  = {}
      tags_all              = {}
      throughput            = 125
      volume_id             = "vol-0397b4e8af02913bd"
      volume_size           = 8
      volume_type           = "gp3"
    }
  }
}

# aws_internet_gateway.main:
resource "aws_internet_gateway" "main" {
  arn      = "arn:aws:ec2:ap-northeast-1:500692702174::internet-gateway/igw-0ce1fa738586039ad"
  id       = "igw-0ce1fa738586039ad"
  owner_id = "500692702174"
  region   = "ap-northeast-1"
  tags     = {
    "Name" = "terraform-igw"
  }
  tags_all = {
    "Name" = "terraform-igw"
  }
  vpc_id = "vpc-08fbcf1e559209d23"
}

# aws_route_table.main:
resource "aws_route_table" "main" {
  arn      = "arn:aws:ec2:ap-northeast-1:500692702174::route-table/rtb-0aec2146e17ad1b3"
  id       = "rtb-0aec2146e17ad1b3"
  owner_id = "500692702174"
  propagating_vgws = []
  region     = "ap-northeast-1"
  routes     = [
    {
      carrier_gateway_id = null
      cidr_block         = "0.0.0.0/0"
      core_network_arn   = null
      destination_prefix_list_id = null
      egress_only_gateway_id = null
      gateway_id         = "igw-0ce1fa738586039ad"
      ipv6_cidr_block     = null
      local_gateway_id    = null
      nat_gateway_id      = null
      network_interface_id = null
      odb_network_arn     = null
      transit_gateway_id  = null
      vpc_endpoint_id     = null
      vpc_peering_connection_id = null
    }
  ],
  tags     = {
    "Name" = "terraform-route-table"
  }
  tags_all = {
    "Name" = "terraform-route-table"
  }
  vpc_id = "vpc-08fbcf1e559209d23"
}

# aws_route_table_association.main:
resource "aws_route_table_association" "main" {
  gateway_id = null
  id         = "rtbassoc-0440d37041bb0a47b"
  region     = "ap-northeast-1"
  route_table_id = "rtb-0aec2146e17ad1b3"
  subnet_id    = "subnet-0932d341a834ceea9"
}

# aws_security_group.ec2:
resource "aws_security_group" "ec2" {
  arn      = "arn:aws:ec2:ap-northeast-1:500692702174::security-group/sg-077c0ad33d177d7df"
  description = "Managed by Terraform"
  egress     = [
    {
      cidr_blocks = [
        "0.0.0.0/0",
      ]
      description = null
      from_port   = 0
      ipv6_cidr_blocks = []
      prefix_list_ids = []
      protocol    = "-1"
      security_groups = []
      self        = false
      to_port     = 0
    }
  ],
  id = "sg-077c0ad33d177d7df"
  ingress = [
    {
      cidr_blocks = [
        "0.0.0.0/0",
      ]
      description = null
      from_port   = 22
      ipv6_cidr_blocks = []
      prefix_list_ids = []
      protocol    = "tcp"
      security_groups = []
    }
  ]
}

```

```

name_prefix      = null
owner_id         = "508692702174"
region           = "ap-northeast-1"
revoke_rules_on_delete = false
tags             = {}
tags_all         = {}
vpc_id           = "vpc-08fbcf1e559209d23"
}

# aws_subnet.main:
resource "aws_subnet" "main" {
  arn                                = "arn:aws:ec2:ap-northeast-1:508692702174:subnet/subnet-0932d341a834ceea9"
  assign_ipv6_address_on_creation   = false
  availability_zone                 = "ap-northeast-1a"
  cidr_block                        = "10.0.1.0/24"
  customer_owned_ipv4_pool          = null
  enable_dns64                     = false
  enable_lni_at_device_index        = 0
  enable_resource_name_dns_a_record_on_launch = false
  enable_resource_name_dns_aaaa_record_on_launch = false
  id                                = "subnet-0932d341a834ceea9"
  ipv6_cidr_block                   = null
  ipv6_cidr_block_association_id    = null
  ipv6_native                       = false
  map_customer_owned_ip_on_launch  = false
  map_public_ip_on_launch          = true
  outpost_arn                      = null
  owner_id                         = "508692702174"
  private_dns_hostname_type_on_launch = "ip-name"
  region                           = "ap-northeast-1"
  tags                              = {
    "Name" = "terraform-subnet"
  }
  tags_all                         = {
    "Name" = "terraform-subnet"
  }
  vpc_id                           = "vpc-08fbcf1e559209d23"
}

# aws_vpc.main:
resource "aws_vpc" "main" {
  arn                                = "arn:aws:ec2:ap-northeast-1:508692702174:vpc/vpc-08fbcf1e559209d23"
  assign_generated_ipv6_cidr_block   = false
  cidr_block                        = "10.0.0.0/16"
  default_network_acl_id            = "acl-0292be08e362a889d"
  default_route_table_id            = "rtb-09cbbc56e42534610"
  default_security_group_id         = "sg-0fb5f7d50a53ff622"
  dhcp_options_id                   = "dopt-01b4955bc82d09f96"
  enable_dns_hostnames              = false
  enable_dns_support                 = true
  enable_network_address_usage_metrics = false
  id                                = "vpc-08fbcf1e559209d23"
  instance_tenancy                  = "default"
  ipv6_association_id               = null
  ipv6_cidr_block                   = null
  ipv6_cidr_block_network_border_group = null
  ipv6_ipam_pool_id                 = null
  ipv6_netmask_length               = 0
  main_route_table_id               = "rtb-09cbbc56e42534610"
  owner_id                         = "508692702174"
  region                           = "ap-northeast-1"
  tags                              = {
    "Name" = "terraform-vpc"
  }
}

```



2. Using terraform create an s3 bucket and host a static website in it.
Answer:

```

# aws_s3_bucket_policy.website will be created
+ resource "aws_s3_bucket_policy" "website" {
+   bucket = (known after apply)
+   id     = (known after apply)
+   policy = (known after apply)
+   region = "ap-northeast-1"
+ }

# aws_s3_bucket_public_access_block.website will be created
+ resource "aws_s3_bucket_public_access_block" "website" {
+   block_public_acls = false
+   block_public_policy = false
+   bucket            = (known after apply)
+   id               = (known after apply)
+   ignore_public_acls = false
+   region           = "ap-northeast-1"
+   restrict_public_buckets = false
+ }

# aws_s3_bucket_website_configuration.website will be created
+ resource "aws_s3_bucket_website_configuration" "website" {
+   bucket            = (known after apply)
+   id               = (known after apply)
+   region           = "ap-northeast-1"
+   routing_rules     = (known after apply)
+   website_domain    = (known after apply)
+   website_endpoint  = (known after apply)

+   error_document {
+     key = "error.html"
+   }

+   index_document {
+     suffix = "index.html"
+   }

+   routing_rule (known after apply)
+ }

Plan: 4 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ website_endpoint = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_s3_bucket.website: Creating...
aws_s3_bucket.website: Creation complete after 7s [id=terraform-static-website-2026-unique]
aws_s3_bucket_policy.website: Creating...
aws_s3_bucket_public_access_block.website: Creating...
aws_s3_bucket_website_configuration.website: Creating...
aws_s3_bucket_public_access_block.website: Creation complete after 1s [id=terraform-static-website-2026-unique]
aws_s3_bucket_website_configuration.website: Creation complete after 1s [id=terraform-static-website-2026-unique]
aws_s3_bucket_policy.website: Creation complete after 1s [id=terraform-static-website-2026-unique]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.

Outputs:

website_endpoint = "terraform-static-website-2026-unique.s3-website-ap-northeast-1.amazonaws.com"
skpandey@Ss-MacBook-Air terraform-q2 %

```

- Using terraform create an IAM user that has the administrative access and ec2 full access.
Answer:

```

Changes to Outputs:
+ iam_user_name = "terraform-admin-user"

Note: You didn't use the -out option to save this plan, so Terraform can't
guarantee to take exactly these actions if you run "terraform apply" now.
skpandey@Ss-MacBook-Air terraform-q3 % terraform apply

Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_iam_user.admin_user will be created
+ resource "aws_iam_user" "admin_user" {
+ arn                = (known after apply)
+ force_destroy      = false
+ id                 = (known after apply)
+ name                = "terraform-admin-user"
+ path                = "/"
+ tags_all           = (known after apply)
+ unique_id           = (known after apply)
}

# aws_iam_user_policy_attachment.admin_access will be created
+ resource "aws_iam_user_policy_attachment" "admin_access" {
+ id                = (known after apply)
+ policy_arn        = "arn:aws:iam::aws:policy/AdministratorAccess"
+ user               = "terraform-admin-user"
}

# aws_iam_user_policy_attachment.ec2_full_access will be created
+ resource "aws_iam_user_policy_attachment" "ec2_full_access" {
+ id                = (known after apply)
+ policy_arn        = "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
+ user               = "terraform-admin-user"
}

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ iam_user_name = "terraform-admin-user"

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_iam_user.admin_user: Creating...
aws_iam_user.admin_user: Creation complete after 1s [id=terraform-admin-user]
aws_iam_user_policy_attachment.ec2_full_access: Creating...
aws_iam_user_policy_attachment.admin_access: Creating...
aws_iam_user_policy_attachment.ec2_full_access: Creation complete after 1s [id=terraform-admin-user/arn:aws:iam::aws:policy/AmazonEC2FullAccess]
aws_iam_user_policy_attachment.admin_access: Creation complete after 1s [id=terraform-admin-user/arn:aws:iam::aws:policy/AdministratorAccess]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

iam_user_name = "terraform-admin-user"
skpandey@Ss-MacBook-Air terraform-q3 %

```

- Using terraform create an ec2 instance and pass a file called provisioners.sh and the .sh file should contain steps to create a webserver using nginx.

Answer:

```

+ cidr_blocks      = [
+   + "0.0.0.0/0",
+ ]
+ from_port        = 80
+ ipv6_cidr_blocks = []
+ prefix_list_ids  = []
+ protocol         = "tcp"
+ security_groups  = []
+ self             = false
+ to_port          = 80
+ # (1 unchanged attribute hidden)
},
]
+ name              = "terraform-nginx-sg"
+ name_prefix       = (known after apply)
+ owner_id          = (known after apply)
+ region            = "ap-northeast-1"
+ revoke_rules_on_delete = false
+ tags_all          = (known after apply)
+ vpc_id            = (known after apply)
}

Plan: 1 to add, 1 to change, 0 to destroy.

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_security_group.nginx: Creating...
aws_security_group.nginx: Creation complete after 5s [id=sg-09d4a4dfa952443ce]
aws_instance.websserver: Modifying... [id=i-02fc0b1e5c65a18a8]
aws_instance.websserver: Modifications complete after 4s [id=i-02fc0b1e5c65a18a8]

Apply complete! Resources: 1 added, 1 changed, 0 destroyed.

Outputs:
public_ip = "3.113.201.73"
skpandey@Ss-MacBook-Air terraform-q4 % curl http://3.113.201.73
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
skpandey@Ss-MacBook-Air terraform-q4 % █

```

5. Find all the options on the command - plan init


```
skpandey@Ss-MacBook-Air terraform-q4 % terraform init -help
Usage: terraform [global options] init [options]
```

Initialize a new or existing Terraform working directory by creating initial files, loading any remote state, downloading modules, etc.

This is the first command that should be run for any new or existing Terraform configuration per machine. This sets up all the local data necessary to run Terraform that is typically not committed to version control.

This command is always safe to run multiple times. Though subsequent runs may give errors, this command will never delete your configuration or state. Even so, if you have important information, please back it up prior to running this command, just in case.

Options:

```
[ -backend=false          Disable backend or HCP Terraform initialization
                           for this configuration and use what was previously
                           initialized instead.

                           aliases: -cloud=false

-backend-config=path      Configuration to be merged with what is in the
                           configuration file's 'backend' block. This can be
                           either a path to an HCL file with key/value
                           assignments (same format as terraform.tfvars) or a
                           'key=value' format, and can be specified multiple
                           times. The backend type must be in the configuration
                           itself.

-force-copy               Suppress prompts about copying state data when
                           initializing a new state backend. This is
                           equivalent to providing a "yes" to all confirmation
                           prompts.

-from-module=SOURCE       Copy the contents of the given module into the target
                           directory before initialization.

-get=false                Disable downloading modules for this configuration.

-input=false              Disable interactive prompts. Note that some actions may
                           require interactive prompts and will error if input is
                           disabled.

-lock=false               Don't hold a state lock during backend migration.
                           This is dangerous if others might concurrently run
                           commands against the same workspace.

-lock-timeout=0s          Duration to retry a state lock.

-no-color                 If specified, output won't contain any color.

-json                     If specified, machine readable output will be
                           printed in JSON format.

-plugin-dir                Directory containing plugin binaries. This overrides all
                           default search paths for plugins, and prevents the
                           automatic installation of plugins. This flag can be used
                           multiple times.

-reconfigure              Reconfigure a backend, ignoring any saved
                           configuration.
```

```
-get=false                Disable downloading modules for this configuration.

-input=false              Disable interactive prompts. Note that some actions may
                           require interactive prompts and will error if input is
                           disabled.

-lock=false               Don't hold a state lock during backend migration.
                           This is dangerous if others might concurrently run
                           commands against the same workspace.

-lock-timeout=0s          Duration to retry a state lock.

-no-color                 If specified, output won't contain any color.

-json                     If specified, machine readable output will be
                           printed in JSON format.

-plugin-dir                Directory containing plugin binaries. This overrides all
                           default search paths for plugins, and prevents the
                           automatic installation of plugins. This flag can be used
                           multiple times.

-reconfigure              Reconfigure a backend, ignoring any saved
                           configuration.

-migrate-state             Reconfigure a backend, and attempt to migrate any
                           existing state.

-upgrade                  Install the latest module and provider versions
                           allowed within configured constraints, overriding the
                           default behavior of selecting exactly the version
                           recorded in the dependency lockfile.

-lockfile=MODE             Set a dependency lockfile mode.
                           Currently only "readonly" is valid.

-ignore-remote-version     A rare option used for HCP Terraform and the remote backend
                           only. Set this to ignore checking that the local and remote
                           Terraform versions use compatible state representations, making
                           an operation proceed even when there is a potential mismatch.
                           See the documentation on configuring Terraform with
                           HCP Terraform or Terraform Enterprise for more information.

-test-directory=path       Set the Terraform test directory, defaults to "tests".

-var 'foo=bar'             Set a value for one of the input variables in the root
                           module of the configuration. Use this option more than
                           once to set more than one variable.

-var-file=filename         Load variable values from the given file, in addition
                           to the default files terraform.tfvars and *.auto.tfvars.
                           Use this option more than once to include more than one
                           variables file.

-enable-pluggable-state-storage-experiment [EXPERIMENTAL]
                           A flag to enable an alternative init command that allows use of
                           pluggable state storage. Only usable with experiments enabled.

-create-default-workspace [EXPERIMENTAL]
                           This flag must be used alongside the -enable-pluggable-state-storage-
                           experiment flag with experiments enabled. This flag's value defaults
                           to true, which allows the default workspace to be created if it does
                           not exist. Use -create-default-workspace=false to disable this behavior.
```

```
skpandey@Ss-MacBook-Air terraform-q4 %
```

```
skpandey@Ss-MacBook-Air terraform-q4 % terraform plan -help
Usage: terraform [global options] plan [options]
```

Generates a speculative execution plan, showing what actions Terraform would take to apply the current configuration. This command will not actually perform the planned actions.

You can optionally save the plan to a file, which you can then pass to the "apply" command to perform exactly the actions described in the plan.

Plan Customization Options:

The following options customize how Terraform will produce its plan. You can also use these options when you run "terraform apply" without passing it a saved plan, in order to plan and apply in a single command.

-destroy	Select the "destroy" planning mode, which creates a plan to destroy all objects currently managed by this Terraform configuration instead of the usual behavior.
-refresh-only	Select the "refresh only" planning mode, which checks whether remote objects still match the outcome of the most recent Terraform apply but does not propose any actions to undo any changes made outside of Terraform.
-refresh=false	Skip checking for external changes to remote objects while creating the plan. This can potentially make planning faster, but at the expense of possibly planning against a stale record of the remote system state.
-replace=resource	Force replacement of a particular resource instance using its resource address. If the plan would've normally produced an update or no-op action for this instance, Terraform will plan to replace it instead. You can use this option multiple times to replace more than one object.
-target=resource	Limit the planning operation to only the given module, resource, or resource instance and all of its dependencies. You can use this option multiple times to include more than one object. This is for exceptional use only.
-var 'foo=bar'	Set a value for one of the input variables in the root module of the configuration. Use this option more than once to set more than one variable.
-var-file=filename	Load variable values from the given file, in addition to the default files terraform.tfvars and *.auto.tfvars. Use this option more than once to include more than one variables file.

Other Options:

-compact-warnings	If Terraform produces any warnings that are not accompanied by errors, shows them in a more compact form that includes only the summary messages.
-detailed-exitcode	Return detailed exit codes when the command exits. This will change the meaning of exit codes to: 0 - Succeeded, diff is empty (no changes) 1 - Errored 2 - Succeeded, there is a diff
-generate-config-out=path	(Experimental) If import blocks are present in configuration, instructs Terraform to generate HCL

-refresh=false	Skip checking for external changes to remote objects while creating the plan. This can potentially make planning faster, but at the expense of possibly planning against a stale record of the remote system state.
-replace=resource	Force replacement of a particular resource instance using its resource address. If the plan would've normally produced an update or no-op action for this instance, Terraform will plan to replace it instead. You can use this option multiple times to replace more than one object.
-target=resource	Limit the planning operation to only the given module, resource, or resource instance and all of its dependencies. You can use this option multiple times to include more than one object. This is for exceptional use only.
-var 'foo=bar'	Set a value for one of the input variables in the root module of the configuration. Use this option more than once to set more than one variable.
-var-file=filename	Load variable values from the given file, in addition to the default files terraform.tfvars and *.auto.tfvars. Use this option more than once to include more than one variables file.

Other Options:

-compact-warnings	If Terraform produces any warnings that are not accompanied by errors, shows them in a more compact form that includes only the summary messages.
-detailed-exitcode	Return detailed exit codes when the command exits. This will change the meaning of exit codes to: 0 - Succeeded, diff is empty (no changes) 1 - Errored 2 - Succeeded, there is a diff
-generate-config-out=path	(Experimental) If import blocks are present in configuration, instructs Terraform to generate HCL for any imported resources not already present. The configuration is written to a new file at PATH, which must not already exist. Terraform may still attempt to write configuration if the plan errors.
-input=true	Ask for input for variables if not directly set.
-lock=false	Don't hold a state lock during the operation. This is dangerous if others might concurrently run commands against the same workspace.
-lock-timeout=0s	Duration to retry a state lock.
-no-color	If specified, output won't contain any color.
-out=path	Write a plan file to the given path. This can be used as input to the "apply" command.
-parallelism=n	Limit the number of concurrent operations. Defaults to 10.
-state=statefile	A legacy option used for the local backend only. See the local backend's documentation for more information.

```
skpandey@Ss-MacBook-Air terraform-q4 %
```


Assignment 10

TOPIC : AWS Assignment S3

Date: 10/08/2026

Question 01: Write a bash script to create a s3 bucket. The bucket name should be created using a random function where you have no idea of the name of the bucket. Use some commands to retrieve the bucket name and cp a script file to it. Display the contents of the bucket and delete it.

Answer:

```
skpandey@Ss-MacBook-Air ~ % mkdir assignment10
skpandey@Ss-MacBook-Air ~ % cd assignment10
skpandey@Ss-MacBook-Air assignment10 % pwd
/Users/skpandey/assignment10
skpandey@Ss-MacBook-Air assignment10 % nano q1.sh
skpandey@Ss-MacBook-Air assignment10 % chmod +x q1.sh
skpandey@Ss-MacBook-Air assignment10 % ./q1.sh
make_bucket: mybucket-1787602855
upload: ./script.txt to s3://mybucket-1787602855/script.txt
Bucket contents:
2026-08-25 01:51:00          9 script.txt
delete: s3://mybucket-1787602855/script.txt
remove_bucket: mybucket-1787602855
skpandey@Ss-MacBook-Air assignment10 % █
```

```
UW PICD 5.09 File: q1.sh
#!/bin/bash
BUCKET="mybucket-$(date +%s)"
aws s3 mb s3://$BUCKET
echo "Hello S3" > script.txt
aws s3 cp script.txt s3://$BUCKET
echo "Bucket contents:"
aws s3 ls s3://$BUCKET
aws s3 rb s3://$BUCKET --force
```

Question 02: Write a bash script to modify the access control policy of an existing bucket.

Answer:

```
skpandey@Ss-MacBook-Air assignment10 % aws s3 mb s3://mybucket-policy-$(date +%s)
make_bucket: mybucket-policy-1787603014
skpandey@Ss-MacBook-Air assignment10 % nano policy.json
skpandey@Ss-MacBook-Air assignment10 % aws s3api put-bucket-policy --bucket mybucket-policy-1787603014 --policy file://policy.json
aws: [ERROR] An error occurred (AccessDenied) when calling the PutBucketPolicy operation: User: arn:aws:iam::500692702174:user/saumya is not authorized to perform: s3:PutBucketPolicy on resource: "arn:aws:s3:::mybucket-policy-1787603014" because public policies are prevented by the BlockPublicPolicy setting in S3 Block Public Access.
skpandey@Ss-MacBook-Air assignment10 % nano policy.json
skpandey@Ss-MacBook-Air assignment10 % aws s3api put-bucket-policy --bucket mybucket-policy-1787603014 --policy file://policy.json
skpandey@Ss-MacBook-Air assignment10 % aws s3api get-bucket-policy --bucket mybucket-policy-1787603014
{
  "Policy": "{\n\"Version\": \"2012-10-17\", \"Statement\": [{\n\"Effect\": \"Allow\", \"Principal\": {\n\"AWS\": \"arn:aws:iam::500692702174:user/saumya\"}, \"Action\": \"s3:GetObject\", \"Resource\": \"arn:aws:s3:::mybucket-policy-1787603014/*\"}]]}"
}
skpandey@Ss-MacBook-Air assignment10 % █
```

```
UW PICD 5.09 File: policy.json
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::500692702174:user/saumya"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::mybucket-policy-1787603014/*"
    }
  ]
}
```

Question 03: Write a bash script to create a bucket and host a static website in it. Is the website accessible through curl and through browser?

Answer:

```
skpandey@Ss-MacBook-Air assignment10 % nano q3.sh
skpandey@Ss-MacBook-Air assignment10 % chmod +x q3.sh
skpandey@Ss-MacBook-Air assignment10 % ./q3.sh
make_bucket: mywebsite-1787603308
upload: ./index.html to s3://mywebsite-1787603308/index.html
Website: http://mywebsite-1787603308.s3-website-ap-northeast-1.amazonaws.com
skpandey@Ss-MacBook-Air assignment10 % aws s3api put-public-access-block --bucket mywebsite-1787603308 --public-access-block-configuration BlockPublicAcl=false,IgnorePublicAcl=false,BlockPublicPolicy=false,RestrictPublicBuckets=false
skpandey@Ss-MacBook-Air assignment10 % nano website-policy.json
skpandey@Ss-MacBook-Air assignment10 % aws s3api put-bucket-policy --bucket mywebsite-1787603308 --policy file://website-policy.json
skpandey@Ss-MacBook-Air assignment10 % curl http://mywebsite-1787603308.s3-website-ap-northeast-1.amazonaws.com
<h1>Hello from S3</h1>
skpandey@Ss-MacBook-Air assignment10 %
```

```
UW PICO 5.09 File: q3.sh

#!/bin/bash

BUCKET="mywebsite-${date +%s}"

aws s3 mb s3://$BUCKET

echo "<h1>Hello from S3</h1>" > index.html

aws s3 cp index.html s3://$BUCKET

aws s3 website s3://$BUCKET --index-document index.html

echo "Website: http://$BUCKET.s3-website-ap-northeast-1.amazonaws.com"
```

```
UW PICO 5.09 File: website-policy.json

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::mywebsite-1787603308/*"
    }
  ]
}
```



Hello from S3

Question 04: Using AWS CLI create an Ec2 instance. Use some command to display the publicdns address of the ec2 instance. Using the publicdns ssh into the instance through the script. Write another remote script that should be copied to the ec2 instance. This script should contain steps to create a bucket and attach an access point to it.

Answer:

```

assignment10 — ubuntu@ip-172-31-16-249: ~ — ssh -i ~/Downloads/terraf...

+-----+
|                CidrIp                |
+-----+
|  49.42.135.148/32                    |
+-----+

skpandey@Ss-MacBook-Air assignment10 % ./q4.sh
Instance ID: i-03b335e18a64d75cc
Public DNS: ec2-18-180-226-232.ap-northeast-1.compute.amazonaws.com
ssh: connect to host ec2-18-180-226-232.ap-northeast-1.compute.amazonaws.com port
22: Operation timed out
lost connection
^C
skpandey@Ss-MacBook-Air assignment10 % curl ifconfig.me
curl: (28) Failed to connect to ifconfig.me port 80 after 75041 ms: Operation time
d out
skpandey@Ss-MacBook-Air assignment10 % aws ec2 describe-security-groups --group-id]
s sg-0469fcb003b8da5bf --query "SecurityGroups[0].IpPermissions[0].IpRanges" --out
put table

-----
|DescribeSecurityGroups|
+-----+
|                CidrIp                |
+-----+
|  49.42.135.148/32                    |
+-----+

ubuntu@ip-172-31-16-249:~$ aws sts get-caller-identity
{
  "UserId": "AIDAXJE44KPPB4UNT4E3E",
  "Account": "500692702174",
  "Arn": "arn:aws:iam::500692702174:user/saumya"
}
ubuntu@ip-172-31-16-249:~$ ./remote.sh
make_bucket: accesspoint-1787635650
Bucket created: accesspoint-1787635650
{
  "AccessPointArn": "arn:aws:s3:ap-northeast-1:500692702174:accesspoint/myaccesspoint",
  "Alias": "myaccesspoint-bgfokqt7oj8q8338xm9etdpnm8bwapn1a-s3alias"
}
Access point created
ubuntu@ip-172-31-16-249:~$

```


PROJECT

Three-Tier Application on AWS using Terraform and Docker

Problem Statement:

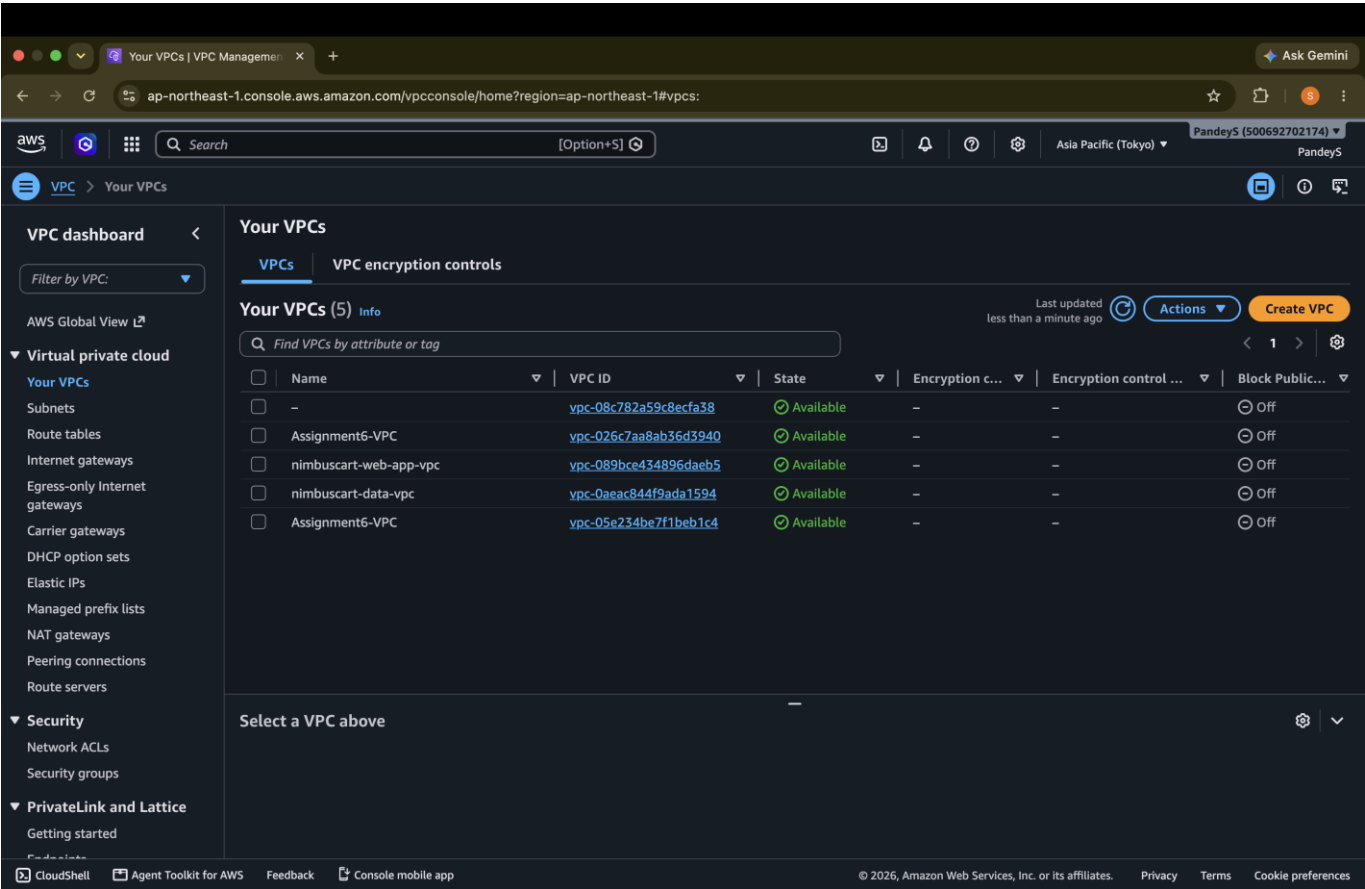
The project requires building NimbusCart, a lightweight product-catalog application, to demonstrate a three-tier architecture on AWS provisioned entirely through Terraform. The system consists of a static frontend served by nginx on a public web-tier EC2 instance, which reverse-proxies API calls to a containerized REST API running on a private app-tier instance. The REST API, built with a framework of the student's choice (Flask, Express, or FastAPI), exposes health-check, item-listing, and item-creation endpoints, and persists product data in a single products table within a PostgreSQL or MySQL database hosted in an isolated data-tier VPC that is reachable only through VPC peering. The API is responsible for creating its own database schema on first boot, with no manual database setup allowed. The entire infrastructure - VPCs, subnets, EC2 instances, VPC peering, routing, and application deployment - must be automated through a single bash script that runs Terraform init, plan, and apply, backed by remote state stored in S3 with DynamoDB locking.

Three-Tier Application on AWS using Terraform and Docker

VPC :

Figure 1: NimbusCart VPC

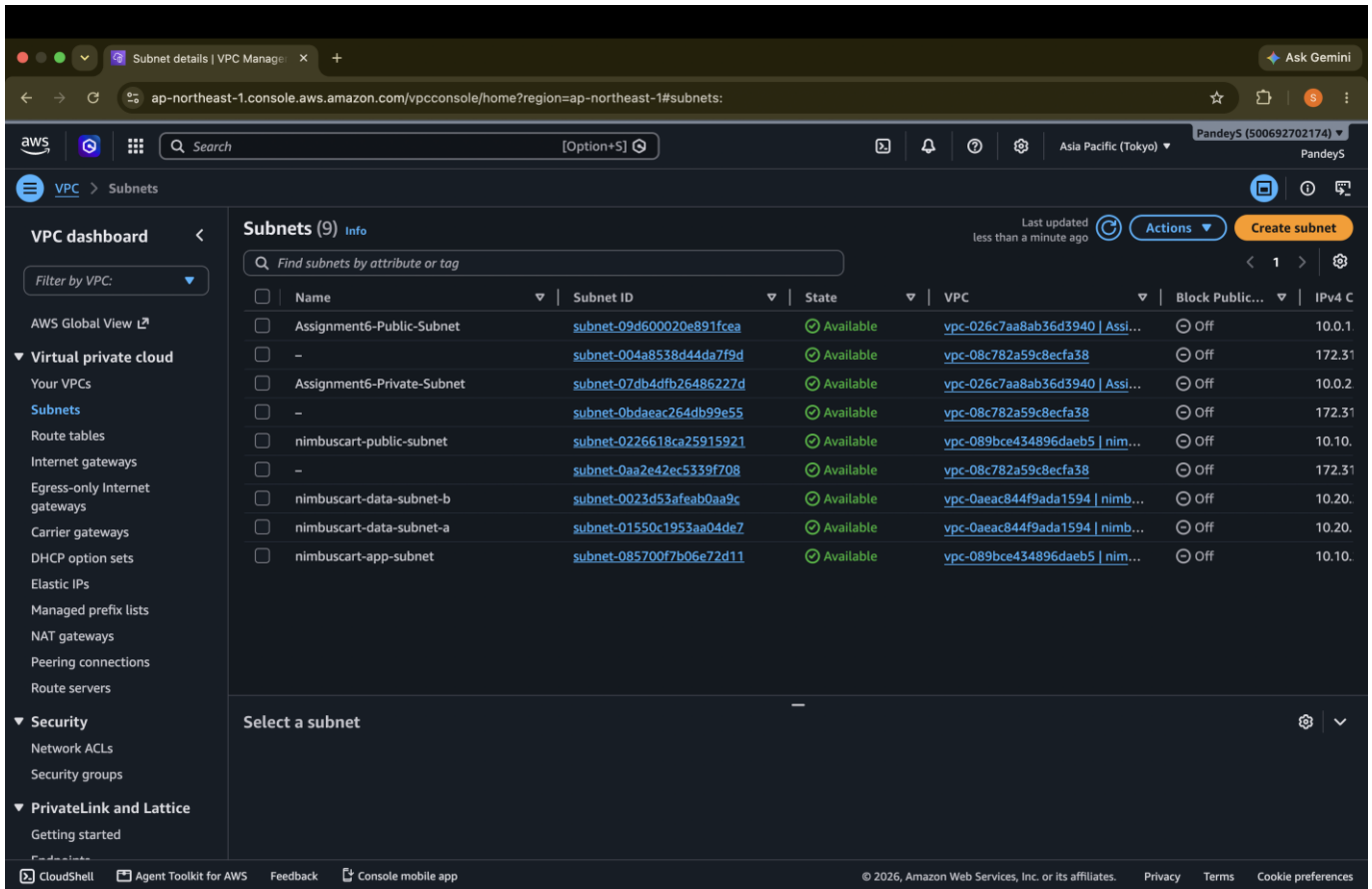
Shows the VPC created for the NimbusCart application.



Subnets:

Figure 2: NimbusCart Subnets

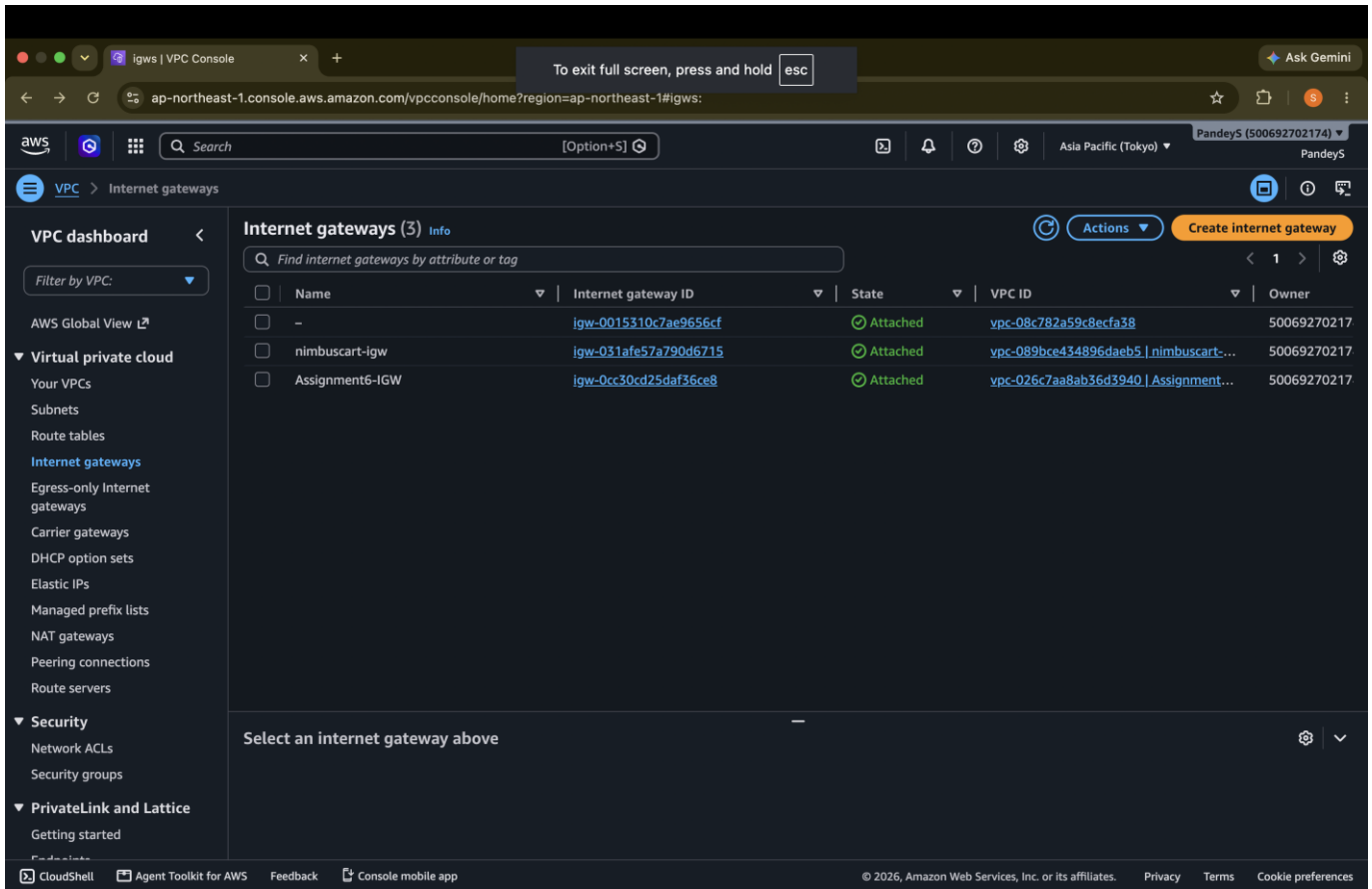
Shows the public, application, and database subnets configured within the VPC.



Internet Gateway:

Figure 3: Internet Gateway

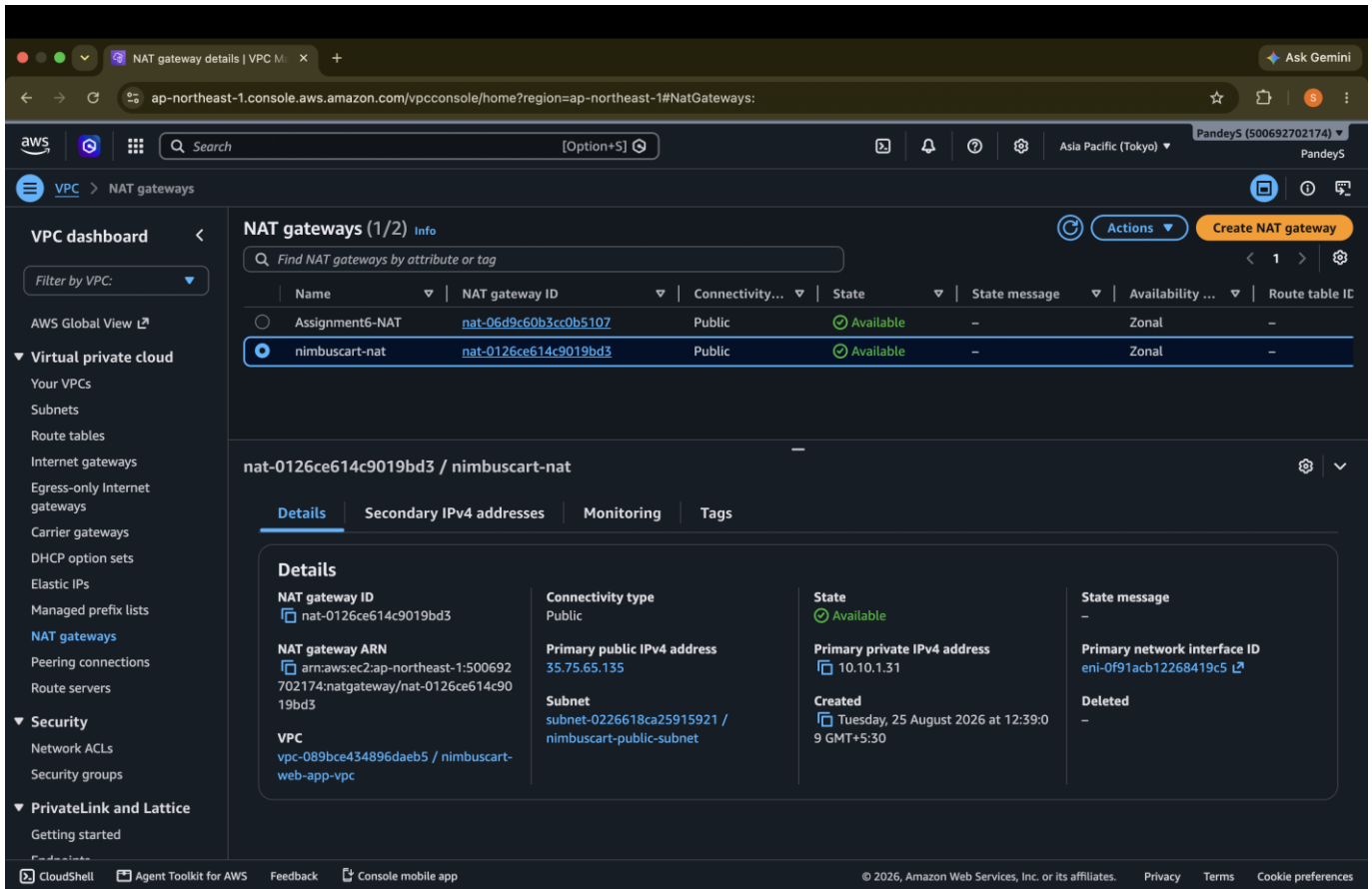
Shows the Internet Gateway attached to the NimbusCart VPC for internet connectivity.



NAT Gateway:

Figure 4: NAT Gateway

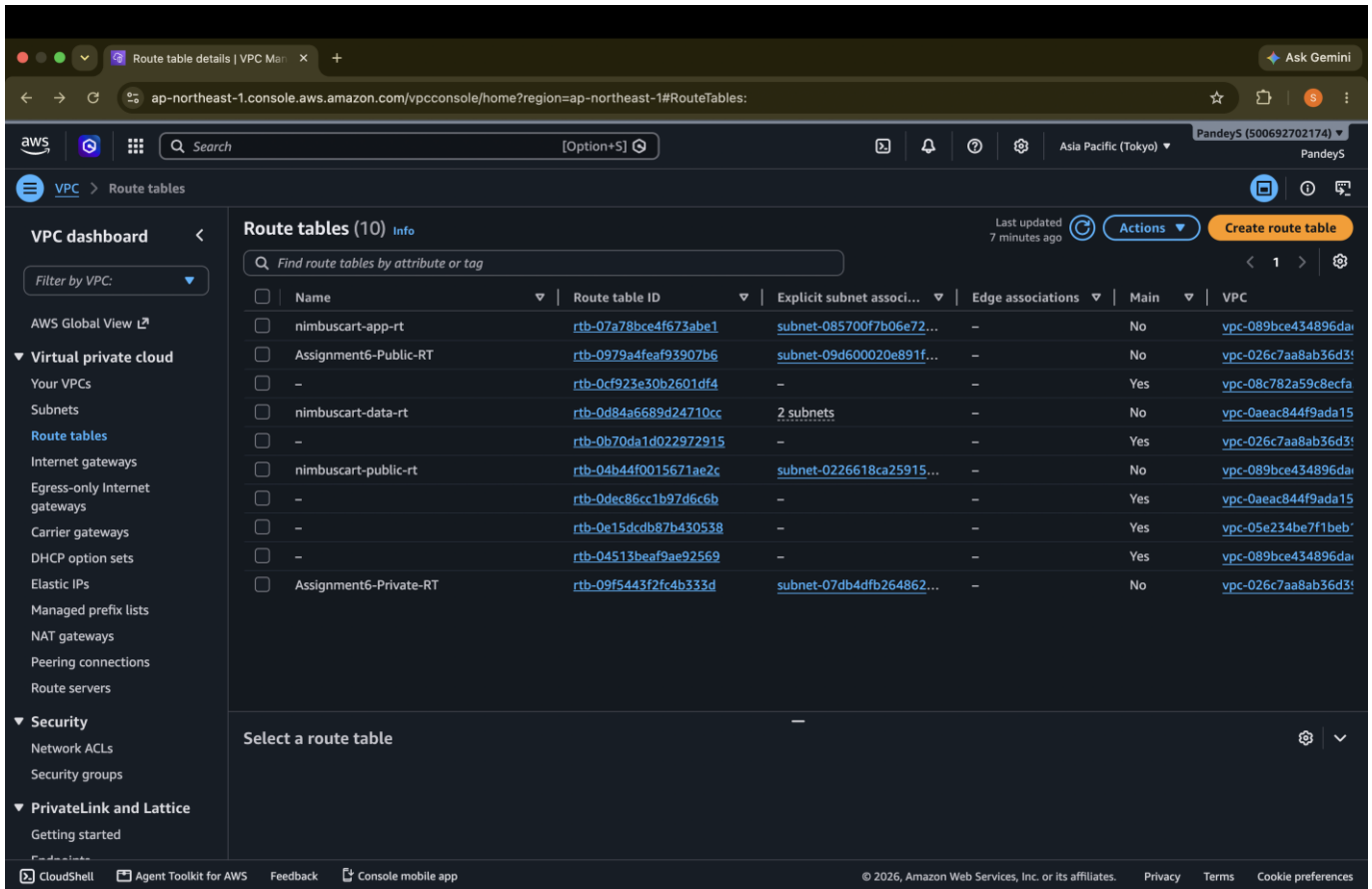
Shows the NAT Gateway used to provide outbound internet access to resources in the private subnet.



Route Tables:

Figure 5: Route Tables

Shows the routing configuration for public and private subnets, including Internet Gateway and NAT Gateway routes.



Security Groups:

Figure 6: Security Groups

Shows the security rules configured for the web, application, and database tiers to control network access.

Security Groups

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

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Getting started

Endpoints

Endpoint services

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Resource configurations

Resource gateways

Target groups

Domain verifications

DNS firewall

Rule groups

Security Groups (1/15) Info

Find security groups by attribute or tag

	Name	Security group ID	Security group name	VPC ID	Description
☒	-	sg-05d293937c28a5808	nimbuscart-web-sg	vpc-089bce434896daeb5	Managed
☐	-	sg-07880b5272ad7e9e4	nimbuscart-db-sg	vpc-0aeac844f9ada1594	Managed
☐	-	sg-0b69d519855fcb8e8d	Assignment6-Public-SG	vpc-026c7aa8ab36d3940	Assignment6-Public-SG
☐	-	sg-0464596ce89f96a4b	Assignment7-Web2-SG	vpc-026c7aa8ab36d3940	Assignment7-Web2-SG
☐	-	sg-0469fcb003b8da5bf	dockersg	vpc-08c782a59c8ecfa38	allow ssh
☐	-	sg-0a90ffd84d0162b70	Assignment7-Web1-SG	vpc-026c7aa8ab36d3940	Assignment7-Web1-SG
☐	-	sg-0ef6792ca24dbbc10	default	vpc-05e234be7f1beb1c4	default VPC
☐	-	sg-0ac646e593b3a9c95	default	vpc-026c7aa8ab36d3940	default VPC
☐	-	sg-06c91224e57652e65	Assignment6-Private-SG	vpc-026c7aa8ab36d3940	Assignment6-Private-SG

sg-05d293937c28a5808 - nimbuscart-web-sg

Details

Security group name

nimbuscart-web-sg

Security group ID

sg-05d293937c28a5808

Description

Managed by Terraform

VPC ID

vpc-089bce434896daeb5

Owner

500692702174

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

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Security Groups

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

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Security Groups (1/15) Info

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	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-05d293937c28a5808	nimbuscart-web-sg	vpc-089bce434896daeb5	Managed
☒	-	sg-07880b5272ad7e9e4	nimbuscart-db-sg	vpc-0aeac844f9ada1594	Managed
☐	-	sg-0b69d519855fcb8e8d	Assignment6-Public-SG	vpc-026c7aa8ab36d3940	Assignment6-Public-SG
☐	-	sg-0464596ce89f96a4b	Assignment7-Web2-SG	vpc-026c7aa8ab36d3940	Assignment7-Web2-SG
☐	-	sg-0469fcb003b8da5bf	dockersg	vpc-08c782a59c8ecfa38	allow ssh
☐	-	sg-0a90ffd84d0162b70	Assignment7-Web1-SG	vpc-026c7aa8ab36d3940	Assignment7-Web1-SG
☐	-	sg-0ef6792ca24dbbc10	default	vpc-05e234be7f1beb1c4	default VPC
☐	-	sg-0ac646e593b3a9c95	default	vpc-026c7aa8ab36d3940	default VPC

sg-07880b5272ad7e9e4 - nimbuscart-db-sg

Details

Security group name

nimbuscart-db-sg

Security group ID

sg-07880b5272ad7e9e4

Description

Managed by Terraform

VPC ID

vpc-0aeac844f9ada1594

Owner

500692702174

Inbound rules count

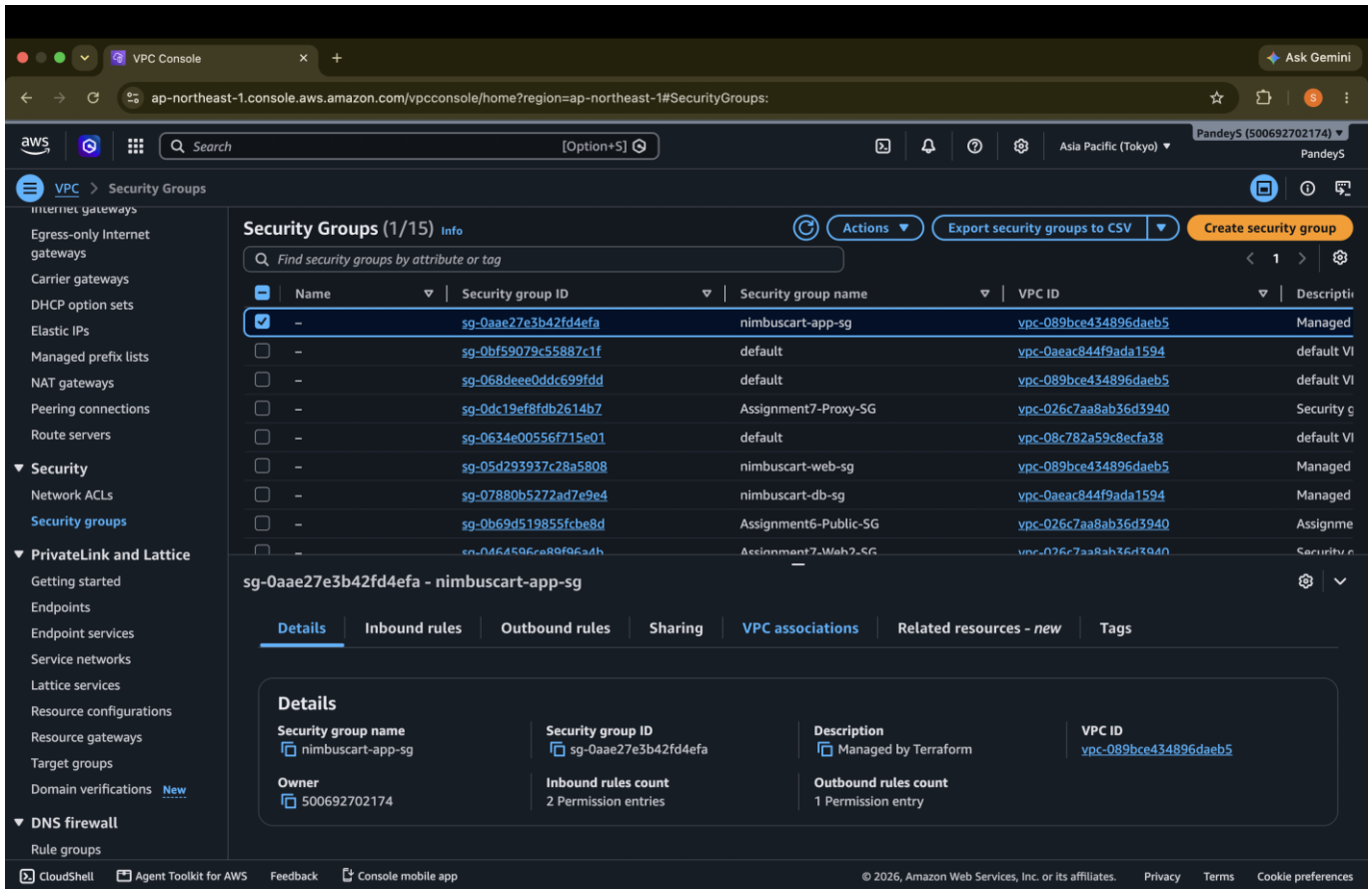
1 Permission entry

Outbound rules count

1 Permission entry

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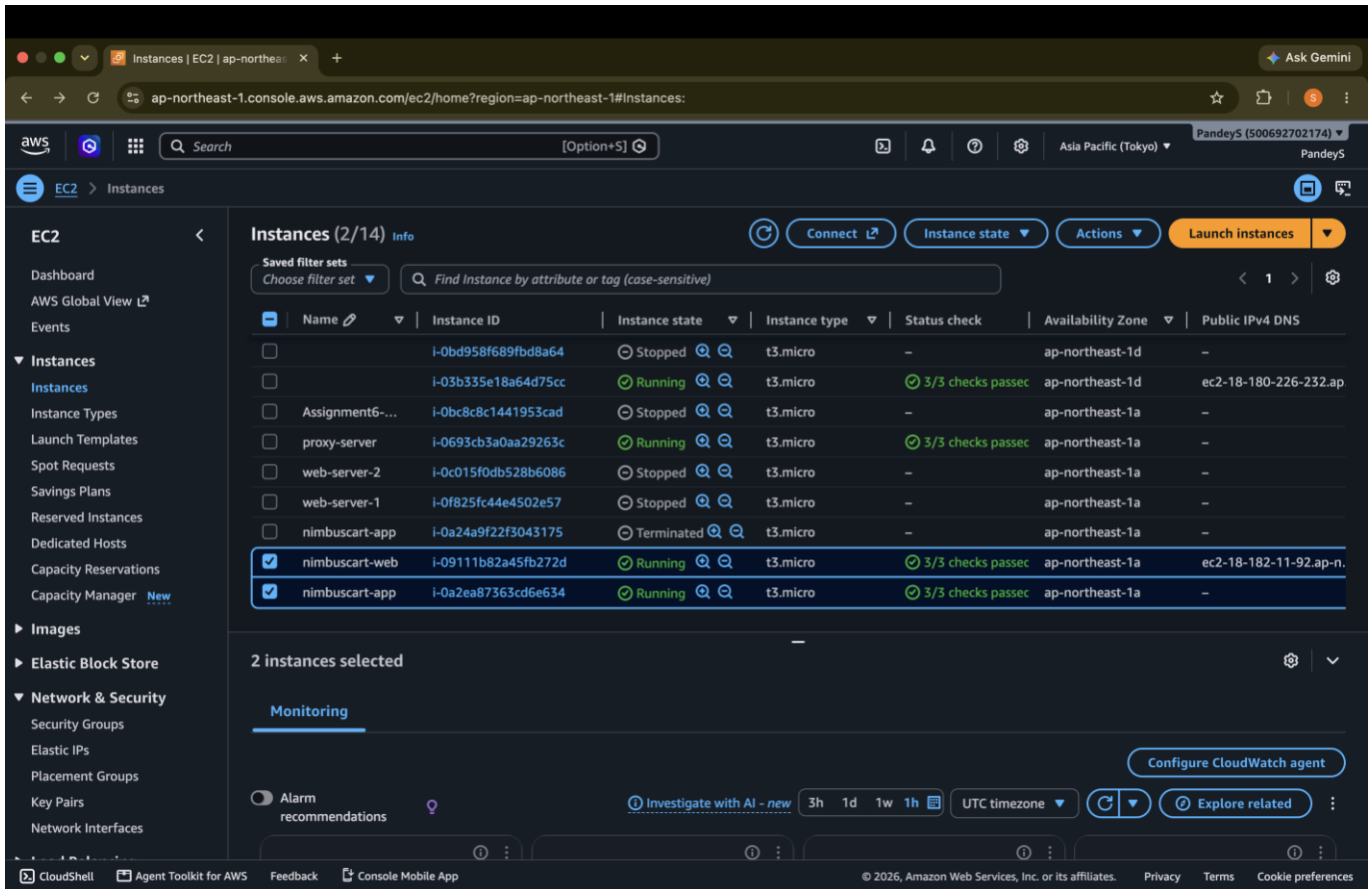
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EC2 Instances:

Figure 7: EC2 Instances

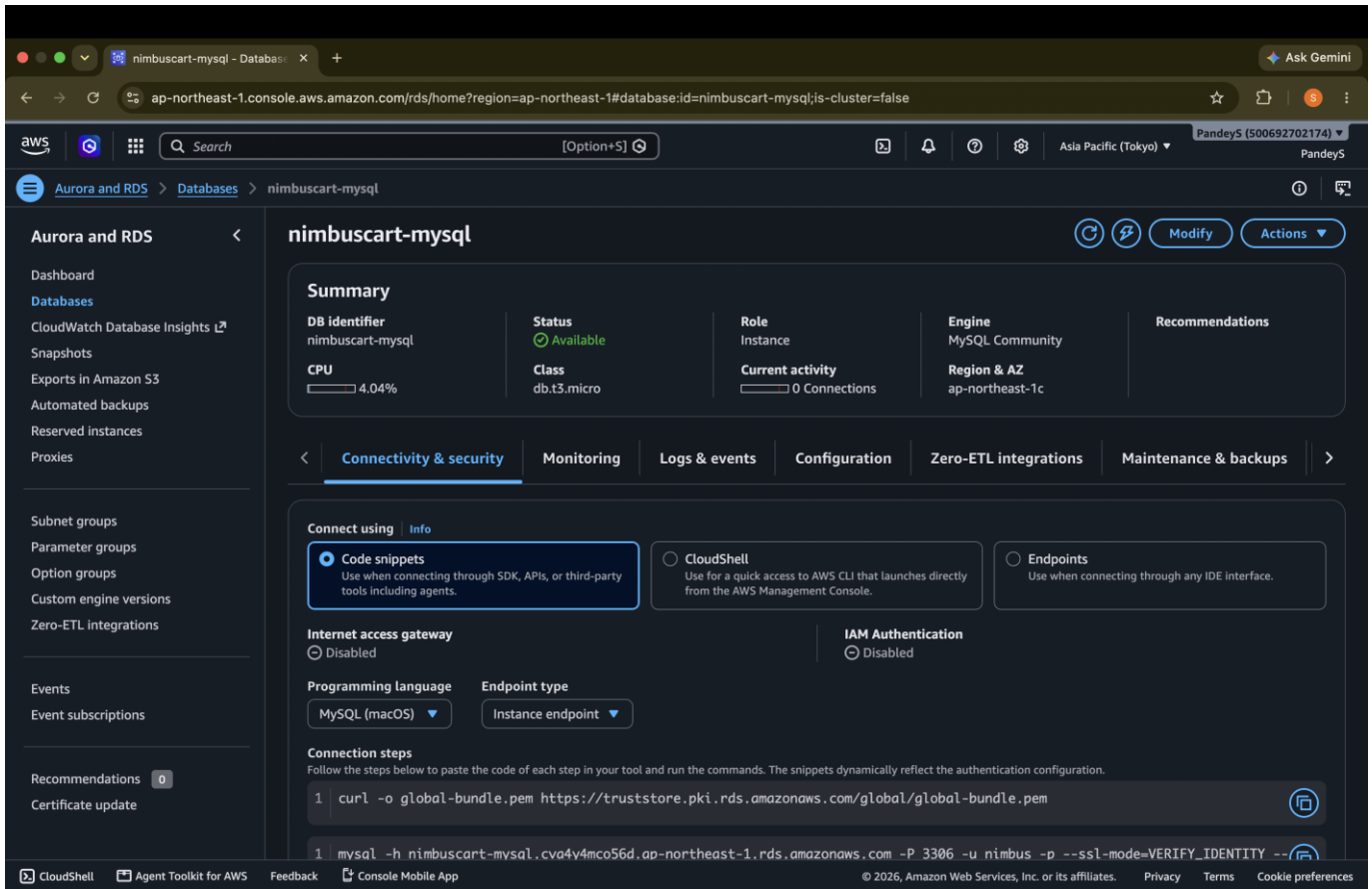
Shows the running Web and Application EC2 instances used to host the NimbusCart application.



RDS Database:

Figure 8: Amazon RDS MySQL Database

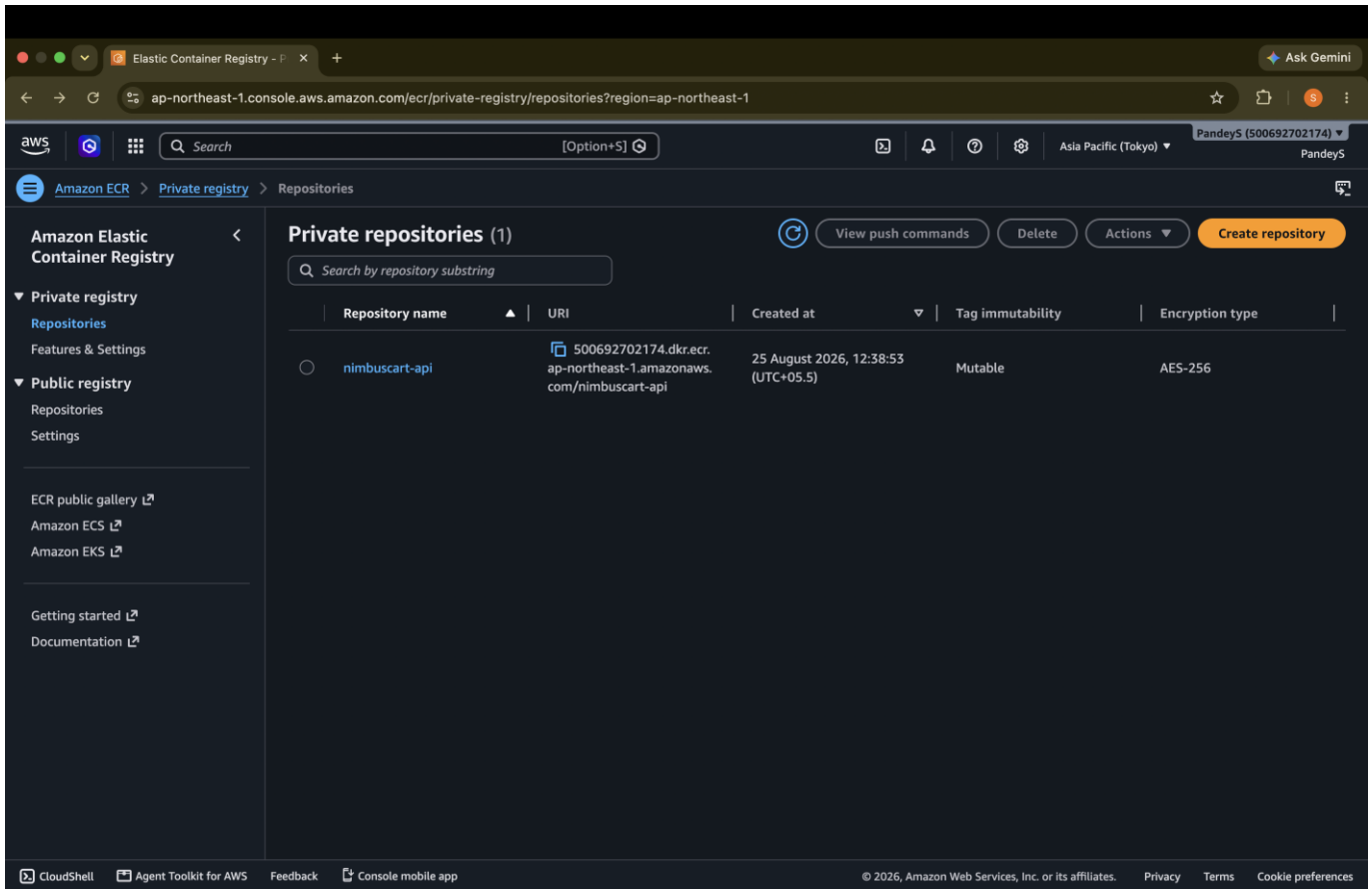
Shows the NimbusCart MySQL database running on Amazon RDS and its configured endpoint.



ECR Repository:

Figure 9: Amazon ECR Repository

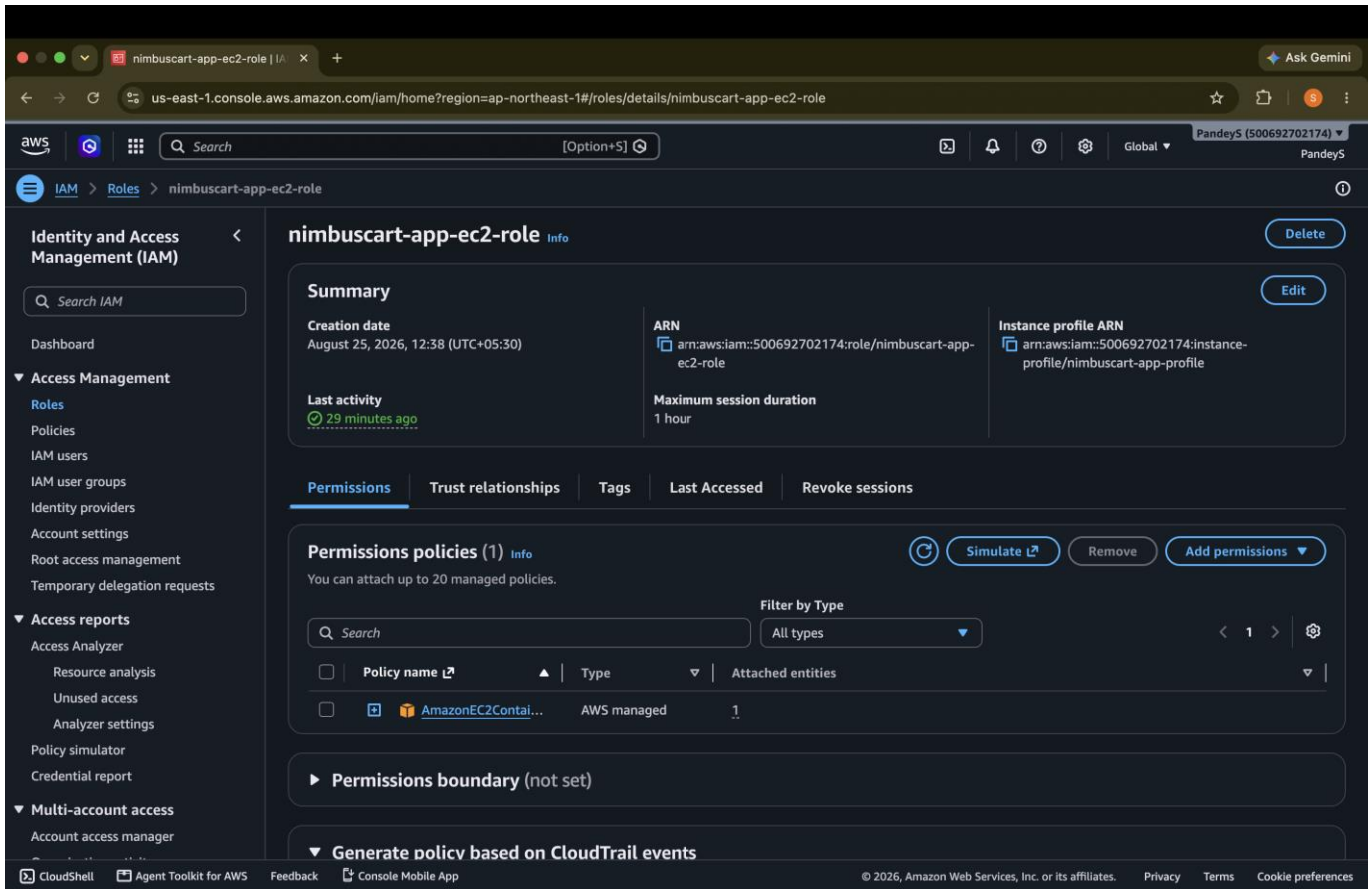
Shows the Docker container image repository used to store the NimbusCart API image.



IAM Role:

Figure 10: IAM Role for Application EC2

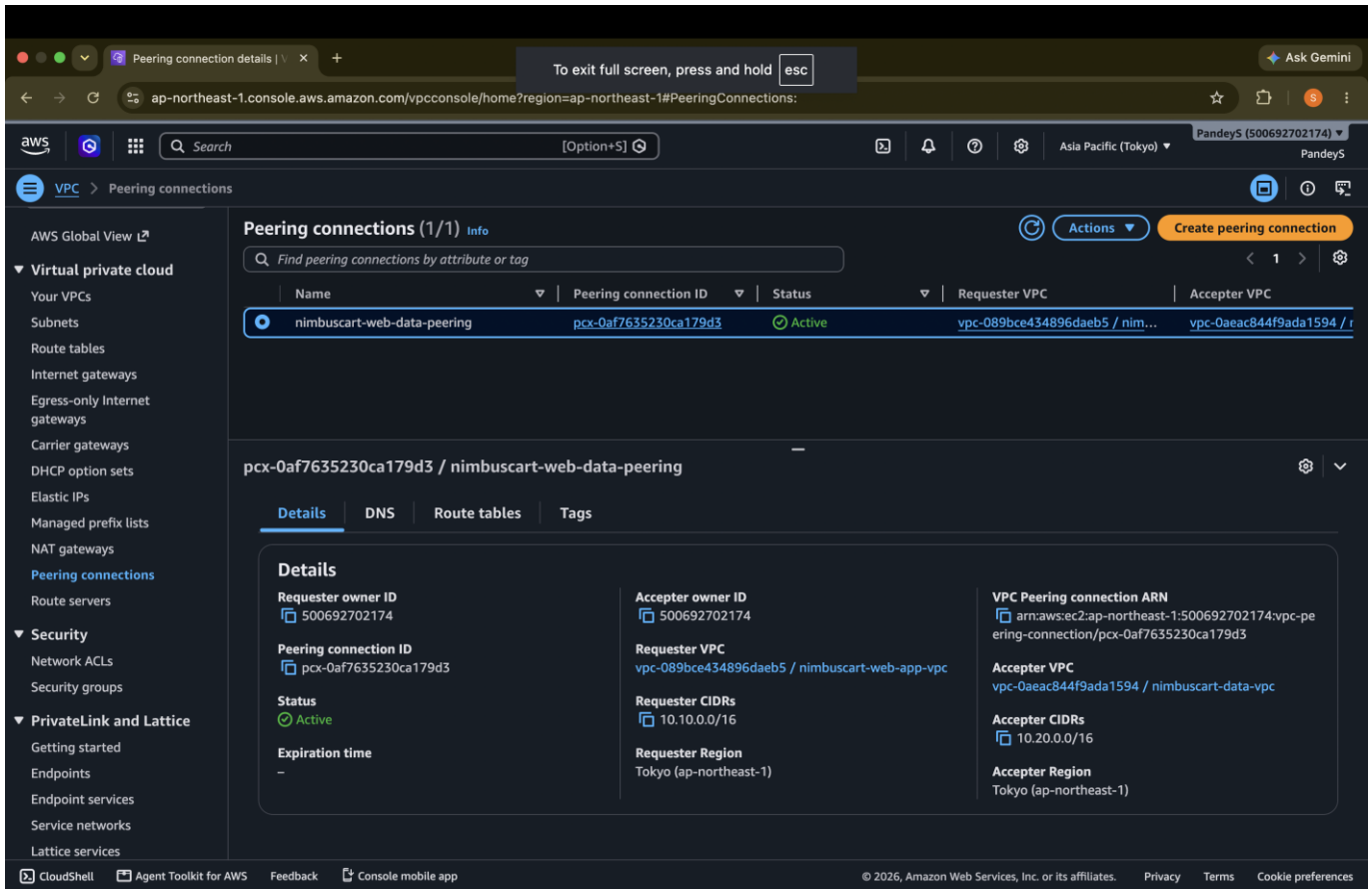
Shows the IAM role attached to the application EC2 instance for required AWS service permissions.



VPC Peering:

Figure 11: VPC Peering Connection

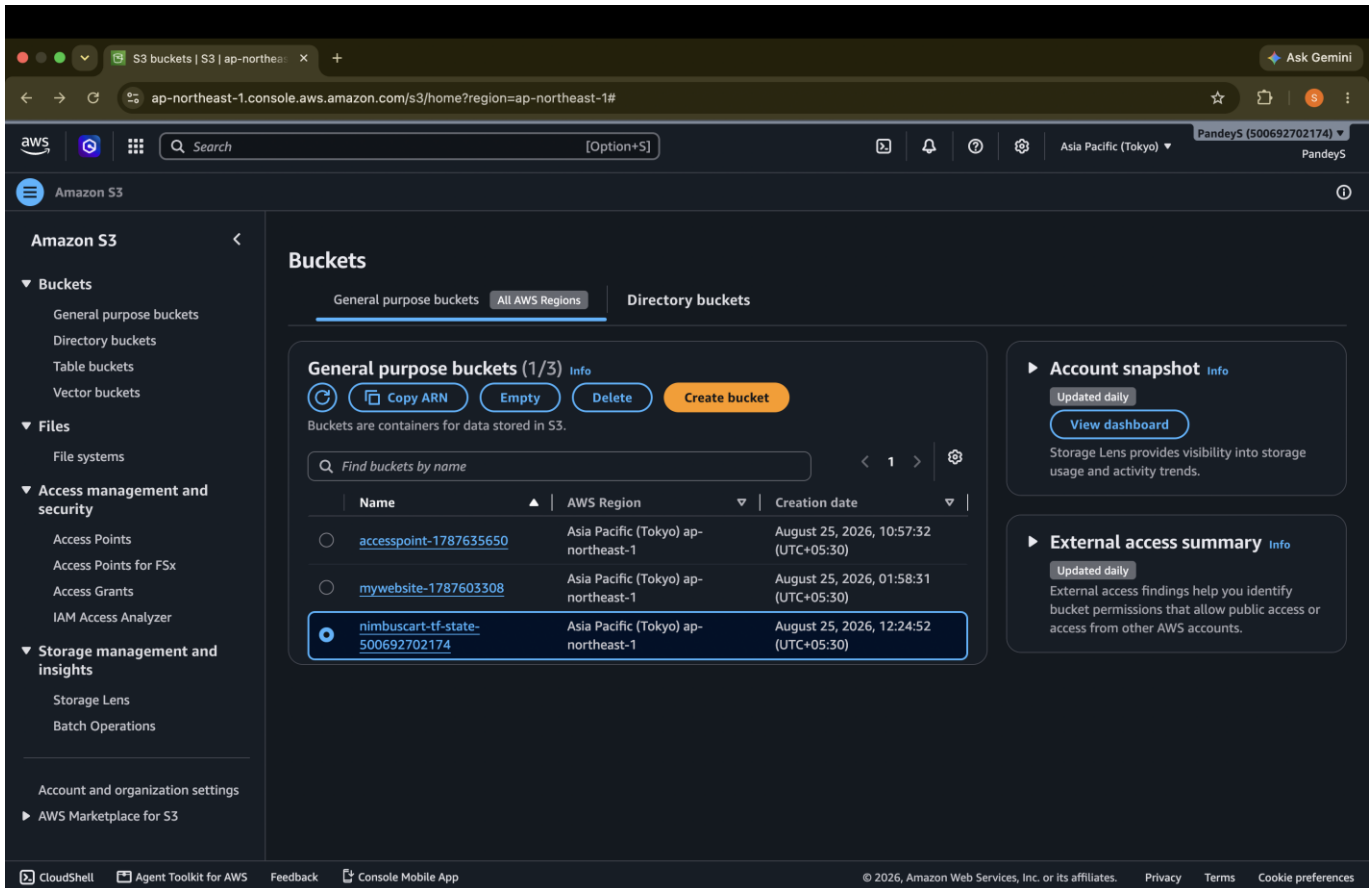
Shows the active VPC peering connection configured for communication between the required VPC networks.



S3 Bucket:

Figure 12: Amazon S3 Bucket

Shows the S3 bucket created as part of the NimbusCart AWS infrastructure.



TerraForm Output, Docker Container, API Health Check :

Figure 13: Terraform Deployment Outputs

Shows the Terraform outputs including the database endpoint, ECR repository, web server public IP, NAT Gateway IP, and VPC peering connection.

Figure 14: Running NimbusCart API Container

Shows the NimbusCart API Docker container running on the application EC2 instance.

Figure 15: Application Health Check

Shows a successful HTTP 200 response from the deployed NimbusCart API, confirming that the application is running correctly.

EXPLORER

NIMBUSCARTPROJECT

app

screenshots

terraform

terraform.lock.hcl

backend.tf

main.tf

outputs.tf

terraform.tfvars

terraform.tfvars.ex...

variables.tf

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REPORT.md

script.sh

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

Expanded Security Maintenance for Applications is not enabled.

72 updates can be applied immediately.

64 of these updates are standard security updates.

To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.

Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

Last login: Tue Aug 25 12:25:08 2026 from 10.10.1.57

ubuntu@ip-10-10-2-196:~\$ sudo docker ps

CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES

b3ccd94c9681 nimbus-api:latest "python app.py" 7 minutes ago Up 2 minutes 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp nimbus-api

ubuntu@ip-10-10-2-196:~\$ curl -i http://localhost:5000/health

HTTP/1.1 200 OK

Server: Werkzeug/3.1.8 Python/3.12.14

Date: Tue, 25 Aug 2026 12:28:54 GMT

Content-Type: application/json

Content-Length: 16

Access-Control-Allow-Origin: *

Connection: close

{"status":"ok"}

ubuntu@ip-10-10-2-196:~\$ exit

logout

Connection to 10.10.2.196 closed.

ubuntu@ip-10-10-1-57:~\$ exit

logout

Connection to 18.182.11.92 closed.

skpandey@SS-MacBook-Air terraform % terraform output

db_endpoint = "nimbuscart-mysql.cva4y4mco56d.ap-northeast-1.rds.amazonaws.com"

ecr_repository_url = "500692702174.dkr.ecr.ap-northeast-1.amazonaws.com/nimbuscart-api"

frontend_url = "http://18.182.11.92"

nat_gateway_public_ip = "35.75.65.135"

peering_connection_id = "pcx-baf7635230ca179d3"

web_public_ip = "18.182.11.92"

Warning: Deprecated Parameter

The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.

skpandey@SS-MacBook-Air terraform % aws ec2 describe-instances \

--region ap-northeast-1 \

--filters "Name=instance-state-name,Values=running" \

--query 'Reservations[].Instances[].[ID:InstanceId,PrivateIP:PrivateIpAddress,State:State.Name]' \

--output table

DescribeInstances

table with 3 columns: ID, PrivateIP, State

skpandey@SS-MacBook-Air terraform % sudo docker ps

Password:

sudo: a password is required

skpandey@SS-MacBook-Air terraform %

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EXPLORER

main.tf

NIMBUSCARTPROJECT

app

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terraform

terraform.lock.hcl

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```
resource "aws_security_group" "db" {
  name = "${var.project_name}-db-sg"
  vpc_id = aws_vpc.data.id

  ingress {
    from_port = 3306
    to_port = 3306
    protocol = "tcp"
    cidr_blocks = [var.app_subnet_cidr]
  }

  egress {
    from_port = 0
    to_port = 0
    protocol = "-1"
    cidr_blocks = ["10.20.0.0/16"]
  }
}
```

PROBLEMS

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DEBUG CONSOLE

TERMINAL

Press CTRL+C to quit

ubuntu@ip-10-10-2-196:~\$ curl -i http://localhost:5000/health

HTTP/1.1 200 OK

Server: Werkzeug/3.1.8 Python/3.12.14

Date: Tue, 25 Aug 2026 12:26:22 GMT

Content-Type: application/json

Content-Length: 16

Access-Control-Allow-Origin: *

Connection: close

{"status":"ok"}

ubuntu@ip-10-10-2-196:~\$ exit

logout

Connection to 10.10.2.196 closed.

ubuntu@ip-10-10-1-57:~\$ exit

logout

Connection to 18.182.11.92 closed.

skpandey@SS-MacBook-Air terraform % curl -i http://18.182.11.92/health

HTTP/1.1 200 OK

Server: nginx/1.24.0 (Ubuntu)

Date: Tue, 25 Aug 2026 12:27:32 GMT

Content-Type: application/json

Content-Length: 16

Connection: keep-alive

Access-Control-Allow-Origin: *

{"status":"ok"}

skpandey@SS-MacBook-Air terraform %

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main.tf

terraform > main.tf

main.tf

224
225
226
227
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238
239
240

```
resource "aws_security_group" "db" {  
  name = "${var.project_name}-db-sg"  
  vpc_id = aws_vpc.data.id  
  
  ingress {  
    from_port = 3306  
    to_port = 3306  
    protocol = "tcp"  
    cidr_blocks = [var.app_subnet_cidr]  
  }  
  
  egress {  
    from_port = 0  
    to_port = 0  
    protocol = "-1"  
    cidr_blocks = ["10.20.0.0/16"]  
  }  
}
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

zsh - terraform

https://ubuntu.com/aws/pro

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ubuntu@ip-10-10-2-196:~\$ curl -i http://localhost:5000/health
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.14
Date: Tue, 25 Aug 2026 12:28:54 GMT
Content-Type: application/json
Content-Length: 16
Access-Control-Allow-Origin: *
Connection: close

{ "status": "ok" }
ubuntu@ip-10-10-2-196:~\$

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Spaces: 2

UTF-8

LF

Plain Text

Go Live